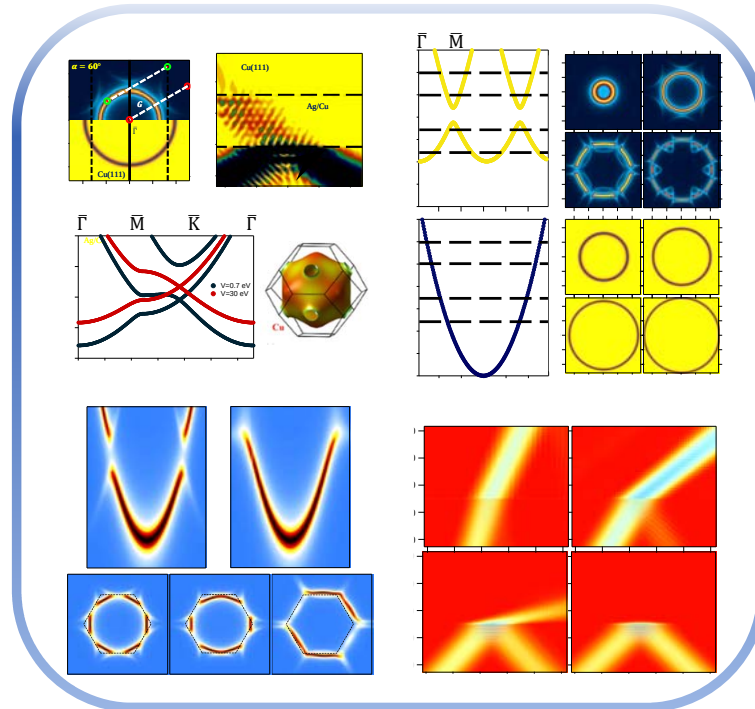




**Al-Azhar University
Faculty of Science
Physics Department**

Theoretical Routes Toward Surface States Nanoelectronics



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Submitted for the degree of PhD in Theoretical Physics

Egypt, 2017

Theoretical Routes Toward Surface States Nanoelectronics

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A thesis submitted in total fulfillment of the requirements for
the degree of Doctor of Philosophy

**Al-Azhar University
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Egypt
2017

Abstract

THE bulk band structure of metal single crystals often exhibit energy gaps at some high symmetry points in the Brillouin zone (BZ). Therefore, for specific surface terminations, the surface band structure reveals bulk-projected energy gaps, similar to semiconductors, where new electronic states might exist. These are called “*surface states*” when their full dispersion is positioned inside the projected gaps. Noble metal surfaces, for example, host such Shockley-type surface states characterized by free-electronic parabolic dispersions. Their Fermi wavelength is typically 1-3 nm and, therefore, two dimensional (2D) periodic structures within this order are required to fabricate surface state based devices. Techniques, such as self-assembly, made the fabrication of these nanostructures possible.

USING electron plane wave expansion (PWE) method, we present detailed and systematic theoretical analysis of electron scattering in the presence of 2D hexagonal and square superlattices on metal surfaces. In both cases, the surface state band minima were found to shift upward in energy with the magnitude of the scattering potential or for reduced periodicity. Most importantly, energy gaps are opened at the BZ boundaries depending on the scattering potential, the periodicity, and the shape of the scatterers. These findings were obtained after the full analysis of the calculated band structures, constant energy surfaces (CES), and local density of states (LDOS).

REFRACTION of electromagnetic waves across the interface separating two optically dissimilar homogeneous and non-dispersive media is best described by Snell’s law. This law is a consequence of the, more general, momentum conservation constrains at the particular interface, which requires that the parallel component of the wave vector at the interface are conserved. Given the well-established wave-particle duality, matter waves

also exhibit refraction at the interface between two different media. Therefore, refraction of surface electrons, for example, can occur at the interface separation two electronically dissimilar materials. The important quantity is the wave vector difference (Δk) at the particular energy, which is solely decided by the dispersion relations.

USING combined wavevector-space analysis and real-space simulations obtained by EPWE and electron boundary element method (EBEM), respectively, we investigate electron reflection-refraction for homogeneous-homogeneous and homogeneous-nanostructured metal interfaces. For both interfaces, we showed that surface electrons, indeed, refract following Snell's law, and a total internal reflection occurs beyond a critical angle. Interestingly, additional extraordinary optical phenomena were demonstrated for the homogeneous-nanostructured interface, such as negative refractions, self-collimation, and superprism effect. The homogeneous-nanostructured system presented in the thesis is the lateral interface between the dislocation-1 ML Ag/Cu(111) system and Cu(111), which could be fabricated experimentally .

Many Thanks To

“Allah”

And To

My Parents

My Wife

My Sons

My Brothers

My Closest Friends

For their support and caring attitude which kept me going until the end
of this work.

Acknowledgements

Thanks to “**Allah**” for giving me the ability and patience to do this work.

The thesis would not have been possible without the contribution, guidance and help from several people who extended their valuable assistance in the preparation and completion of this study.

Firstly, I would like to express my sincere gratitude to my advisors Prof. “**Mohammed Mahmoud El Okr**”, Prof. of solid state physics, Faculty of Science, Al-Azhar University for the continuous support of my Ph.D study and related research, for his patience, motivation, and immense knowledge. His guidance helped me in all the time of research and writing of this thesis.

I would like to express my special appreciation and thanks to my advisor Professor Dr. “**Osama Mohammed Yassin**”, Asst. Prof of theoretical physics, Faculty of Science Al-Azhar University, you have been a tremendous mentor for me. I would like to thank you for encouraging my research and for allowing me to grow as a research scientist. Your advice on both research as well as on my career have been invaluable.

I would like to express my deepest gratitude to my advisor Dr. “**Zakaria Mohamed Abd-Elfattah**”, lecturer of solid state physics, Faculty of Science, Al-Azhar University, for suggesting the point of this thesis, continues guidance, supervision and encouragement during the fulfillment of the present thesis. I could not have imagined having a better advisor and mentor for my Ph.D study.

I gratefully acknowledge Prof. “**F. J. Garcra de Abajo**” for providing the EPWE and EBEM codes.

I would also like to acknowledge fruitful discussions with Prof. “**J. E. Ortega**” and his careful reading of the manuscript that came out from the thesis.

Parts of the simulations presented in the thesis were run at the Oberon cluster belong to the “**Material Physics Center (CFM)**”, San Sebastian, Spain.

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1

General Introduction

Low-dimensional systems are, nowadays, one of the key areas in the modern condensed matter research. They provide a large number of diverse physical phenomena, which occur in spatial dimensions less than three. The low dimensionality intricately impacts the physics and often gives rise to exotic phenomena which are not, otherwise, available in three dimensional (3D) systems. A vast basic low-dimensional systems that have been of interest recently are, for example, quantum dots (zero dimensional)^[1], quantum wires^[2] and quantum wells (or barriers)^[3] (one dimensional), Thin films^[4,5], surfaces^[6,7], Graphene^[8] and Topological Insulators^[9] (two dimensional). All of these examples are central themes in “*nanotechnology*”. The concepts that seeded nanotechnology were first discussed in 1959 by renowned physicist Richard Feynman in his talk “*There’s Plenty of Room at the Bottom*”^[10], in which he described the possibility to synthesis nanostructures via direct manipulation of atoms. The definition of nanotechnology has generally been extended to include features as large as 100 nm, i.e., it is naturally interdisciplinary field that require knowledge from fields such as surface science, semiconductor physics, nanofabrication, organic chemistry, molecular biology, molecular engineering, etc^[11]. Surface science, in particular, is one of the major branches that provides the fundamental physics/science behind the low-dimensional nanotechnology.

In recent years, there has been considerable interest in the surface region of solid materials, which constitutes model of two dimensional (2D) systems. The surfaces of given 3D crystals generally have electronic and magnetic properties, for example, which are quite different from those of bulk counterpart^[12–14]. Similarly, thin films in the nanoscale region exhibit thickness-dependent properties, such as variation of optical band gap in semiconductors^[15] or Plasmon frequency in metals^[16]. It is now well es-

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established that the change of the electronic properties of materials when the size shrinks to the nanoscale stems from the quantum confinement, when the electron's wavelength became comparable to the object dimension(s). Additionally, for low dimensional systems, the surface-to-volume ratio is enhanced, making the properties of the materials mainly dominated by surface effects (contributions) together with a significant suppressing of the bulk properties. For 2D systems, in particular, the symmetry that a bulk possesses is broken at the surface. This broken symmetry renders the energetic of the surface different from the bulk counterpart^[17], which in turns might lead to the emergent of new electronic states characteristic for the surface, as shown next.

Metals have bands which cross the Fermi level at selected momentum-regions of the Brillouin zone. In other regions, however, gaps exist at the Fermi energy similar to those in insulators and semiconductors. When a 3D crystal is terminated by a specific surface orientation, new electronic states, namely “*surface states*”, can exist in these “*partial gaps*” (projected band gap) in metals. Here a partial gap means that there are restricted regions of the surface Brillouin zone (SBZ) that have no bulk states^[18]. One of the most well-known partial gaps in noble metals is the “*neck*” in the Fermi surface in the (111) direction. For all wavevector (k) values along the (111) direction, the central portion of this neck has no allowed states: i.e., a partial gap when projected to the (111) surface. A surface state has been found to exist in this gap. Several groups have measured these Shockley type surface states, which are characterized by a Fermi wavelength of the order of 1-3 nm^[19–24].

The present thesis concerns, in part, the calculation of the electronic structure of these Shockley-type surface states emerging at the surface of 2D metallic systems.

In the previous discussion on metal surfaces and the emerging surface states, the in-plane atomic periodicity of the surface is considered to be the same as the underneath layers of the bulk materials. This is the case for some metal substrate, where the top-most surface layer undergoes vertical relaxation^[25] only towards the surface while maintaining the lateral atomic periodicity. Generally speaking, the free energy of the surface might, thermodynamically, triggers different surface termination. Reconstruction^[26] and charge redistribution^[27] are such natural processes, where the in-plane periodicity changes to allow surface energy minimization. In many cases, the overall reconstruction process yields new surface periodicity much larger than the bulk atomic-scale lattice parameter^[28], and are, therefore, termed “*lateral superlattices*” or “*superstructures*”^[29]. These superlattices typically have nanoscale periodicities and

are also termed 2D “*nanostructures*”. Although these superlattices are naturally created on many surfaces, the fabrication of tunable superlattices of different periodicity, shapes, etc, goes beyond the natural reconstruction process.

Nowadays, by employing several surface nanofabrication tools, it is possible to grow a wide variety of materials in different configurations on metallic substrate surfaces thus; unique systems can be fashioned for practical applications or for scientific study. Both, the chemical and physical properties of substances can change in presence of surface nanostructure compared to the featureless surface^[30].

Some of the most important tools used in the nanofabrication research are self-assembly^[31–33] and lithography^[34–36] techniques. The self-assembly (including epitaxial and non-epitaxial) of nanostructures is an attractive topic in nanoscience^[37]. The concept of self-assembly is categorized as a “*bottom-up*” approach of nanostructure fabrication, in contrast to lithography which is a “*top-down*” approach^[38,39]. In a self-assembly process, inorganic or organic materials are deposited onto a given substrate with amounts as small as atomic sub-monolayer. The final pattern produced by this process depends on many factors, such as the interaction between the adsorbates (atoms/molecules), their interaction with the substrate, and/or a combination of both. For example, the use of pre-patterned surfaces to guide adsorption processes^[40], through engineering the substrate- adsorbate interaction, has been demonstrated on a variety of spontaneously nanostructured substrates such as dislocation networks^[41], vicinal surfaces^[42], and self-ordered biphase systems^[43,44]. Given the wide varieties of materials and substrate, the delicate balance between adsorbate-adsorbate and adsorbate-substrate interactions^[45] could be finely tuned to achieve the desired molecular or metallic nanoarchitecture assemblies on crystal surfaces. Such self-assembled nanopatterned surfaces found promising applications in all fields of nanoscience, particularly nanoelectronics^[46], chemical sensing^[47], and as template surfaces for selective molecular assembly^[48,49]. Their fabrication has progressed rapidly such that patterns with complex unit cells and almost of any arbitrary geometry could be fabricated^[50].

Electronically, these nanopatterns on noble substrate act as defects or scattering potentials for the, otherwise, free electrons. The interaction of those free electrons with 2D periodic arrays of scatterers modifies the parabolic dispersions of their Shockley surface states at different rates^[51,52]. For example, the surface state band minimum shifts towards higher binding energy together with the opening of energy gaps at the “*smaller*” superstructure Brillouin zones. These modifications could be tuned through both the barrier height and superlattice periodicity of these metallic nanopatterns.

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This has been reported for varieties of inorganic or organic superstructures induced on noble metal surfaces, such as the 1D array of steps in vicinal surfaces^[4], and the 2D hexagonal or honeycomb arrays in 1 ML Ag/Cu(111)^[53] and DPDI/Cu(111)^[54]. Additionally, the shape of the scattering barrier was also found to modify the final dispersion of the surface state, according to symmetry considerations. For example, a triangle scattering barrier in 1 ML Ag/Cu(111) system induces a band gap opening at the corner of the hexagonal Brillouin zone^[55]. This gap is completely absent in similar systems for a hexagonal or circular scatterer, such as the moiré 1 ML Ag/Cu system^[53,56].

In these regards, a major part of the thesis forethought theoretical investigation of electron scattering problem in 2D hexagonal and square superlattices formed on metal surfaces.

With the current advance in surface science, it is now possible to image these superstructures and probe the corresponding electronic surface structure. Since the overall focus is to correlate the resultant geometric/atomic and the electronic structure, comparison with existing data from experimental techniques is an important prerequisite. The most common primary experimental tools, in these regards, are scanning tunneling microscopy (STM)^[57], which is a powerful technique used to atomically resolve the images of surfaces, and angle-resolved photoemission spectroscopy (ARPES)^[58], which reveals the electronic structure of solid surfaces. STMs can reach a lateral resolution of sub-Angstrom^[59], where individual atoms within materials are routinely imaged and manipulated^[60]. In its spectroscopic mode, i.e., scanning tunneling spectroscopy (STS), the local density of state at any point in the surface could be obtained^[61]. On the other hand, ARPES is an active and a rapidly progressing technique that is used to reveal the relationships between electronic, magnetic, and chemical structures of novel materials by precisely measuring their electronic energy and momentum distributions^[62–64], i.e., the full band structure is measured.

Throughout the thesis, and in line with band structure calculations, simulations of ARPES and STS data will be given following different computational approaches.

Band structure and associated dispersion relations are the pillars of theoretical analysis in solid state physics, defining the essential characteristics of electronic states of a system^[65]. Its numerical calculations provide a relation between the energy and ultimately the momentum of the allowed propagating states in a solid. Typically, the allowed states cluster into bands of propagating states, separated by band gaps in which

no propagating states exist. The band structure describes the dispersive effect of a material on the wave propagating through it. Electronic structure calculation methods can be classified into two main categories^[66], namely the “*ab initio methods*” and “*the empirical methods*”. The *ab initio* methods, such as Hartree-Fock^[67,68] and Density Functional Theory (DFT)^[69,70], calculate the electronic structure from first principles, and sometimes require massively parallel computers, i.e., computationally expensive. Using some empirical parameters to fit the experimental data such as, the band minima and effective mass (obtained from photoemission measurements, ARPES) the ground-state and excited-state electronic dispersion can be calculated by the second category of the empirical methods. The *empirical methods* are computationally less expensive than *ab initio* calculations as the electronic structure can be calculated by solving a one-electron Schrödinger equation. While many of these empirical methods such as, the tight-binding method^[71,72], provides, the exact ground-state properties, some other ones, such as the Electron Plane Wave Expansion method (EPWE)^[73–75] and the Electron boundary element method (EBEM)^[76,77] are the most suitable approaches for studying excited-state properties of extended systems. The problem under investigation in this thesis is the interaction of free electrons, commonly present at metal surfaces, with two-dimensional (2D) periodic arrays of scatterers. It is, therefore, a natural choice to employ the nearly free electron approach in our calculations. Within this framework, the electronic wave functions are determined by the time-independent Schrödinger equation with the periodic potential $U(r)$, included as a weak perturbation of the free electron gas. This equation could, then, be solved through different computational methods. The Electron Plane Wave Expansion (EPWE) is once such an approach that is well suited to obtain the electronic properties in the case of extended periodic systems.

Throughout the thesis, EPWE method shall be used to calculate the band structure (dispersion), simulated ARPES data, and local DOS for the concerned hexagonal and square metallic superlattices.

It should be stressed that the band structure picture of electrons in solids involves the wave nature of electrons. This wave-particle duality cruelly established for light in the beginning of the twentieth century and, later, for electrons, has set great analogue between them and brought together their corresponding physics^[78]. A number of physical phenomena have, since then, found to hold for both electrons and electromagnetic (EM) waves^[79]. For example, the electronic superlattices just discussed have the optical analogous counterpart in photonic crystals, where the propagation of

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EM waves is a consequence of the multiple diffraction through the strongly modulated periodic lattice made of dielectric scatterers^[80]. For electronic lattice/superlattice, it is the modulation of the scattering potential landscape that affects the electron wave propagation. With such an analogy between EM and electrons, varieties of optical phenomenon could have the electronic counterpart. For example, at the lateral interface between two electronically dissimilar metal surfaces, i.e., with difference dispersion parameters of their surface states, electron refraction/reflection is expected, in similar way to light refraction/reflection at the interface between two homogeneous dielectric media having different refractive index. The important quantity is the presence of a wavevector difference (k) for both media at the concerned energy. In view of the wavevector-space analysis, the reflection-refraction takes place because the two media have Constant Energy Surfaces (CESs) of different k -radius, and the refracted beam follow the direction that conserves the parallel component of the wavevector. Therefore, the propagation of electron beam at the interface between two surfaces is solely determined by their band structures and the size/shape of their CESs. The CESs are always circular for free electrons, i.e., in non-patterned “*homogeneous*” surfaces, while they take complicated shapes for the nanostructured surfaces. The propagation of an electron plane wave or beam through such interfaces could, therefore, provides a wealth of ordinary and extraordinary optical-like phenomenon, such as total internal reflection^[81], positive and/or negative refractions^[82,83], self-collimation^[84], and superprism effect^[85].

Although the propagation of electrons through these interface could be extract from the wavevector space analysis obtained from EPWE methods, its real-space simulation requires the solution of Schrödinger equation for finite geometries in the presence of electron sources. Such simulations could be obtained by employing the Electron Boundary Element method (EBEM) which is appropriate to handle arbitrarily shaped nanostructures or electronic interfaces.

The thesis presents EBEM simulations of electron refraction/reflection at the interface between two electronically dissimilar materials having surface states with different dispersion parameters, and demonstrates the corresponding possible applications.

This thesis is structured in four chapters, apart from chapter two, each chapter contains original work starting with a detailed introduction to the problem studied and ending with concluding remarks and the possible further directions.

The **current chapter** introduces the context of the thesis and presents the motivations and the main objectives of the study.

Chapter Two, entitled “*Theoretical Background*”, contains the theoretical basics of band structure calculations and measurements. A detailed description of the Electron Plane Wave Expansion (EPWE) and Boundary Element Method (EBEM) will be given. Information on the EPWE/EBEM package, the input structure, and additional numerical considerations are provided. Finally, the theory behind Angle Resolved Photoemission (ARPES) and Scanning Tunneling Microscopy (STM) Spectroscopy (STS) are briefly discussed.

Chapters Three and four contain the results and discussion on the investigated systems and their applications.

Chapters Three, entitled “*Electronic Structure of 2D Metallic Superlattices*”, will start with a brief introduction and a literature survey on electronic structure of the experimentally reported patterned surfaces. Results and discussion of this chapter will be divided into two parts. (**Part 1: Hexagonal Superlattice**) presents detailed and systematic theoretical analysis of electron scattering in 2D hexagonal networks on metal surfaces. Information regarding the full band structure, constant energy surfaces (CES), and local density of states (LDOS) are obtained by EPWE method. The effect of periodicity and the magnitude of the periodic potential are discussed by systematically varying these scattering parameters. Results demonstrate the correlation between both and their influences on the electronic properties of hexagonally-patterned metal surfaces. The main finding of this part is the opening of energy gaps at the boundaries of the superlattice Brillouin zone. Furthermore, the effect of superlattice symmetry is investigated. The form of these energy gaps were found to change from a partial to a fully direct band gap depending on the overall rotational symmetry of the lattice. All these calculations are also performed for 2D square superlattice (**Part 2: Square Superlattice**). The main difference finding is that, such square superlattices trigger the opening of an indirect band gap at the boundaries of the superlattice Brillouin zone.

Chapters Four, entitled “*Surface State Nanoelectronics Applications*”, starts with a brief introduction and a literature survey on wave-particle duality and a number of physical phenomena that could equally hold for both electrons and electromagnetic waves. Numerical considerations required to properly simulate electron refraction/reflection at metallic interfaces are stated. The results and discussion of this

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chapter will be divided into two parts. **Part 1**, entitled “**Homogeneous-Homogeneous Interfaces**” presents real-space simulation of electron reflection-refraction at the lateral interface between two “*homogeneous*” metals with different potential barriers featuring surface states characterized by isotropic constant energy surfaces. The main finding of this part is the decent agreement obtained between such real-space EBEM simulations and the wavevector-space analysis obtained using EPWE method. The electron refraction was found to take place following Snell’s law and Fresnel coefficients, and total internal reflections occur beyond energy-dependent critical angles. In **Part 2**, namely “**Homogeneous-Nanostructured Interfaces**” electron refraction is investigated at the interface between homogeneous medium, with circular CESs, and a nanostructured medium characterized by highly featured and gapped CESs. Specifically, it presents simulations on a real system, namely the interface between the homogeneous Cu(111) and the nanostructured 1 ML Ag/Cu(111). The attractive findings in this part is that varieties of ordinary and extraordinary optical phenomena have been observed for a Gaussian electron beam hitting this interface, such as total internal reflection, positive and/or negative refractions, self-collimation, and superprism effect. The real-space simulations of these phenomena as obtained by EBEM agree well with the behavior expected from EPWE-based wavevector-space analysis performed at different binding energies.

2

Theoretical Background

2.1 Introduction

Band structure is one of the most important concepts in solid state physics. It provides the electronic energy levels/bands in “*ideal*” crystal structures, which are characterized by two quantum numbers, namely the Bloch wavevector k and the band index n ^[86]. Here, the Bloch wavevector is an element of the reciprocal space (in units of 1/length) and the energy of the electron $E_n(k)$ is a continuous function of k , so that one obtains a continuous range of energies referred to as the energy band. Most of the electrical, optical, and magnetic properties of crystals can be explained in terms of their band structures. Of particular importance is the location of the Fermi energy, below which all energy bands are occupied at zero temperature. If the Fermi energy is located in a full band gap, the material is insulating (or semiconducting) while it is metallic otherwise. Unfortunately, the band structure of solids cannot be solved analytically due to the huge number of constituent electrons. However, numerical solutions could be obtained, thanks to the rapidly progressing area of research, namely computational solid state physics. The modern methods for calculating the electronic band structure of solids allow predicting many important properties of solids. The majority of these methods involve the development of quite complicated computer codes. Nowadays some of such programs are available in the market and can be purchased and used by researchers.

Electronic band structure calculation methods can be grouped into two general categories^[66]. The first category consists of *ab initio* methods, such as Hartree-Fock and Density Functional Theory (DFT), which calculate the electronic structure from first principles, i.e., without the need for empirical fitting parameters. In general, these methods utilize a variational approach to calculate the ground state energy of

2. THEORETICAL BACKGROUND

a many-body system, where the system is defined at the atomic level. These calculations were originally performed on systems containing few atoms. Today, such calculations could be performed using approximately 1000 atoms but are computationally expensive, sometimes requiring massively parallel computers. In contrast to *ab initio* approaches, the second category consists of the *empirical methods*, such as the Electron Plane Wave Expansion method (EPWE)^[73–75] (based on the nearly free electron approach), the Orthogonalized Plane Wave (OPW)^[87], the tight-binding^[88] (also known as the Linear Combination of Atomic Orbitals (LCAO) method), and the local/non-local^[89–91] empirical pseudopotential methods (EPM). These methods involve empirical parameters to fit the experimental data such as the band-to-band transitions at specific high-symmetry points derived from optical absorption experiments or the band minima and effective mass obtained from photoemission measurements. The appeal of these methods is that the electronic structure can be calculated by solving a one-electron Schrödinger equation. Thus, *empirical methods* are computationally less expensive than *ab initio* calculations and provide a relatively easy means of generating the electronic band structure. Throughout the present work this type of calculation will be used.

As mentioned previously, this thesis concerns the interaction of free electrons, commonly present at metal surfaces, with two-dimensional (2D) periodic arrays of scatterers (i.e., periodic potentials). The approximation which starts from the wave functions of the free atoms is known as the tight binding approximation or the LCAO. In general this approximation is quite good to describe the inner electronic shells of atoms and relatively localized bands^[86]. However, it is not convenient for the description of the delocalized electrons, such as the conduction electrons in metals. This is because the origin of the free electrons is completely different to the nature of the localized atomic states. In the tight-binding model we assume the opposite limit to that used for the nearly-free-electron approach, i.e., the potential is so large that the electrons spend most of their lives bound to ionic cores, only occasionally summoning the quantum-mechanical wherewithal to jump from atom to atom. Hence the nearly free electron approach is a more natural starting point to describe metal surfaces concerned here, while the tight-binding approach is the obvious starting point for covalently bonded crystals, such as graphene^[92], or for the more localized electrons in metals, such as the *d* electrons in transition metals^[65]. Based upon that, in our calculations we will follow the nearly free electron approach and there are two computational methods that will be used. These two methods have been implemented by Prof. Javier Garca de Abajo in

his numerical technique which is called (BEM) code. One of these is the Electron Plane Wave Expansion (EPWE) method which refers to a computational technique commonly used in electromagnetics to solve the Maxwell's equations by formulating an eigenvalue problem out of the equation. This method is popular among the photonic crystal community as a method of solving the band structure (i.e., the dispersion relation) of specific photonic crystal geometries. EPWE is traceable to the analytical formulations, and is useful in calculating modal solutions of Maxwell's equations over an inhomogeneous or periodic geometry. It is specifically tuned to solve problems in time-harmonic forms, with non-dispersive media. For the electronic band structure calculations, we employed the EPWE part of the BEM package which was recently demonstrated to be a valuable tool for the analysis of the band structure and density of states in metallic interfacial dislocation and self-assembled molecular networks^[73–75,93–95]. The program code used here permits to extend the particle-in-a-box model to infinite 2D systems by defining an elementary cell and using periodic boundary conditions. Within the EPWE code, solutions of the Schrödinger equation are represented as a linear combination of plane waves and a satisfactory convergence is achieved with a basis set consisting of G_{max} waves. The second computational method is the Electron Boundary Element Method (EBEM), which represents a scalar variant of the electromagnetic boundary element method extensively used for solving Maxwell's equations and optical response for arbitrary shapes and finite systems. It is suitable for the regions of constant potential that tessellate a plane, with the boundaries between regions being abrupt. The method proved to be a valuable tool for the analysis of confinement effects in several recent cases. The eigenstates of the systems are expressed in terms of auxiliary sources on the region boundaries, whose propagation is described by Green's functions. The combined EPWE and EBEM methods, allow us to study the electronic structure, local density of states (LDOS), the photoemission intensity, and the scattering problems for both periodic and finite systems, respectively.

On experimental side, the band structure of solids is commonly obtained by Angle-Resolved Photoemission Spectroscopy (ARPES), which is an experimental technique to observe the distribution of the electrons (more precisely, the density of single-particle electronic excitations) in the reciprocal space of solids^[96]. The technique is a refinement of ordinary photoemission spectroscopy, studying photoemission of electrons from a sample achieved usually by illumination with soft X-rays. ARPES is one of the most direct methods of studying the electronic structure of the surface of solids. This means that information can be gained on both the energy and momentum of an electron,

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resulting in detailed information on the band dispersion and Fermi surface^[19]. The surface structure and local density of states (LDOS), on the other hand, are commonly obtained by employing the Scanning Tunneling Microscopy (STM) and Spectroscopy (STS), respectively. STM is developed in 1981 and its inventors, Gerd Binnig and Heinrich Rohrer (at IBM Zurich), earned the Nobel Prize in Physics in 1986^[97,98]. The STM allows imaging surfaces with atomic resolution, typically 0.1 nm and 0.01 nm in lateral and vertical resolutions, respectively^[99]. With this resolution, individual atoms within materials are routinely imaged and manipulated. In its voltage scanning mode, the dI/dV measurements provide the site specific LDOS.

In the following we give detailed description of the underlying physics of the plane wave expansion method, the boundary element method, and the experimental techniques (ARPES and STM).

2.2 Nearly Free Electron Approach

2.2.1 General Overview

The free electron model provides good insights into the heat capacity, thermal conductivity, electrical conductivity, magnetic susceptibility, and electrodynamics of metals^[65]. But the model fails to answer other fundamental questions such as the distinction between metals, semimetals, semiconductors, and insulators; the occurrence of positive values of the Hall coefficient; the relation of conduction electrons in the metal to the valence electrons of free atoms; and many transport properties, particularly magneto transport. We need a less naive theory, and fortunately it turns out that almost any simple attempt to improve upon the free electron model is enormously profitable. Every solid contains electrons. The important question for electrical conductivity is how the electrons respond to an applied electric field. We shall see that electrons in crystals are arranged in energy bands, separated by regions in energy for which no wavelike electron orbitals exist. Such forbidden regions are called “*energy gaps*” or “*band gaps*”, and result from the interaction of the conduction electron waves with the ion cores of the crystal. The crystal behaves as an insulator if the allowed energy bands are either filled or empty, for then no electrons can move in an electric field. The crystal behaves as a metal if one or more bands are partly filled, say between 10 and 20 percent filled. The crystal is a semiconductor or a semimetal if one or two bands are slightly filled or slightly empty^[65]. To understand the difference between insulators and conductors, we must extend the free electron model to take account of

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the periodic lattice of the solid. The possibility of a band gap is the most important new property that emerges. We shall encounter other quite remarkable properties of electrons in crystals. For example, they respond to applied electric or magnetic fields as if the electrons were endowed with an effective mass m_e , which may be larger or smaller than the free electron mass, or may even be negative. On the free electron model the allowed energy values are distributed essentially continuously from zero to infinity Fig. 2.1 (a), that is

$$E = \frac{\hbar^2 k^2}{2m} \quad (2.1)$$

where, for periodic boundary conditions over a cube of side L ,

$$k_x, k_y, k_z = 0; \pm \frac{2\pi}{L}; \pm \frac{4\pi}{L}; \dots \quad (2.2)$$

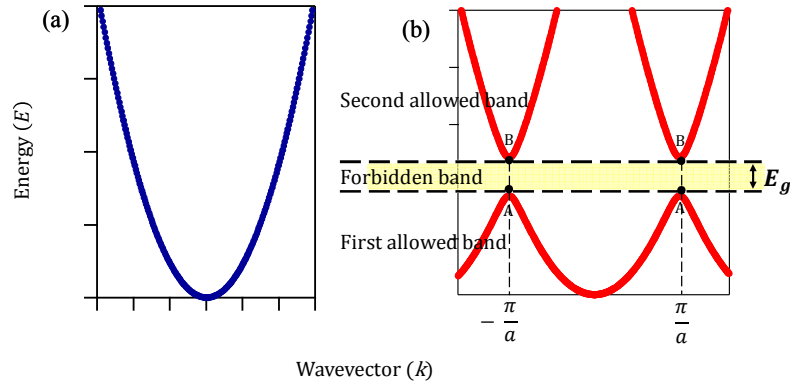


Figure 2.1: Plot of energy E versus wavevector k - (a) Plot of energy E versus wavevector k for a free electron. (b) Plot of energy versus wavevector for an electron in a monatomic linear lattice of lattice constant a . The energy gap E_g , shown is associated with the first Bragg reflection at $k = \pm\pi/a$; other gaps are found at higher energies at $(\pm n\pi/a)$, for integral values of n .

The free electron wavefunctions are of the form:

$$\psi_k(r) = e^{ik \cdot r} \quad (2.3)$$

They represent running waves and carry momentum ($p = \hbar k$).

The band structure of a crystal can often be explained by the nearly free electron model for which the band electrons are treated as perturbed only weakly by the periodic potential of the ion cores. This model answers almost all the qualitative questions about the behavior of electrons in metals. We know that Bragg reflection is a characteristic

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feature of wave propagation in crystals, and that Bragg reflection of electron waves in crystals is the cause of energy gaps. At Bragg reflection wavelike solutions of the Schrödinger equation do not exist, as in Fig. 2.1(b). These energy gaps are of decisive significance in determining whether a solid is an insulator or a conductor^[65]. We explain physically the origin of energy gaps in the simple problem of a linear solid of lattice constant a . The low energy portions of the band structure are shown qualitatively in Fig. 2.1, in (a) for entirely free electrons and in (b) for electrons that are nearly free, but with an energy gap at $k = \pm\pi/a$. The Bragg condition $(k+G)^2 = k^2$ for diffraction of a wave of wavevector k becomes in one dimension

$$k = \pm \frac{G}{2} = \pm \frac{n\pi}{a} \quad (2.4)$$

where ($G = 2\pi n/a$) is a reciprocal lattice vector and n is an integer. The first reflections and the first energy gap occur at $k = \pm\pi/a$. The region in k -space between $-\pi/a$ and π/a is the first Brillouin zone of this lattice. Other energy gaps occur for other values of the integer n . The wavefunctions at $k = \pm\pi/a$ are not the traveling waves $\exp(i\pi x/a)$ or $\exp(-i\pi x/a)$ of free electrons. At these special values of k the wavefunctions are made up of equal parts of waves traveling to the right and to the left. When the Bragg reflection condition ($k = \pm\pi/a$) is satisfied by the wavevector, a wave traveling to the right is Bragg-reflected to travel to the left, and vice versa. Each subsequent Bragg reflection will reverse the direction of travel of the wave. A wave that travels neither to the right nor to the left is a standing wave: it doesn't go anywhere.

The time-independent state is represented by standing waves. We can form two different standing waves from the two traveling waves

$$e^{\pm i\pi x/a} = \cos(\pi x/a) \pm i \sin(\pi x/a)$$

So that the standing waves are:

$$\begin{aligned} \psi(+) &= e^{i\pi x/a} + e^{-i\pi x/a} = 2 \cos\left(\frac{\pi x}{a}\right) \\ \psi(-) &= e^{i\pi x/a} - e^{-i\pi x/a} = 2i \sin\left(\frac{\pi x}{a}\right) \end{aligned} \quad (2.5)$$

The standing waves are labeled (+) or (-) according to whether or not they change sign when $-x$ is substituted for x . Both standing waves are composed of equal parts of right- and left-directed traveling waves.

2.2.2 Origin and Magnitude of the Energy Gap

The two standing waves $\psi(+)$ and $\psi(-)$ pile up electrons at different regions, and therefore the two waves have different values of the potential energy in the field of the ions of the lattice^[86]. This is the origin of the energy gap. The ion cores bear a net positive charge because the atoms are ionized in the metal, with the valence electrons taken off to form the conduction band. The potential energy of an electron in the field of a positive ion is negative, so that the force between them is attractive. The wavefunctions at the Brillouin zone boundary $k = \pi/a$ are $(\sqrt{2}\cos(\pi x/a))$ and $(\sqrt{2}\sin(\pi x/a))$, normalized over unit length. Let us suppose that the potential energy of an electron in the crystal at point x is

$$U(x) = U \cos \frac{2\pi x}{a}$$

The first-order energy difference between the two standing wave states is

$$E_g = \int_0^1 dx U(x) [|\psi(+)|^2 - |\psi(-)|^2] = U \quad (2.6)$$

We see that the gap is equal to the Fourier component of the crystal potential.

2.2.3 Bloch Functions

F. Bloch proved the important theorem that the solution of the Schrödinger equation for a periodic potential must be of a special form:

$$\psi_k(r) = u_k(r) e^{ik \cdot r} \quad (2.7)$$

where $u_k(r)$ has the period of the crystal lattice with $u_k(r) = u_k(r + T)$. Here T is a translation vector of the lattice. Thus, the Bloch theorem reads:

The eigenfunctions of the wave equation for a periodic potential are the product of a plane wave $\exp(ik \cdot r)$ times a function $u_k(r)$ with the periodicity of the crystal lattice.

A one-electron wavefunction of the form Eq. 2.7 is called a Bloch function and can be decomposed into a sum of traveling waves, as we see later^[86].

2.2.4 Kronig-Penney Model

A periodic potential for which the wave equation can be solved in terms of elementary functions is the square-well array of Fig. 2.2 The wave equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + U(x)\psi = E\psi \quad (2.8)$$

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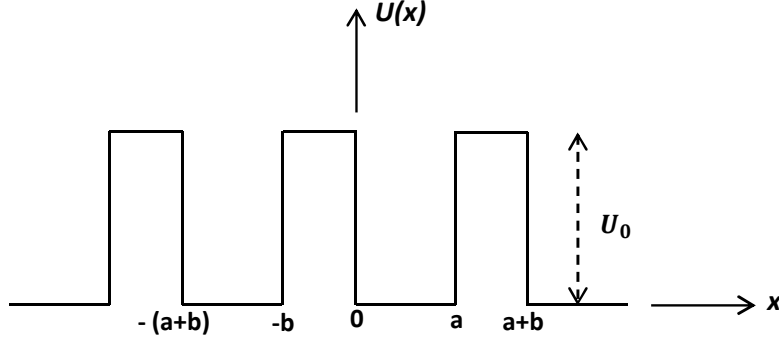


Figure 2.2: Kronig-Penney potential barrier - Square-well periodic potential as introduced by Kronig and Penney.

Where $U(x)$ is the potential energy and E is the energy eigenvalue. In the region $0 < x < a$ in which $U = 0$, the eigenfunction is a linear combination,

$$\psi = Ae^{iKx} + Be^{-iKx} \quad (2.9)$$

of plane waves traveling to the right and to the left, with energy

$$E = \frac{\hbar^2 K^2}{2m} \quad (2.10)$$

In the region $-b < x < 0$ within the barrier the solution is of the form

$$\psi = Ce^{Qx} + De^{-Qx} \quad (2.11)$$

With

$$U_0 - E = \frac{\hbar^2 Q^2}{2m} \quad (2.12)$$

We want the complete solution to have the Bloch form Eq. 2.7. Thus the solution in the region $[a < x < a + b]$ must be related to the solution 2.11 in the region $[-b < x < 0]$ by the Bloch theorem:

$$\psi(a < x < a + b) = \psi(-b < x < 0)e^{ik(a+b)} \quad (2.13)$$

which serves to define the wavevector k used as an index to label the solution. The constants A, B, C, D are chosen so that ψ and $d\psi/dx$ are continuous at $x = 0$ and $x = a$. These are the usual quantum mechanical boundary conditions in problems that involve square potential wells.

At $x = 0$,

$$A + B = C + D \quad (2.14)$$

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$$iK(A - B) = Q(C - D) \quad (2.15)$$

With Q from Eq. 2.11. At $x = a$, with the use of Eq. 2.13 for $\psi(a)$ under the barrier in terms of $\psi(-b)$,

$$Ae^{iKa} + Be^{-iKa} = (Ce^{-Qb} + De^{Qb})e^{ik(a+b)} \quad (2.16)$$

$$ik(Ae^{iKa} - Be^{-iKa}) = (Ce^{-Qb} - De^{Qb})e^{ik(a+b)} \quad (2.17)$$

The four equations (2.14) to (2.17) have a solution only if the determinant of the coefficients of A, B, C, D vanishes, yielding

$$[(Q^2 - K^2)/2QK] \sinh(Qb) \sin(Ka) + \cosh(Qb) \cos(Ka) = \cos k(a + b)$$

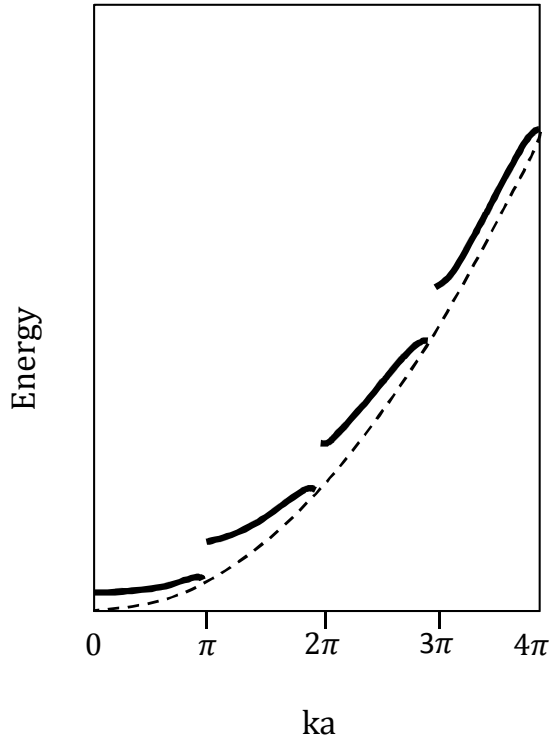


Figure 2.3: Kronig-Penney Dispersion - Plot of energy vs. wavenumber for the Kronig-Penney potential, with $P = 3\pi/2$. Notice the energy gaps at $ka = \pi, 2\pi, 3\pi$,

It is rather tedious to obtain this equation. The result is simplified if we represent the potential by the periodic delta function obtained when we pass to the limit $b = 0$ and $U_0 = \infty$ in such a way that $Q^2ba/2 = P$, a finite quantity. In this limit $Q \gg K$ and $Qb \ll 1$. Then the last equation reduces to

$$\frac{P}{Ka} \sin(Ka) + \cos(Ka) = \cos(ka) \quad (2.18)$$

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The ranges of K for which this equation has solution and its corresponding values of the energy are plotted in Fig. 2.3. Note the energy gaps at the zone boundaries. The wavevector k of the Bloch function is the important index, not the K in Eq. 2.9, which is related to the energy by Eq. 2.10.

2.2.5 Wave equation of electron in a periodic potential: The Central Equation

We considered before the approximate form we expect for the solution of the Schrödinger equation if the wavevector is at a zone boundary (i.e., at $k = \pm \pi/a$). We treat in detail the wave equation for a general potential, at general values of k . The single particle Schrödinger equation is

$$\begin{aligned} (H - E_{nk})\psi_{nk} &= 0, \\ \left(-\frac{\nabla^2}{2} + V_s(r) - E_{nk}\right)\psi_{nk} &= 0 \end{aligned} \quad (2.19)$$

We can solve Eq. 2.19 at given wavevector, k , and band index, n , to find the energy, E_{nk} , and the single particle wavefunction, ψ_{nk} .

In a crystal, the eigenstates, ψ_{nk} , obey Bloch's theorem, and can be written as:

$$\psi_{nk}(r) = e^{ik \cdot r} u_{nk}(r) \quad (2.20)$$

where u_{nk} has the periodicity of the lattice. Thus, $u_{nk}(r + T) = u_{nk}(r)$, where T is the translation vector ($T = n_1 a_1 + n_2 a_2 + n_3 a_3$ and a_1, a_2, a_3 are the unit cell vectors). Since the function u_{nk} have the periodicity of the lattice we can expand it as a Fourier series, $u_{nk} = \sum_G C_G(k) e^{iG \cdot r}$, so that

$$\psi_{nk} = \sum_G C_G(k) e^{i(k+G) \cdot r} \quad (2.21)$$

Where the set of G are reciprocal lattice vectors of the crystal, and $C_G(k)$ are the expansion coefficients of the single particle wave function at each k . The potential $V_s(r)$ must also have the periodicity of the lattice, and could be Fourier expanded as follow:

$$V_s(r) = \sum_{G'} V_{G'}(k) e^{iG' \cdot r} \quad (2.22)$$

Before substituting into the Schrödinger equation (2.19) we note that, both the kinetic and potential energy terms (multiplied by the wavefunction) constituting the Hamiltonian are given by;

$$-\frac{\nabla^2}{2} \psi_{nk} = \sum_G C_G \frac{|k + G|^2}{2} e^{i(k+G) \cdot r} \quad (2.23)$$

and

$$V_s(r)\psi_{nk} = \sum_{G'} \sum_Q V_{G'} C_Q(k) e^{i(G'+Q+k).r} \quad (2.24)$$

Substituting Eq. 2.23 and Eq. 2.24 into Eq. 2.19, we obtain the following expression;

$$\sum_G \left(\frac{|k+G|^2}{2} - E_{nk} \right) e^{i(k+G).r} + \sum_{G'} \sum_Q V_{G'} C_Q(k) e^{i(G'+Q+k).r} = 0 \quad (2.25)$$

It is now advantageous to rearrange the double sum involving $V_{G'}$ so that the plane wave part in the double sum becomes $e^{i(k+G).r}$. To do this, we write Q in terms of another reciprocal lattice vector, $G = G' + Q$ and replace Q with $G - G'$. Because the sums are over an infinite set of reciprocal lattice vectors, for each fixed G' , we can replace the infinite sum over Q with an infinite sum over the equivalent set of vectors, G . Then,

$$\sum_G \left(\frac{|k+G|^2}{2} - E_{nk} \right) C_G(k) e^{i(k+G).r} + \sum_{G'} \sum_G V_{G'} C_{G-G'}(k) e^{i(k+G).r} = 0 \quad (2.26)$$

or,

$$\sum_G e^{i(k+G).r} \left\{ \left(\frac{|k+G|^2}{2} - E_{nk} \right) C_G(k) + \sum_{G'} V_{G'} C_{G-G'}(k) \right\} = 0 \quad (2.27)$$

The left hand side of this equation can only be zero if, for each G , the term inside the curly brackets $\{\dots\}$ is zero. Hence,

$$\left(\frac{|k+G|^2}{2} - E_{nk} \right) C_G(k) + \sum_{G'} V_{G'} C_{G-G'}(k) = 0 \quad (2.28)$$

This equation is called the “*central equation*”. The form above is similar to that given by Tanner^[100]. We can also rearrange the central equation in a number of different ways. For convenient, we write the central equation in the form used by Martin^[101]. To do so, we must rewrite the infinite sum over reciprocal lattice vectors in Eq. 2.28. We choose a new reciprocal lattice vector, $G'' = G - G'$, and replace the infinite sum over G' with an equivalent infinite sum over G'' . Then,

$$\left(\frac{|k+G|^2}{2} - E_{nk} \right) C_G(k) + \sum_{G''} V_{G-G''} C_{G''}(k) = 0 \quad (2.29)$$

Because both terms on the left now contain a coefficient that looks like C_Q we can place both terms within the sum over G'' and factoring out the $C_{G''}$.

$$\sum_{G''} \left[\left(\frac{|k+G|^2}{2} - E_{nk} \right) \delta(G - G'') + V_{G-G''} \right] C_{G''}(k) = 0 \quad (2.30)$$

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Where the delta function $\delta(G - G'')$ ensures we only get the required $G'' = G$ terms in the first term in the sum over G'' . In the following we comment on the possible ways of solving the central equation of the form given in Eq. 2.30.

2.2.6 Solving the central equation: Plane Wave Expansion (PWE) method

From Eq. 2.30 it should be clear that the central equation is an infinite set of coupled simultaneous equations - with one equation for each possible reciprocal lattice vector. We can solve the central equation numerically or, in some very simple cases, analytically. In all cases we must truncate the infinite sum over reciprocal lattice vectors at some point. When solving the central equation it is common to take $V_0 = 0$. The V_0 Fourier component is the average value of the potential and we can always set the average of the potential to zero by adding or subtracting a constant offset to the energies.

2.2.7 Truncating the series: Example with 3 terms

To explicitly see how Eq. 2.30 becomes a set of coupled simultaneous equations let us imagine that the infinite sum over reciprocal lattice vectors is restricted to just three possibilities, $-g, 0$, and g . In this case G in Eq. 2.30 has three possible values $(-g, 0, g)$ and, in the sum over reciprocal lattice vectors, G'' must sum over the three possible values $(-g, 0, g)$. The central equation will therefore give us three simultaneous equations each containing three terms. For example, if we select $(G = -g)$ we can write the first of the three simultaneous equations as,

$$\sum_{G''=-g,0,g} \left[\left(\frac{|k-g|^2}{2} - E_{nk} \right) \delta(-g - G'') + V_{-g-G''} \right] C_{G''}(k) = 0 \quad (2.31)$$

In the first term in the sum, $(G'' = -g)$ and, because we have picked $(G = -g)$, we get a contribution from the part of the equation containing the delta function. In this case, when $(G'' \neq -g)$, we obviously only retain the terms containing the Fourier components of the potential $V_{G-G''}$. Remembering that $V_0 = 0$ we can therefore write Eq. 2.31 as,

$$\left(\frac{|k-g|^2}{2} - E_{nk} \right) C_{-g}(k) + V_{-g} C_0(k) + V_{-2g} C_g(k) = 0 \quad (2.32)$$

Similarly, we can write down another equation for the case when $(G = 0)$. In this case we obviously do not get a contribution from the delta function term if $(G'' = -g)$

or ($G'' = g$), and

$$V_g C_{-g}(k) + \left(\frac{|k|^2}{2} - E_{nk}\right) C_0(k) + V_{-g} C_g(k) = 0 \quad (2.33)$$

Finally, we get the third equation in the set by choosing ($G = g$),

$$V_{2g} C_{-g}(k) + V_g C_0(k) + \left(\frac{|k+g|^2}{2} - E_{nk}\right) C_g(k) = 0 \quad (2.34)$$

We can write this set of simultaneous equations as a matrix equation,

$$\begin{bmatrix} 0.5|k-g|^2 - E_{nk} & V_{-g} & V_{-2g} \\ V_g & 0.5|k|^2 - E_{nk} & V_{-g} \\ V_{2g} & V_g & 0.5|k+g|^2 - E_{nk} \end{bmatrix} \begin{bmatrix} C_{-g}(k) \\ C_0(k) \\ C_g(k) \end{bmatrix} = 0 \quad (2.35)$$

and this only has non-trivial solutions if the determinant vanishes,

$$\begin{vmatrix} 0.5|k-g|^2 - E_{nk} & V_{-g} & V_{-2g} \\ V_g & 0.5|k|^2 - E_{nk} & V_{-g} \\ V_{2g} & V_g & 0.5|k+g|^2 - E_{nk} \end{vmatrix} = 0 \quad (2.36)$$

2.2.8 General form

In shorthand form we can write our matrix equation (2.35) as,

$$[H - EI]C = 0 \quad (2.37)$$

where I is the unit matrix and E are the eigenvalues E_{nk} . From our previous example (with the sum restricted to 3 reciprocal lattice vectors) H is the Hamiltonian matrix,

$$H = \begin{bmatrix} 0.5|k-g|^2 - E_{nk} & V_{-g} & V_{-2g} \\ V_g & 0.5|k|^2 - E_{nk} & V_{-g} \\ V_{2g} & V_g & 0.5|k+g|^2 - E_{nk} \end{bmatrix} \quad (2.38)$$

and C is the eigenvector that contains the expansion coefficients,

$$C = \begin{bmatrix} C_{-g}(k) \\ C_0(k) \\ C_g(k) \end{bmatrix} \quad (2.39)$$

The notation in Eq. 2.37 is completely general. So, to find the single particle energies, we must always find the values of E so that the determinant,

$$|H - EI|C = 0 \quad (2.40)$$

In our previous example H was a 3×3 matrix, but we could have restricted the infinite series in the central equation to any number of terms. For example, if we had included N reciprocal lattice vectors we would have an $N \times N$ matrix for the Hamiltonian.

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2.2.9 The empty lattice

If we set $V_s(r) = 0$ then all of the off-diagonal terms in the Hamiltonian matrix are zero and we get a simple set of uncoupled equations that immediately give us the allowed energies,

$$E_{nk} = \frac{1}{2}|k + G_n|^2 \quad (2.41)$$

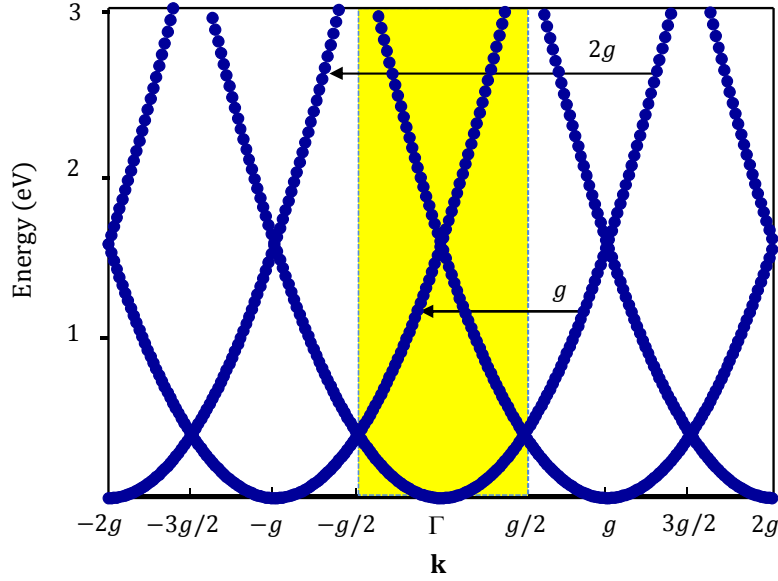


Figure 2.4: Illustration of the reduced zone scheme. - The parts of the $E - k$ curve outside the first Brillouin zone are folded back into the first Brillouin zone by translating by an integer number of reciprocal lattice vectors. The gray rectangle denote the edges of the first Brillouin zone at $g/2$ and $-g/2$.

These are essentially the familiar free electron solutions, $E = |P|^2/2m$, but, in a periodic crystal wavevectors k and $k + G$ are equivalent. We can use this equivalence to simplify the way in which we represent the states: any wavevector k' that is larger than a reciprocal lattice vector, G , can be represented by a shorter wavevector inside the first Brillouin zone, $k = k' - G$, because of the periodicity of the reciprocal lattice. In the reduced zone scheme we represent all of the possible E_{nk} in terms of wavevectors within the first Brillouin zone. This is illustrated schematically in Fig. 2.4.

2.2.10 Weak periodic potential: solution at ($k = g/2$)

First we assume that the potential is weak and periodic such that $V_s(r) = V \cos(g \cdot r)$, only the V_g Fourier components of the potential are non-zero, and $V_{-g} = V_g$. Then,

2.2 Nearly Free Electron Approach

as V is small, we might imagine that the wavefunctions $\psi_{nk} = \sum_G C_G e^{i(k+G) \cdot r}$, will be similar to the free electron solutions, $\psi_{nk} = e^{i(k \cdot r)}$, and it should therefore be possible to restrict the infinite sum in Eq. 2.30 to just a few terms. As we are interested in solutions near the Brillouin zone boundary, where $k = g/2$, we can restrict this sum to cover just two reciprocal lattice vectors. Near $k = g$ we assume that the $G = 0$ and $G = -g$ components are the most important so that,

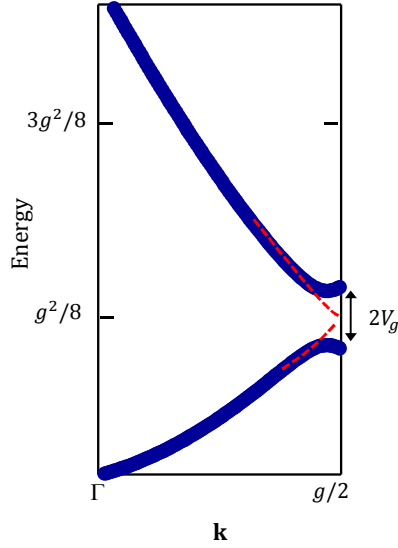


Figure 2.5: Weak periodic potential dispersion. - $E(k)$ relation in the presence of a weak periodic potential (solid line). The faint dotted line shows the empty lattice solutions. The weak periodic potential introduces a gap of $2V_g$ at the Brillouin zone boundary where $k = g/2$.

$$\psi_{nk} = C_0 e^{ik \cdot r} + C_{-g} e^{i(k-g) \cdot r} \quad (2.42)$$

The Hamiltonian matrix is then just a 2×2 matrix and, by retaining only the $G = -g$ and $G = 0$ terms in Eq. 2.35 we can find the allowed energies from,

$$\begin{vmatrix} 0.5|k - g|^2 - E_{nk} & V_g \\ V_g & 0.5|k|^2 - E_{nk} \end{vmatrix} = 0 \quad (2.43)$$

If we use the shorthand $\lambda_k = 0.5|k|^2$ then the allowed energies are obtained by solving the resultant quadratic equation $(\lambda_{k-g} - E)(\lambda_k - E) - V_g^2 = 0$. The solutions for the allowed energies near $k = g/2$ are then,

$$E = 0.5(\lambda_k + \lambda_{k-g}) \pm 0.5[(\lambda_k - \lambda_{k-g})^2 + 4V_g^2]^{\frac{1}{2}} \quad (2.44)$$

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At the zone boundary, $k = g/2$ and $\lambda_k = \lambda_{(k-g)} = |g|^2$. So, at the zone boundary there are two possible values of the energy,

$$E = \frac{|g|^2}{8} \pm V_g \quad (2.45)$$

A periodic potential, no matter how small, therefore opens up a gap at the Brillouin zone boundary of $2V_g$. This is illustrated in Fig. 2.5.

2.2.11 Numerical solution

In principle, from Eq. 2.30, the Hamiltonian matrix is infinite in extent, but to solve the central equation we must always truncate the sum over reciprocal lattice vectors at some point. By doing this we restrict the reciprocal lattice vectors in our Fourier expansion of the single-particle wavefunction (Eq. 2.21) to be less than some maximum length G_{max} ,

$$\psi_{nk} = \sum_{G_{|G| < |G_{max}|}} C_G(k) e^{i(k+g) \cdot r} \quad (2.46)$$

As with any Fourier series, we can include more and more terms in the expansion until the wavefunction is as accurate as we need it to be. How do we know how many terms to include? As always, we can use the variational theorem to help us decide. The variational theorem tells us that each E_{nk} will decrease as G_{max} increases and our approximate form for the wavefunction improves. We can therefore examine the convergence of E_{nk} with G_{max} to determine how many reciprocal lattice vectors we must include to obtain the single-particle energies to a given accuracy. Many computer codes have been developed to solve systems of equations, like the central equation, numerically. To solve Eq. 2.30 we must essentially give our computer code just two pieces of information. First we need to specify how many reciprocal lattice vectors we would like to include in our expansion of the single particle wavefunction (this sets the size of the $N \times N$ Hamiltonian in Eq. 2.37. Second we need to supply the Fourier components of the potential, V_G , that set each element of the Hamiltonian matrix in Eq. 2.37. There exist many sophisticated numerical routines which will then diagonalize this matrix-essentially, calculate and report the values of E that satisfy $|H - EI| = 0$. For most of our purposes a value of $G_{max} = 10$ is usually sufficient for energy convergence. The wavefunctions are accordingly expanded over $G_{max}^2 + 1$ (i.e., 101 plane wave are employed in the expansion).

2.2.12 The Local Density Of State (LDOS)

We want to know the highest occupied energy that we get when filling up the states. If the number of electrons to be distributed is very large, we can use the same geometric construction as for the vibrational states. Suppose that we have N electrons in our enclosed volume, such that the conduction electron density is $n = N/V = N/L^3$. These can be accommodated on the $N/2$ states with the lowest energy since we can have two electrons per state^[65]. We have to fill all states that lie within a sphere of radius n_{max} or, alternatively, a sphere with radius k_{max} such that

$$\frac{N}{2} = \frac{4}{3}\pi n_{max}^3,$$

which gives,

$$n_{max} = \left(\frac{3N}{8\pi}\right)^{\frac{1}{3}} \quad (2.47)$$

From this, we can calculate the energy of the highest occupied electron states to

$$E_{max} = \frac{\hbar^2 k_{max}^2}{2m_e} = \frac{\hbar^2}{2m_e} \left(\frac{2\pi}{L}\right)^2 n_{max}^2 \quad (2.48)$$

This energy has a special name: It is called the *Fermi energy* E_F . For most metals, it is a few electron volts, that is, in the range of typical chemical binding energies. Similarly, k_{max} is called the *Fermi wavevector* k_F . Also n_{max} is not used very much because it depends on the size of the system. A useful expression that follows from Eq. 2.48 is the relation between Fermi energy E_F and conduction electron density n :

$$E_F = \frac{\hbar^2}{2m_e} (3\pi^2 n)^{\frac{2}{3}} \quad (2.49)$$

Finally, we can calculate the *Density Of States* (DOS) $g(E)$ in the free electron model, that is, the energy-dependent number of available states per energy interval dE . We will need this for the correct description of the situation at finite temperature and for many other things. From Eq. 2.47 and Eq. 2.48, the highest occupied energy for N electrons can be written as

$$E(N) = \frac{\hbar^2}{2m_e} \left(\frac{3\pi^2 N}{V}\right)^{\frac{2}{3}} \quad (2.50)$$

from which we get the total number of states $N(E)$ for a given highest energy E and with this

$$g(E) = \frac{dN}{dE} = \frac{V}{2\pi^2} \left(\frac{2m_e}{\hbar^2}\right)^{\frac{3}{2}} E^{\frac{1}{2}} \quad (2.51)$$

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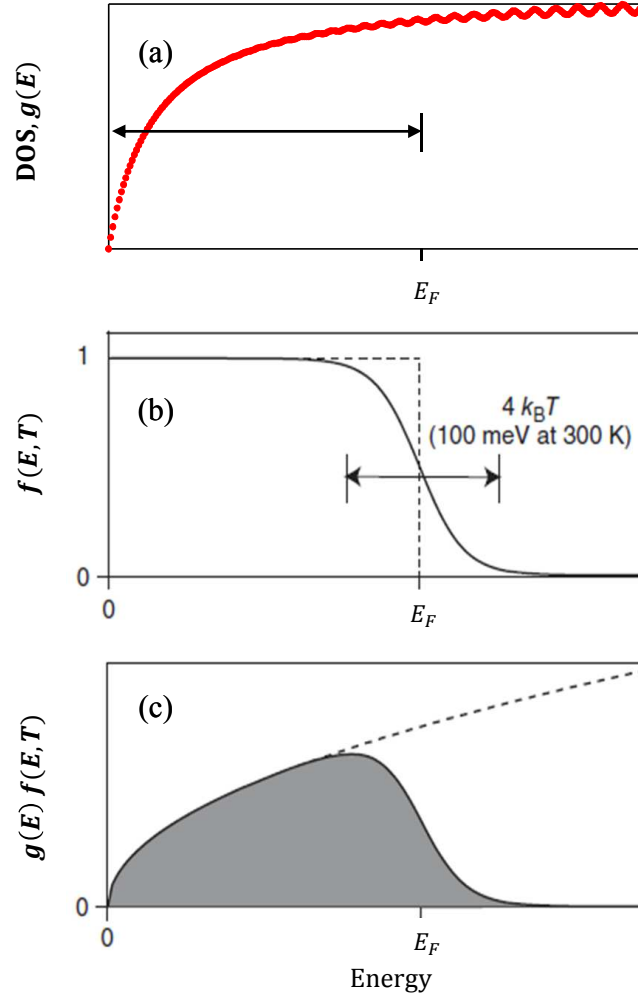


Figure 2.6: Density of states, v.s. Fermi-Dirac distribution function. - (a) Density of states for a free electron gas $g(E)$. (b) Fermi-Dirac distribution function $f(E, T)$ at $T = 0$ (dashed line) and at a finite temperature (solid line). (c) Density of occupied states $g(E)f(E, T)$.

This density of states for the free electron model is shown in Fig. 2.6(a). So far, we have only considered the situation at zero temperature. At any finite temperature, electrons will be thermally excited from their ground state. For fermions, the occupation probability of the states is given by the *Fermi-Dirac* distribution $f(E, T)$

$$f(E, T) = \frac{1}{e^{(E-\mu)/k_B T} + 1} \quad (2.52)$$

where μ is the chemical potential. For metals, we can set $\mu = E_F$ and do not distinguish between the Fermi energy and the chemical potential at all. At zero temperature,

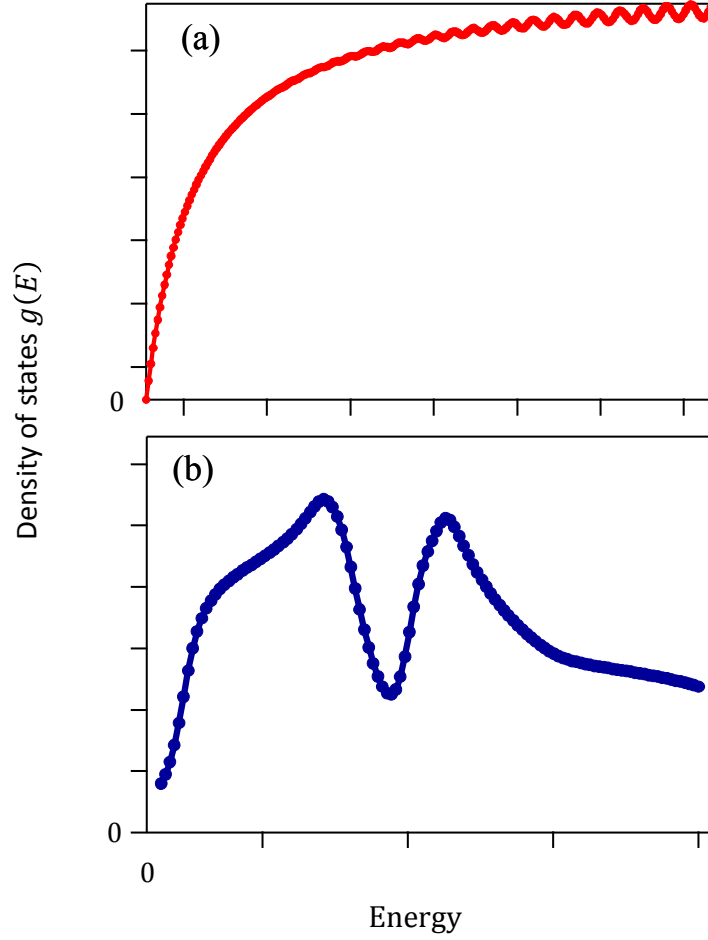


Figure 2.7: Free electron, v.s. nearly free electron LDOS. - (a) Density of states in the free electron model. (b) Qualitative density of states in the nearly free electron model with the appearance of an absolute band gap.

this is exactly true, and at finite temperature, it is a very good approximation. The Fermi-Dirac distribution is shown in Fig. 2.6(b). At zero temperature, it is represented by the dashed line that has a value of 1 for energies smaller than μ and 0 for energies higher than μ , meaning that all states below the chemical potential are occupied and all others are empty. At finite temperature, the Fermi-Dirac distribution develops a “*soft zone*” around μ in which the occupation probability is no longer 1 or 0 but something in between. The soft zone is symmetric around μ (or), and it has a width of about $4k_B T$. At room temperature, $k_B T \approx 25$ meV, such that the soft zone is around 100 meV wide. Finally, the density of occupied electron states at a given energy and temperature can be found by multiplying the density of states $g(E)$ with the Fermi-Dirac distribution

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$f(E, T)$, see Fig. 2.6(c). This definition gives us a way of calculating the chemical potential at any temperature because the total number of electrons N must be given by

$$N = \int_0^\infty g(E)f(E, T)dE \quad (2.53)$$

An even simpler picture for seeing the difference between free electrons and nearly free electron models emerges when we look at their density of states, as in Fig. 2.7. For free electrons, we have calculated the density of states to be proportional to the square root of the energy (Fig. 2.7(a)). For the nearly free electron model, the density of states must be more complicated than this. Gaps appear in the band structure at the Brillouin zone boundaries, and this can (but need not) lead to absolute gaps in the bands. Such a more complicated density of states with an absolute band gap is shown in Fig. 2.7(b).

2.3 Electron boundary element method (EBEM)

Some numerical approaches which are used for studying nanostructured surfaces include plane-wave expansion^[102], finite-difference methods^[103], and finite-element methods^[104]. Plane-wave expansion methods are found to converge relatively slowly in the vicinity of the interfaces of nanostructures where the potentials change rapidly. Finite-difference methods involve discretizing Schrödinger's equation in the two (or more) variables that do not separate. The resulting matrix is be diagonalized then increases roughly with the product of the numbers of points chosen in each direction and can involve relatively large matrices. Finite-element methods bear a resemblance to finite-difference methods. Each of these methods has difficulties in treating highly excited or continuum states, for which the wave functions vary rapidly over the nanostructure. This section will present an alternative method that is efficient and straightforward for the calculation of the wave functions, eigenenergies, and local density of stats for electrons in arbitrarily shaped nanostructures composed of different uniform regions. It is called the “*boundary-element method*” (BEM). This method provides advantages, including improved numerical efficiency, over alternate approaches^[76,77]. In order to

obtain a full understanding of nanostructure systems with arbitrarily shaped building blocks and to provide a detailed interpretation of experimental results for realistic systems it is necessary to have quantitative results for the single-particle electronic states of such structures. In the case of quantum wells and superlattices, Schrödinger's equation generally separates, and such results can readily be obtained from one-dimensional

2.3 Electron boundary element method (EBEM)

calculations. For lower-dimensional structures Schrödinger's equation generally does not separate; to date either simple approximations or numerically extensive methods have been used. EBEM present a numerical approach for calculating quantitatively the single electronic properties of 2D structures of general shape that is relatively efficient and conveniently gives results for high-lying and for continuum states. In general, electronic states are determined by a partial differential equation, whose solution satisfies certain boundary conditions. For the cases of a wave confined to or excluded from a region of finite spatial extent it is known^[105] that by using *Green's-function* techniques one may transform the system of a differential equation and boundary conditions into an integral equation over only the internal interfaces of the system; these equations can then be discretized and solved numerically^[106]. The technique can be extended to inhomogeneous systems such as heterostructures, which involve the quantum-mechanical motion of an electron from one homogeneous region to another.

Using the effective-mass approximation^[107] Schrödinger's equation can be written in the form

$$\left(-\frac{\hbar^2}{2}\nabla \cdot m(r)^{-1}\nabla + U(r) - E\right) \psi(r) = 0 \quad (2.54)$$

when $m(r)$ and $U(r)$ are the *carrier mass* and the *potential energy* operator, respectively. Each of $m(r)$ and $U(r)$ is constant in the several regions of the nanostructure i.e., piecewise uniform. At the interface separating two different regions, both the wave function ψ and its inverse-mass weighted normal derivative are continuous. If there is no magnetic field, the Hamiltonian is time reversible and the wave functions can be expressed as real quantities. Within a particular region V of the nanostructure through which m and U are uniform, the equation governing the Green's function is

$$\left(-\nabla^2 + \frac{2m}{\hbar^2}(U - E)\right)G(r, r'; E) = \delta(r - r') \quad (2.55)$$

where E is complex and usually with only a small imaginary component. Following the usual convention, the particular (unique) form chosen for G satisfies

$$\lim_{|r-r'|\rightarrow\infty} G(r, r'; E)$$

if $\text{Im}(E) > 0$. For uniform regions ($m(r)$ and $U(r)$ are constants) the results for the Green's function are Tabulated in many references^[108] for various geometries. The discussion of issues in terms of the number of independent variables that need to be treated numerically in the problem under consideration is very important. Typically this number is that of the largest set of variables that do not separate from one another in the problem. In the present case of patterned 2-D systems, whose lateral extent

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typically greatly exceeds their vertical extent, the mismatch of the sizes ensures that the motion in the growth direction separates from that in the transverse directions owing to the symmetry-breaking induced by the patterning process. The Green's functions lists have number of independent Cartesian variables which correspond essentially to free motion in one-, two-, or three-dimensional space.

Multiplying the complex conjugate of Eq. 2.54 by $2mG(r, r'; E)/\hbar^2$, subtracting the resulting equation from Eq. 2.55 multiplied by $\psi(r)$, integrating the resulting equation over the volume $V(r)$ and using Green's theorem, then

$$\int_s [m\phi(A)G(A, r'; E) - \psi(A)\delta_A G(r, r'; E)]dA = \psi(r') \quad (2.56)$$

Where A indicates a point r_A located on the surface bounding S ,

$$\phi(A) \equiv m^{-1}\delta_A\psi(r), \delta_A \equiv \hat{n} \cdot \frac{\partial}{\partial r}|_A$$

and $\hat{n}$ is the normal vector that points out of the region V at the point A . Based on Eq. 2.56, if the values for, the Green's function everywhere within V , the wave function at all points A on the surface S , and the normal derivative of the wave function for all $A \in S$, then one can calculate ψ everywhere within V . In order to calculate $\psi(A)$ and $\phi(A)$ we evaluate Eq. 2.56 at a point r' that is within V but arbitrarily close to S and we interchange A and A' to obtain the homogeneous integral equation

$$\int_s [B(A, A')\phi(A') + C(A, A')\psi(A')]dA' = 0 \quad (2.57)$$

The kernels are given by

$$B(A, A') \equiv 2mG(A, A'; E), \text{ and} \\ C(A, A') \equiv -2\delta_{A'}G(r', A; E) - \delta(A - A')$$

The piecewise uniform nanostructures considered here the Green's functions involved in Eq. 2.56 and Eq. 2.57 are known and tabulated^[108]. If the composite system consists of N different homogeneous regions, then such an integral equation can be derived for each of the N regions, including both classically allowed and classically forbidden regions, as well as regions of either finite or infinite spatial extent. Because $G(r, r'; E)$ vanishes as either r or r' goes to infinity, the “surface” at infinity does not appear in any of these equations. The entire linear system of integral equations can then be represented as

$$\begin{bmatrix} B_1 & C_1 \\ B_2 & C_2 \\ \cdot & \cdot \\ \cdot & \cdot \\ B_N & C_N \end{bmatrix} \begin{bmatrix} \phi \\ \psi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}, \quad \begin{bmatrix} B_1 & C_1 \\ B_2 & C_2 \\ \cdot & \cdot \\ \cdot & \cdot \\ B_N & C_N \end{bmatrix} \equiv M \quad (2.58)$$

2.3 Electron boundary element method (EBEM)

where B_I and C_I are the I th region's integral operators, which depend parametrically on the energy E . The lower half of the symbolic column vector ψ th contains the values of the wave function on all interfaces, and the upper half ϕ th contains the values of the normal derivatives of the wave function divided by the effective mass m . If the particle's motion is classically forbidden in all regions of infinite spatial extent and is classically allowed in at least one of the finite regions, then the energy spectrum contains a bound portion. The energies of these bound states are determined by finding the nontrivial solutions of Eq. 2.58, which occur for certain discretely spaced (real) values of E . To implement this approach EBEM discretize the integral in Eq. 2.57, thus approximating it as a sum of n terms, to obtain

$$\sum_{j=1}^n (B_{ij}\phi_j + C_{ij}\psi_j) = 0 \quad (2.59)$$

when $B_{ij} \equiv \int_j dA' B(A_i A')$, $C_{ij} \equiv \int_j dA' C(A_i A')$, $\phi_j \equiv \phi(A_j)$, $\psi_j \equiv \psi(A_j)$, and $\int_j$ is the integration over a small boundary element S_j , where $\sum_{j=1}^n S_j = S$, and n is the number of boundary elements on S . The entire set of homogeneous integral equations then is approximated by a homogeneous equation represented again by Eq. 2.58, but for which B_I and C_I are $n_I \times N$ matrices (instead of integral operators), M is a $2N \times 2N$ matrix (instead of an integral operator), ϕ and ψ are N -component column vectors (instead of functions), and $\sum_{I=1}^N n_I = N$, where n_I is the number of boundary elements on S_I , the surface of the I th region. The divisor of 2 in the definition of N originates from the fact that each boundary element serves two different regions.

Only when the determinant of the complex-valued matrix M in Eq. 2.58 vanishes, this equation will have a nontrivial solution. For the bound states, this determinant vanishes for values of E that are discretely spaced in the complex plane. We usually find these energies to be slightly removed from the real axis owing to the approximation associated with discretizing the integral equation. This procedure yields accurate results for the energies of those bound states for which the wave functions, or, more precisely, the wave functions and their normal derivatives evaluated along the boundary S , vary slowly on the scale of a typical boundary element. In order accurately to determine highly excited states the grid of boundary elements needs to be sufficiently fine.

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2.4 Input structure and algorithm

The problem under investigation by EBEM code is the electronic structure of 2D patterned metal surfaces. For that, EBEM code presents solution of Schrödinger's equation in 2D including customizable potential barriers and using the electron boundary element method (EBEM) for finite systems or electron plane wave expansions (EPWE) for periodic systems. First, the algorithm should be getting the input structure. EBEM code is prepared to accept commands from an input file to define the boundary geometry and instructs it to calculate the outputs such as, the band structure, the local density of states (LDOS), photoemission (PE) intensities, and wave functions. As in Fig. 2.8 the input's first step (construct geometry) is achieved by creating a base of potential barrier, then construct the finite geometry by repeating this lattice point (the base) by the translation vector ($\bar{T} = n_1\bar{a}_1 + n_2\bar{a}_2$) of this problem. The boundary extensions (n_1, n_2) of the finite geometry must be quietly enough to express of an infinite system. After constructing the geometry of the concerned problem, the properties of each region are defined (potential, $V_s(r)$), and the effective electron mass (m_{eff}). The current version of the code allows assigning different potentials for all regions, but only a single effective mass for the whole system. The electron spin and anisotropy due to different m_{eff} are not yet included. For adopting the output results we must test and optimize some numerical points such as, the minimum number of discretization points required to suitably describe the boundary properties, maximum length of reciprocal lattice vectors G_{max} involved in the expansion, and optimal real space position of the periodicity origin. The latter ensures all segments of the defined bases are contained within the first unit cell. These parameters are continuously tuned until convergence is achieved for the given calculation. Nonetheless, it is recommended to consider the parameters that achieve the lower onset of convergence, to avoid time lengthy calculations. According to the EPWE earlier analysis in numerical solution (Sec. 2.2.11) the band structure calculations E_{nk} can be achieved along a momentum segment $k_{||}$. Using output command of band structure along this segment, the EBEM code computes list of the Bloch wavevectors k at each band index n , and solves localized eigenvalue problem at each k . Likewise in this thesis the EPWE output commands of the band structure, 1-D and 2-D local density of states (LDOS), photoemission (PE) intensities, and constant energy surface (CES) are used. We must note that the EPWE output commands only works for periodic systems. For situations such as local defects in periodic solids or finite objects, different EBEM output commands is developed. Green's-function techniques are used to derive integral equations that determine these

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electronic properties. The EBEM attempts to use the given boundary conditions to fit boundary values into the integral equation, rather than values throughout the space defined by a partial differential equation. Once this is done, in the post-processing stage, the integral equation can then be used again to calculate numerically the solution directly at any desired point in the interior of the solution domain. Based on that, all real space interfaces calculations in this work have been stimulated by EBEM.

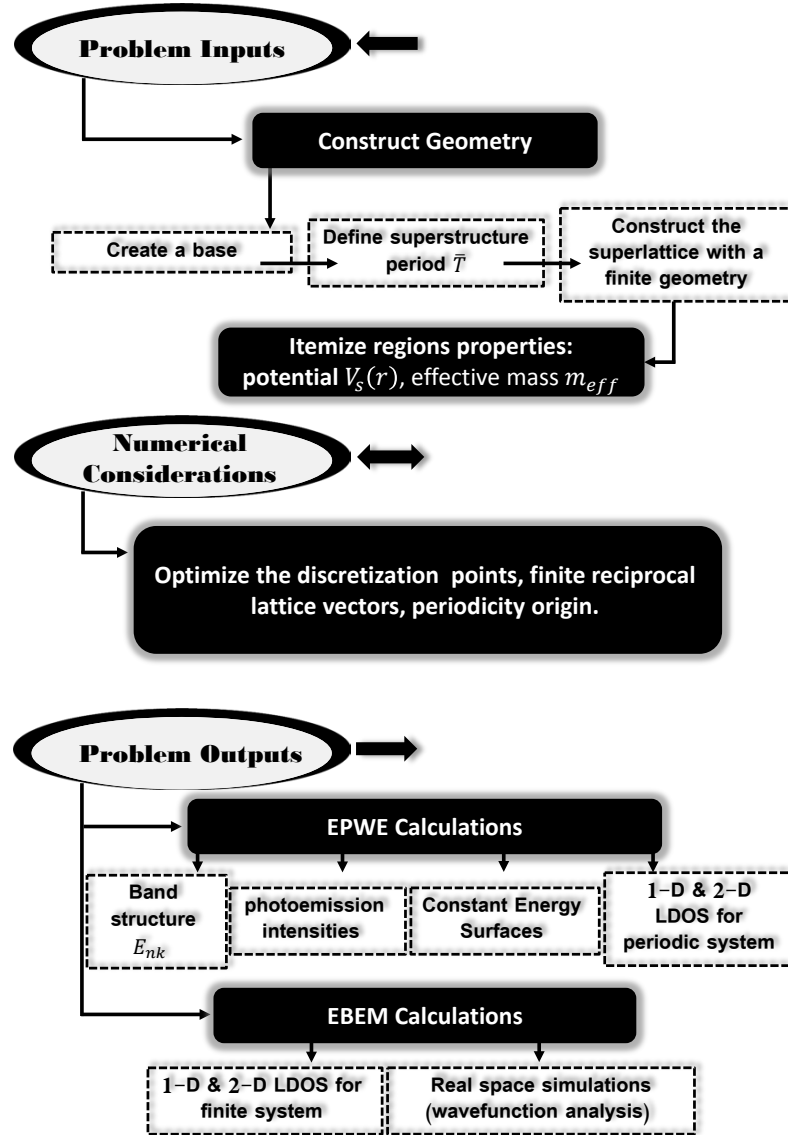


FIG. 2.8. Flowchart of the EBEM-EPWE signature algorithms.

Figure 2.8: Code Flowchart. - Flowchart of the EBEM-EPWE signature algorithms.

2. THEORETICAL BACKGROUND

2.5 Theory behind Angle Resolved Photoemission Spectroscopy

Based on photoelectric effect angle-resolved photoemission spectroscopy (ARPES) can directly detect the band structures of solid materials. This unique advantage, together with its broad applicability to various material systems, has made ARPES a technique that is widely used to study solid state materials^[109]. The photoelectron spectroscopy is based on the “*photoelectric effect*”, experimentally reported by Hertz in 1887^[110] and later explained by Einstein in 1905^[111] as a quantum phenomenon where the particle nature of light (photons) was proposed. For this reason Einstein was later awarded the Noble price in physics in 1921. In his derivation, the relationship between the photon energy $h\nu$ and the maximum kinetic energy E_{kin}^{max} of the photoemitted electrons, i.e., the fundamental photoelectric effect equation, was found to be

$$E_{kin}^{max} = h\nu - Q_0 \quad (2.60)$$

where Q_0 is a characteristic constant of the sample surface which is defined as the work function.

Nowadays, a photoemission experiment for spectroscopic purposes is basically performed in the same way as reported by Hertz. Monochromatized photons, either vacuum ultraviolet (VUV) light, x-ray light, or monochromatized synchrotron radiation, are directed on a sample, and the photoelectrons, liberated by the photoelectric effect, are analyzed with respect to their kinetic energy and/or emission angle using an electrostatic analyzer^[112], Fig. 2.9. Depending on the photon energy used in the photoemission experiment, one usually distinguishes two different kinds of spectroscopies, namely x-ray photoemission spectroscopy (XPS) and ultraviolet photoemission spectroscopy (UPS, ARUPS, or ARPES), when x-ray and ultraviolet radiation, respectively, are used^[19]. XPS provides the investigation of the core-level states at high binding energies and is, therefore, sensitive to the chemical composition. ARPES, on the other hand, investigates the valence band states and the electronic structure. One major advantage of photoemission spectroscopy is the short escape depth of the photoemitted electrons (few Å for the range of the kinetic energies relevant for photoemission experiments), which renders the technique being surface sensitive and, thus, well suited to map the electronic properties of surfaces, surface alloys, thin films, etc.

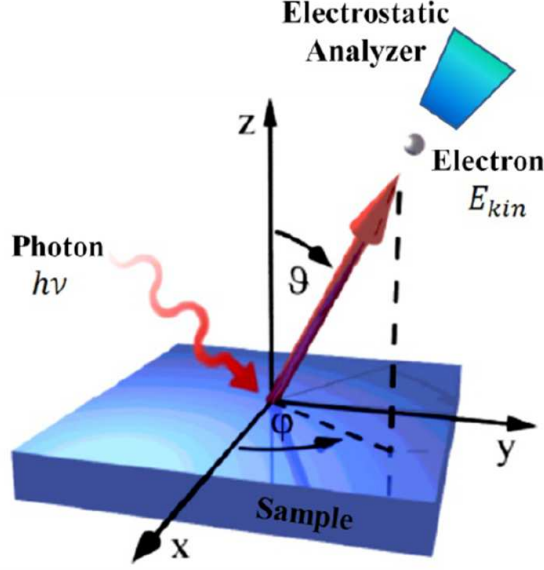


Figure 2.9: Principle of Photoemission. - It consists mainly of a photon source, a sample to be investigated and an electrostatic analyzer. Taken from Ref. ^[113].

2.5.1 Photoemission Theory:

The interaction of the electrons with an oscillating electromagnetic wave field $A(r, t)$ is described by the canonical replacement of the momentum operator p by $p + A$ ^[112]. Assuming that the spatial variation of the light is small on an atomic scale, the resulting perturbation to the original Hamiltonian is $H_{int} = A.p$. According to Fermi's Golden Rule the transition rate between the initial and final one-electron states $|i\rangle$ and $|f\rangle$ with energies E_i and E_f , respectively, can now be written as

$$w_{i \rightarrow f} \propto |\langle f | A.p | i \rangle|^2 \delta(E_f - E_i - h\nu) \quad (2.61)$$

The situation is the same for a solid with a lattice-periodic potential, where the quantum numbers i and f are replaced by the Bloch vectors k_i and k_f for the initial (occupied) and final (empty) state, respectively. The total photocurrent can therefore be written as

$$I_{abs} \propto \sum_{k_f, k_i} |\langle k_f | A.p | k_i \rangle|^2 \delta(E_f - E_i - h\nu) \quad (2.62)$$

In an ARPES experiment, free photoelectrons of specified energy E and momentum K (in vacuum) are measured. This means that only absorption into final states with $E = E_f$ is detected. The momentum component parallel to the surface is a conserved quantity (except for a surface reciprocal lattice vector $G_{||}$: $K_{||} = k_{f||} - G_{||}$). This is

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not the case for the perpendicular component $k_{\perp}$ due to the potential barrier at the surface. The photoelectron current can then be expressed as

$$I(\mathbf{K}, E) \propto \sum_{k_f, k_i} | \langle \mathbf{k}_f | \mathbf{A} \cdot \mathbf{p} | \mathbf{k}_i \rangle |^2 \delta(E_f - E_i - h\nu) \delta(\mathbf{K}_{\parallel} - \mathbf{k}_f - \mathbf{G}_{\parallel}) \delta(E - E_f) \quad (2.63)$$

Due to the Bloch form of the one-electron wavefunctions the matrix element in the

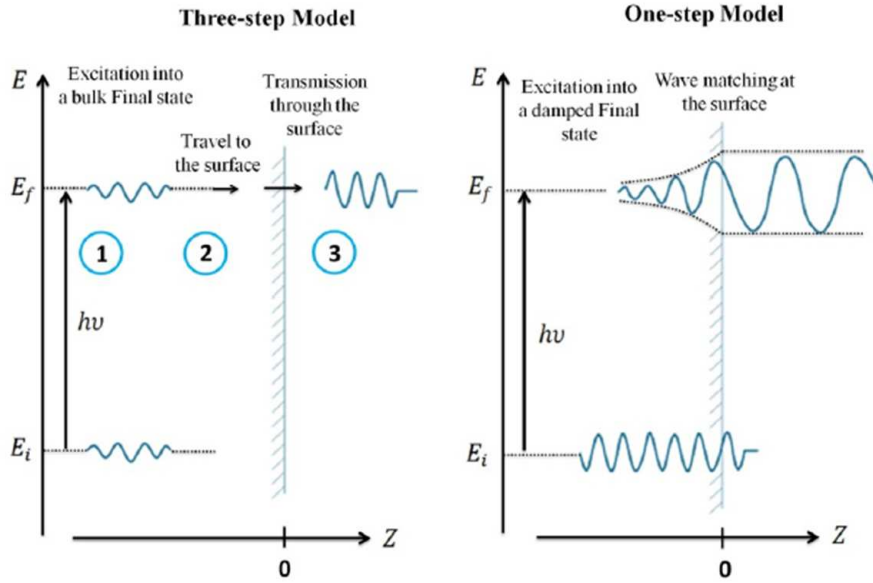


Figure 2.10: Three and One-Step Models - Pictorial representation of three-step and one-step model description of the photoemission process. Adapted from Ref. ^[114].

above equation also conserve the momentum (except for a reciprocal lattice vector $\mathbf{G}$):

$$| \langle \mathbf{k}_f | \mathbf{A} \cdot \mathbf{p} | \mathbf{k}_i \rangle |^2 = | \mathbf{M}_{if} |^2 \delta(\mathbf{k}_f - \mathbf{k}_i - \mathbf{G}) \quad (2.64)$$

where $| \mathbf{M}_{if} |^2$ is the residual matrix element. Hence the expression for the photoemission current can be written as

$$I(\mathbf{K}, E) \propto \sum_{k_f, k_i} | \mathbf{M}_{if} |^2 \delta(E_f - E_i - h\nu) \delta(\mathbf{k}_f - \mathbf{k}_i - \mathbf{G}) \delta(\mathbf{K}_{\parallel} - \mathbf{k}_{f\parallel} - \mathbf{G}_{\parallel}) \quad (2.65)$$

This is actually the expression derived from the frequently used three-step model ^[113,114], illustrated in Fig. 2.10. The model assumes three discrete steps as follows. A photon is absorbed and excites an electron in the solid from an initially occupied state to an unoccupied temporary state. This excited electron is then scattered by the ions and eventually arrives at the surface of the solid. At the surface the excited state penetrates

through the surface into the vacuum^[115]. Those three steps are clearly described in the photoemission current expression, where the first two δ -function takes care of the energy and (parallel) momentum conservation during the photoexcitation deep in the solid, and the last two δ -functions consider the energy and momentum conservation at the solid-vacuum interface. Using this expression, the initial state band dispersions can be mapped out using the energy and momentum of the photoelectrons measured in an ARPES experiment.

2.6 Theory behind Scanning tunneling microscope

A scanning tunneling microscope (STM) is an instrument for imaging surfaces at the atomic level. Its development in 1981 earned its inventors, Gerd Binnig and Heinrich Rohrer (at IBM Zurich), the Nobel Prize in Physics in 1986^[98,116]. For an STM, good resolution is considered to be 0.1 nm lateral resolution and 0.01 nm depth resolution^[117]. With this resolution, individual atoms within materials are routinely imaged and manipulated. The STM can be used not only in ultra-high vacuum but also in air, water, and various other liquid or gas ambient, and at temperatures ranging from near zero Kelvin to a few hundred degrees Celsius. The components of an STM include scanning tip, piezoelectric controlled height and x, y scanner, coarse sample-to-tip control, vibration isolation system, and computer^[118].

STM is based on the quantum mechanical “*tunneling effect*”. When an atomically sharp tip is brought close to a sample surface, at a distance of ~ 1 nm, under a bias voltage, V_t , between the tip and a sample, a small tunneling current, I_t , starts to pass between them. This tunneling current drastically increases with decreasing separation between the tip and the sample^[119]. It is, therefore, possible to precisely evaluate the change in the separation by measuring the tunneling current. The tip is raster-scanned across the surface by using, e.g. a piezoelectric tubes. The atomic corrugation of the surface gives, then, rises to variations in the tunneling current with distance (z) between the sample and the tip. As a rule of thumb, the current reduces an order of magnitude for an increase of the gap distance by ~ 1 Å. This sensitivity leads to a high lateral and vertical resolution: the topmost atom at the tip-apex drags around 90 % of the current (assuming that it protrudes ~ 1 Å further than other tip atoms).

The whole STM operation is usually fully computer-controlled and the scanning parameters like V_t , I_t , and the raster speed are set via an interface, see Fig. 2.11 for

2. THEORETICAL BACKGROUND

clearness. There are basically two modes of operating the STM^[19]. In the “*constant-current mode*”, the current I_t is compared to a preset current by a feedback circuit. It provides a correction voltage to the scanner tubes which adjusts the z position of the tip in order to keep I_t constant. The correction feedback signal is recorded together with the $x - y$ position of the tip and hence the STM image is obtained. In the “*constant-height mode*”, the z -position of the tip is kept constant and the tunneling current is recorded as it varies while raster scanning the sample surface. Generally, the constant-current mode yields better resolution whereas constant-height mode allows faster scanning.

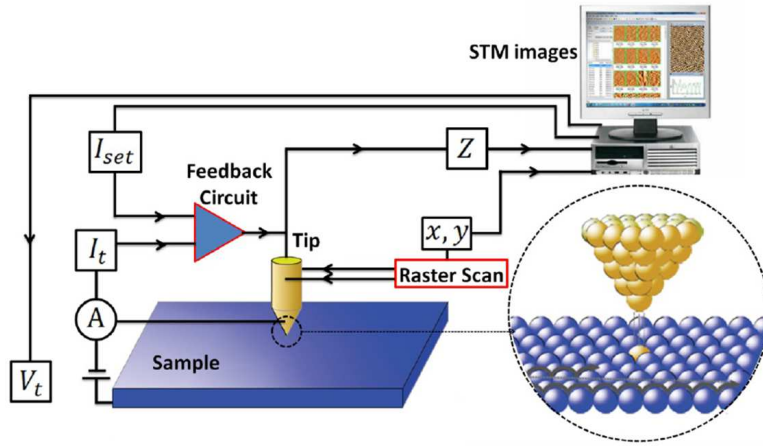


Figure 2.11: STM Operation Principle - Schematic drawing illustrating the operation principle of a fully computer controlled STM. Tip positioning with atomic precision is realized by piezoelectric elements. The current is recorded in a defined number of points while scanning the tip from left to right (gray lines in the zoomed view), but not during the fly-back to the next scan line (dashed line).

An exact theoretical treatment of the tunneling process in STM is virtually impossible for several reasons. It requires a detailed description of the sample and tip states and their evanescence into the tunnel gap; this is not feasible for a low symmetry object like the tip with mostly unknown shape and exact chemical composition^[119]. Moreover, the tip apex structure can even change in the course of an experiment. In a simple approach, the tunneling process in one dimension serves as an introduction to the concept of STM imaging. Assuming a constant potential barrier U in a region $0 < z < d$, the wavefunction $\psi(z)$ describing an electron with energy $E < U$ in the positive z direction, in a classically forbidden region is given by^[120]

$$\psi(z) = \psi(0) e^{-kz} \quad (2.66)$$

2.6 Theory behind Scanning tunneling microscope

where $k = \sqrt{\frac{2m(U-E)}{\hbar}}$ is the wavevector, m is the mass of an electron and $\hbar$ is the Planck's constant divided by 2π . Hence, the probability of observing an electron at the end of the potential barrier ($z = d$) is proportional to $|\psi(z)|^2 = |\psi(0)|^2 e^{-2kz}$.

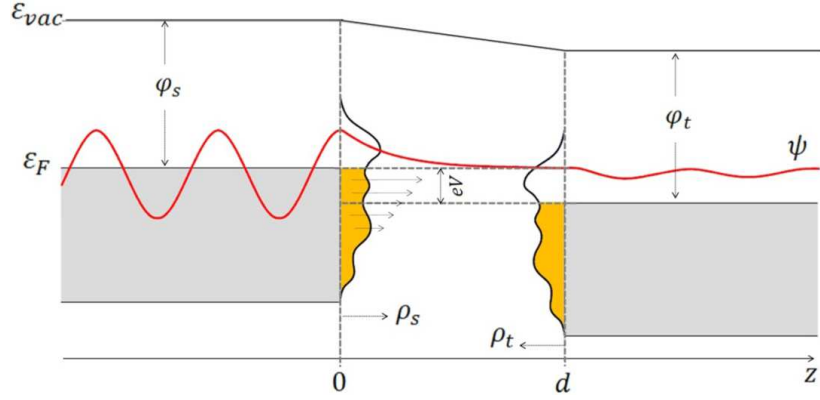


Figure 2.12: Density of States Mapping by STM - Schematic energy diagram for the sample-tip tunnel junction with a width d . A positive bias voltage V is applied to the tip, i.e. tunneling proceeds from occupied sample states to empty tip states (occupied states in the sample/tip are shaded grey).

This exponential decay in the barrier region is illustrated in Fig. 2.12. The simplistic theory presented above is not enough to explain all phenomena observed by STM. In this theory the current at a given energy is proportional to the product of the density of states (DOS) of the tip ρ_t and the local DOS (LDOS) of the sample ρ_s . The expression for the current, I , has the following form^[120,121]:

$$I(V, T, x, y, d) \propto \int_{-\infty}^{\infty} \rho_s(E, x, y) \rho_t(E - eV) \tau(E, V, s) \times [f(E - eV, T) - f(E, T)] dE \quad (2.67)$$

where x and y are the lateral coordinates, f is the Fermi-Dirac function, T is the transmission probability, and τ is the transmission factor. The latter in general depends also on the parallel momentum but in a first approach it can be disregarded without losing any qualitative understanding. Then, the following simplified expression is obtained for the current:

$$I(V, T, x, y, d) \propto e^{-2d\sqrt{\frac{m}{\hbar^2 \sqrt{2\bar{\varphi}}}}} \int_{-\infty}^{\infty} \rho_s(E, x, y) \rho_t(E - eV) \times [f(E - eV, T) f(E, T)] dE \quad (2.68)$$

where $\bar{\varphi} = \frac{\varphi_t + \varphi_s}{2}$ is the average work function.

2. THEORETICAL BACKGROUND

According to this equation, the tunneling current will depend on: the tip-sample distance s (I depends exponentially on d), the applied bias eV , and the energy of the state (the higher the energy of the electron the lower the potential barrier and thus the higher the transmission probability. This is schematically shown in Fig. 2.12 by the length of the horizontal arrows)^[121].

3

Results and Discussion I

3. RESULTS AND DISCUSSION I

Electronic Structure of 2D Metallic Superlattices

3.1 Introduction and Literature Survey

The creation of a surface from a three-dimensional (3D) crystalline material often leads to a change of its electronic and/or structural properties. A crystal surface breaks the periodicity (i.e., the translational symmetry) of the crystal lattice in the direction perpendicular to the surface^[65]. The periodicity in the two-dimensional (2D) surface plane is either preserved or modified. Hence 2D systems, such as the surfaces of crystalline solids, have their own dispersive and/or localized electronic eigenstates. These states are the two dimensional analogues of the electronic states associated with point defects that produce the shallow and deep levels in the band gap of semiconductors^[122]. The defect-like electronic states associated with the termination of a perfect crystal structure at the surface are called “*surface states*” or resonance. These states have properties which are quite different from the bulk electronic structure of the host/substrate. Moreover they are, because of their 2D nature, distinct from most other localized/non-dispersive defect levels and therefore easily recognizable^[123]. Like other types of defect levels, surface states have a profound influence on the physical and chemical properties of a surface. There are different types of electronic states that may occur at the surface of a perfect crystal (Fig. 3.1). These are true surface state, surface resonance, etc^[122]. Obviously the bulk electronic states will extend into the surface region, where they are reflected back by the vacuum barrier, Fig. 3.1(a). True surface states, on the other hand, exist only at the surface and have an exponentially decaying wavefunction both in the direction into the solid and also into the vacuum, Fig. 3.1(b). In energy/momentum space these states are restricted to regions where no bulk states of the same symmetry and quantum number are allowed to exist. If the wavefunction of the surface state mixes with the bulk wavefunction, the true surface state turns into a surface resonance. Therefore, surface resonances are equivalent to bulk wavefunctions with enlarged amplitude at the surface region of the crystal, Fig. 3.1(c). Historically, Shockley surface states at metallic surfaces, first described

3. RESULTS AND DISCUSSION I

by William Shockley^[124], are long known to significantly affect the properties of solid surfaces in many aspects, such as the molecular adsorption mechanisms^[125].

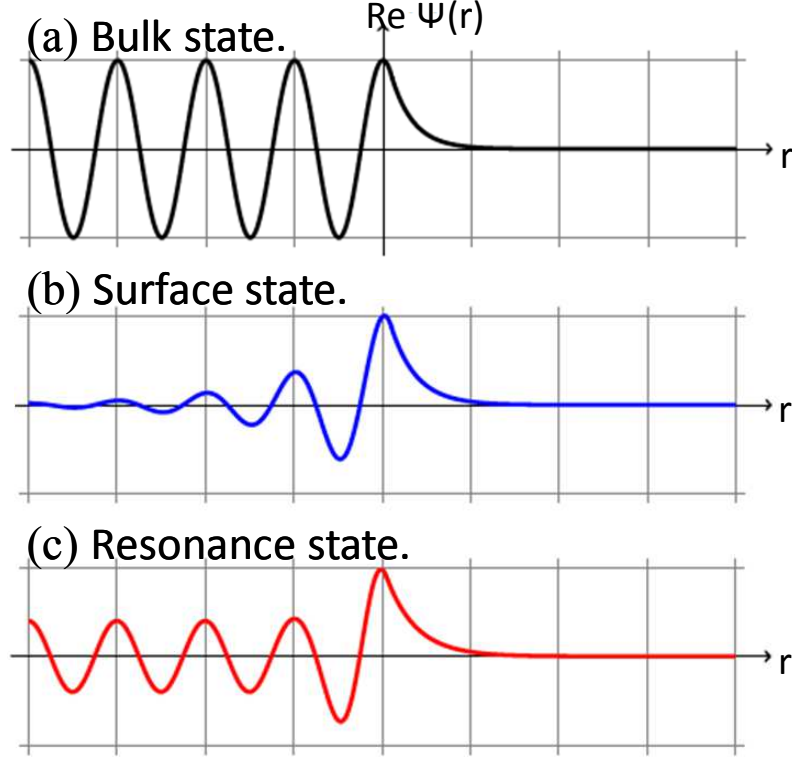


Figure 3.1: Surface states - Selected types of the electronic states at crystal surfaces.

Surface states on noble metal (111) surfaces are among the most often studied surface electronic states. These, Shockley type, surface states are placed inside the bulk projected $\bar{L}$ -band gap at the center of the surface Brillouin zone (Fig. 3.2) and are, therefore, true surface states. There are many theoretical calculations^[126–129] for the bulk band structure of noble metals which show good enough agreements with the experimentally measured ones. These calculated band structures are shown in Fig. 3.2(c) for Cu, Ag, and Au. The three noble metals do have, qualitatively, similar band structure close to the $\bar{L}$ -point. The symmetry points of special interest for the *fcc* noble metals are: $\bar{\Gamma}$ (the center of the BZ), $\bar{X}$ (the midpoint of the diamond square of the BZ boundary), $\bar{L}$ (the center of the hexagonal BZ boundary), and $\bar{K}$ (the midpoint of the edges) as sketched in Fig. 3.2(a). All the three noble metals have the inverted gap character and, therefore, the occurrence of Shockley surface states inside such projected band gap, shown for Cu in Fig. 3.2(d), is possible^[130]. Clearly seen, the sizes

3.1 Introduction and Literature Survey

of the $\bar{L}$ -gaps are different for the three noble metals. More importantly, the energetic position of the lower edges of the $\bar{L}$ -gap is different, where it is close to the Fermi level for Ag than for Cu and Au. Within the free-electron model, the Fermi surface of an *fcc* structure and one conduction electron per atom, as for Cu, is a sphere with a volume equal to half the volume of the BZ and is, therefore, completely contained within the BZ without touching its surface as in Fig. 3.2(b).

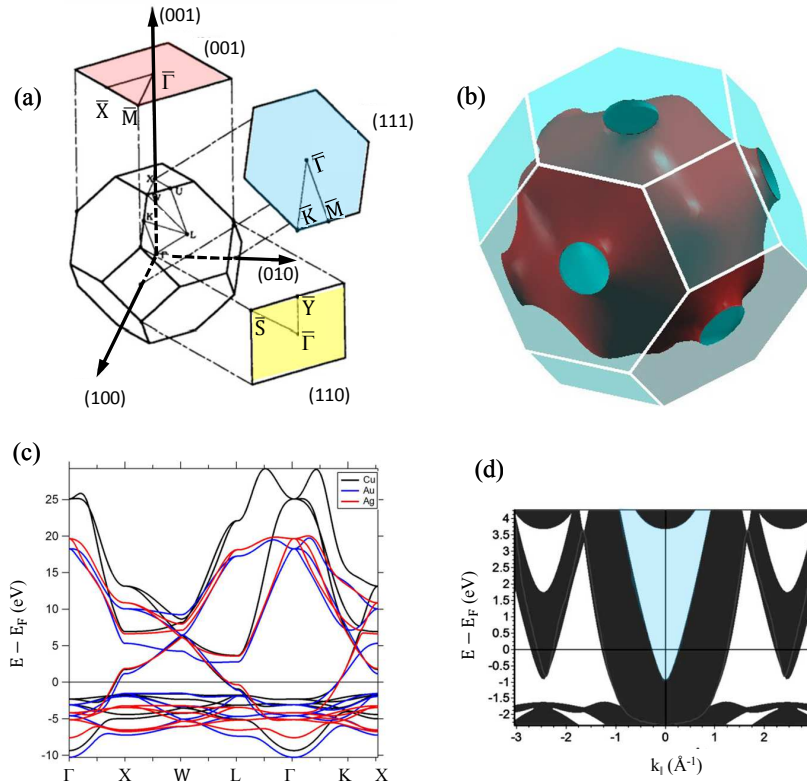


Figure 3.2: Brillouin zones and Bulk band structure of noble metals - (a) First Brillouin zones for the bulk, (100), (110), and (111) planes of fcc crystal. Adapted from Ref.^[19]. (b) A sketch of the Fermi surface of Cu, the polyhedron represents the BZ. Adapted from Ref.^[127]. (c) Calculated band structure of Cu (black), Au (blue), and Ag (red) along the various symmetry axes on the BZ. (d) The projected bulk band gap of Cu on the (111) surface. The Shockley surface states as well as the image potential states are confined in the blue shaded area.

The surface states on the (111) oriented noble metal surfaces are characterized by free-electron parabolic dispersion with different dispersion parameters for the three noble metal surfaces^[20,21,131]. Each one of them has particular energetic position of the surface states band minima (E_0), effective mass (m_e), and the Fermi wavevector (k_F).

3. RESULTS AND DISCUSSION I

These parameters have been accurately determined by angle-resolved photoemission spectroscopy (ARPES) on all the clean (111) noble metal surfaces. ARPES is one of the most direct methods to study the electronic structure of the solid surfaces. It gives information on the energy and momentum of electrons in crystalline solids, resulting in detailed knowledge of the band dispersion and Fermi surface^[19].

The previous free-electron dispersions are related to the defect-free “*clean*” metal surfaces, and are significantly modified in the presence of defects or surface reconstruction. Reconstructions can naturally occur at the surface^[132] or induced by the adsorption of other atoms^[27,28] or thin atomic-layers^[133] onto the surface as the interatomic forces are changed. These reconstructions exhibit long-range “*periodic*” superstructures and can assume a variety of forms when the detailed interactions between different types of atoms and their interaction with the substrate are taken into account. The reconstruction of a surface depends on many factors, such as the composition of the substrate and the adsorbate, the adsorbates coverage, and the ambient conditions (i.e., temperature, gas pressure, etc.)^[122]. Moreover, an ideally ordered surface with complete 2D translational symmetry does not occur in reality^[134,135]. There are always a certain number of different types of defects (e.g. steps, dislocations, vacancies) that coexist on a clean surface and, in turn, scatter the surface electrons. The presence of such defects can be of considerable importance if one was able to assemble them periodically over the surface. This will result in the formation of 1D or 2D lateral superlattices with new (larger) translation symmetry.

Nowadays, using many surface nanofabrication tools such as self-assembly processes, it is possible to tune a vast variety of materials in modulate configurations on metallic substrate surfaces thus; unique systems can be fashioned for practical applications or for scientific study. In self-assembly processes, energy sources such as thermal energy drive the initially disordered components of a system toward a stable, ordered state. The interactions between the components define the final result. One may take advantage of scale-independent self-assembly processes to simplify or tuning the engineering of complex systems over orders of magnitude of size. Two-dimensional nanostructures made of inorganic and organic networks on metallic substrates have received much attention in recent years. Their fabrication has progressed rapidly such that patterns with complex unit cells and almost of any arbitrary geometry can be fabricated, as the real systems of 2D nanopatterned examples^[16,136–138] in Fig. 3.3. These **superstructures** with the plenty different forms of superlattice unit cells, were imaged with

atomic resolution by scanning tunneling microscopy (STM), a common technique to obtain topographic maps of surfaces.

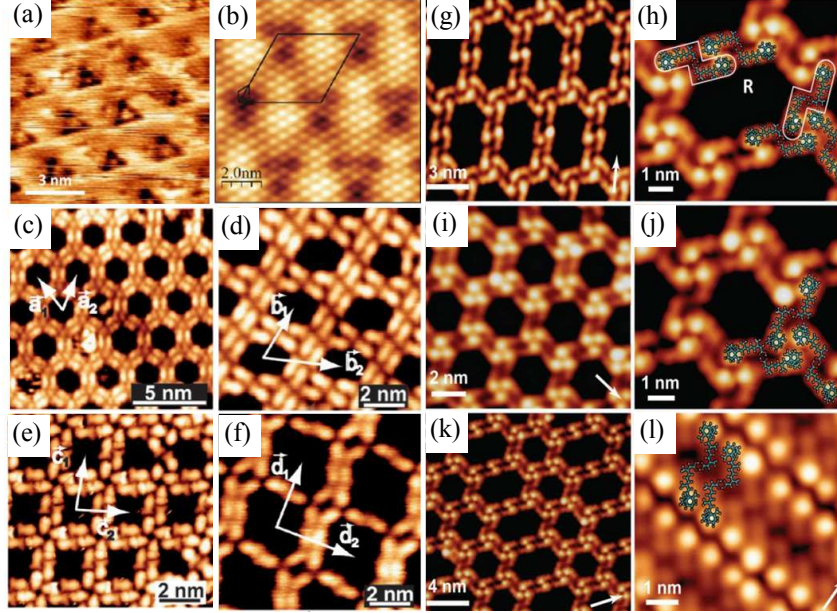


Figure 3.3: STM images - (a) The subsequent deposition of the Ag/Cu(111) dislocation **hexagonal** network. (b) The $GdAu_2$ surface compound, characterized at large scale by a **hexagonal** moiré pattern. Structure models of the open network structures formed by CPhDAT on Au(111), (c) **honeycomb** network with clockwise helicity ($14 \times 14 nm^2$), (d) **rectangular** network ($10 \times 10 nm^2$), (e) **square** open network ($10 \times 10 nm^2$), (f) large open network involving three-fold cyano-Au coordination units indicated by the blue full line ($10 \times 10 nm^2$). Various hierarchical patterns formed by 13-cis-ReA molecules on Au(111), (g and h) elongated hexagonal network, (i and j) Equilateral hexagonal network, (k) Alternative elongated and equilateral hexagonal networks and (l) Close-packed molecular arrays.

The reconstructed Ag/Cu(111) surface is one such an example Fig. 3.3(a). Due to the misfit of lattice parameters for Ag and Cu, a one monolayer (1 ML) of Ag epitaxially grown on Cu(111) induces a reconstruction consisting of a hexagonal **superlattice** made of interfacial dislocations with a lattice parameter ~ 9.5 times the interatomic distance of the Cu(111) surface^[139,140]. The interaction of the surface electrons with these periodic defects (i.e., periodic potential) has led to clear modification of the free electron dispersion as measured with ARPES^[141]. Specifically, its band structure is characterized by energy gaps at the boundary of the reconstructed Brillouin zone, and

3. RESULTS AND DISCUSSION I

the size of such gaps was used to obtain the surface potential or at least its first Fourier components^[56,139]. Therefore, the Ag/Cu system provides a textbook example of the interaction of free electrons with 2D periodic array of scattering potentials. Likewise, the so called “*vicinal surfaces*” simulate 1D periodic array of defects (steps)^[137,142]. The vicinal Au(788) surface, for example, is made up of a uniform array of (111)-oriented terraces of similar width, ~ 3.8 nm, separated by single atomic-height steps. This uniformity makes it possible to study the electronic structure of the resulting step **superlattice** by ARPES. Results showed that for this terrace array the surface state appears to be broken up into one-dimensional quantum-well levels, indicating total electron confinement within the terraces^[143]. Another example of defect-superlattices is found on the ordered array of sodium ion vacancies incorporated into a (100) surface of sodium chloride upon the epitaxial temperature associated with gold being deposited onto that (100) surface^[93]. Likewise, there have been recent achievements in changing the electronic structure of surfaces by adsorption of nanoporous networks whose design principles are based on the concepts of supramolecular chemistry^[94]. An example of the confinement of the Shockley surface state by a honeycomb network prepared by molecular self-assembly was demonstrated by Lobo-Checa et al^[95]. The network was prepared by depositing the perylene derivative 4,9-diaminoperylene-quinone-3,10-diimine (DPDI) molecule on the Cu(111) surface, leading to the formation of a well-ordered honeycomb network with a lattice constant of 2.55 nm^[144,145]. STS measurements revealed that the Cu(111) Shockley state is localized in the pores of the honeycomb network.

Given the wide variety of organic molecules, their complexity, their possible functionalization, and the various degrees of interactions with the substrates, numerous superstructures with different periodicity, crystallographic structure, and symmetry could be obtained. For 2D surface, there are only five possible Bravias superlattices as shown in Fig. 3.4. With the symmetry of the basis additional non- Bravias lattices, such as honeycomb lattice, could be also made. These lateral superstructures (1D, 2D or combination of both) play an important role in many technological processes and applications like catalysis, sensors, electronic devices, computer industry, surface-state nanoelectronics (SSNE), etc^[146]. An example for the SSNE applications is the collimation and guiding of surface electrons in the dislocation networks of Ag/Cu(111) surfaces. Such interesting applications, which provide materials as excellent information carriers for future atomic-scale devices^[146], shall be discussed in details in chapter four of the present thesis.

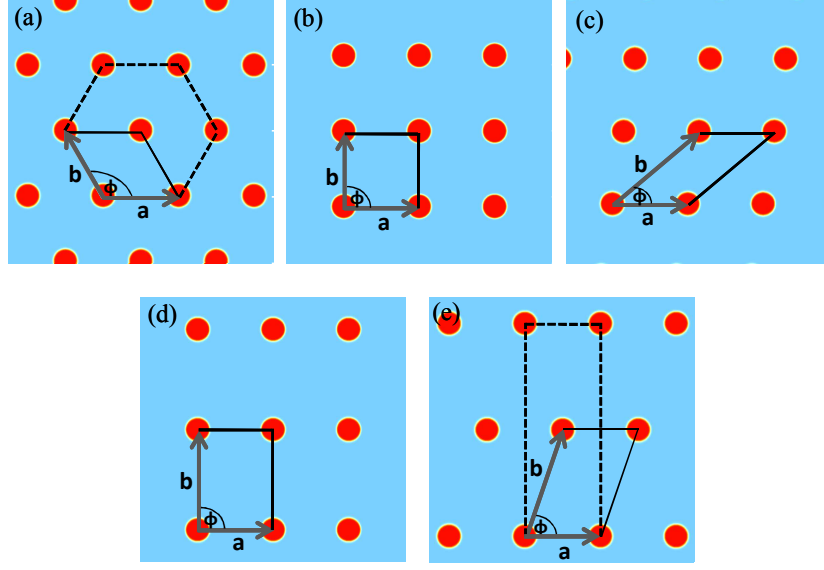


Figure 3.4: The Five 2D possible Bravais lattices. - Translation vectors a and b are shown and the unit cells are hatched. (a) **Hexagonal**: $|a| = |b|, \phi = 120^\circ$, (b) **Square**: $|a| = |b|, \phi = 90^\circ$, (c) **Oblique**: $|a| \neq |b|, \phi \neq 90^\circ$, (d) **Rectangle**: $|a| \neq |b|, \phi = 90^\circ$, (e) **Centered Rectangle**: $|a| \neq |b|, \phi = \cos^{-1} \frac{a_o}{2b}$.

In respect of all previous concepts, the present chapter deals with the interaction of free electrons, commonly present at metal surfaces, with 2D periodic arrays of scatterers arranged in two different choices of the five possible 2D Bravais superlattices. These are hexagonal and square superlattices, Fig. 3.4(a,b), with the highest translational symmetry. The effects of periodicity, periodic potential and symmetry of the bases, i.e., the scatterers, will be investigated. In each case, the band structure, the local density of states, the photoemission intensity, and constant energy surfaces will be calculated using the Electron Plane Wave Expansion (EPWE) method.

3. RESULTS AND DISCUSSION I

Part I: Hexagonal superlattice

3.2 General Overview

The first system to be investigated is a metal surface patterned with a hexagonal array of scattering potentials, as schematically shown in Fig. 3.5(a). The cyan area represents the clean region of a metal surface with a potential $V_0 = 0$, while the red circular regions define the periodic potential barriers each of height $V_1 = 700$ meV. Both the circle diameter and superlattice constant were set to 1 nm and 3 nm, respectively, whereas the effective mass used in our calculations was $0.42 m_e$, i.e. practically the same as reported experimentally for Ag(111) and Cu(111)^[147]. Figure 3.5(b) shows the reciprocal lattice (black hexagon) and first Brillouin Zone (blue hexagon) for the hexagonal superlattice in (a).

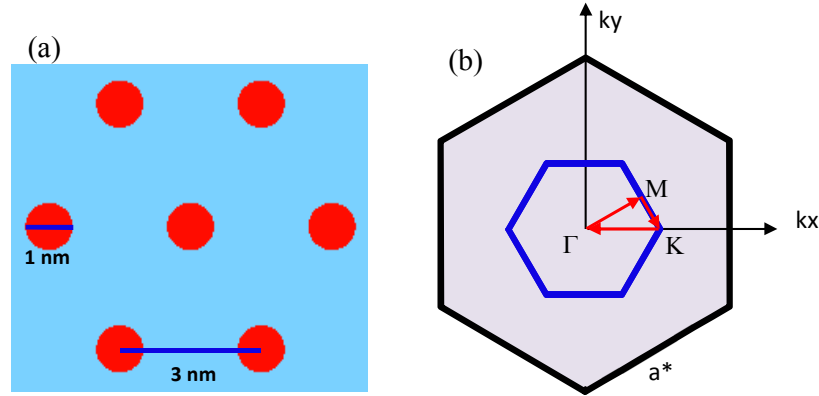


Figure 3.5: Real-Reciprocal geometry of metallic hexagonal lattice - (a) The geometry used in EPWE calculations. The cyan area and red circles represent a hexagonally patterned metal surface by periodic array of circular defects. (b) Its corresponding reciprocal lattice (black hexagon), and first Brillouin Zone (blue hexagon).

The corresponding band structure as calculated by Electron Plane Wave Expansion

3. RESULTS AND DISCUSSION I

method (EPWE) is depicted in Fig. 3.6(a). The calculations are performed along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions (black arrows) in the first BZ (blue hexagon in the inset of Fig. 3.6(a)). In this case surface electrons are experiencing periodic potential barriers of height V_1 . Hence the solution of Eq. 2.30 yields a parabolic dispersion with band gaps opening at the BZ boundaries (i.e., the $\bar{M}$ -points) and a rigid upward shift for the band minimum at $\bar{\Gamma}$ -point. The $\bar{\Gamma}$ -point energy of this Shockley surface state is found at 42 meV, and the $\bar{M}$ -point gap size amount to ~ 65 meV. We note that, the energetic position of the $\bar{\Gamma}$ -point is only sensitive to the average/effective potential per unit cell, whereas the $\bar{M}$ -point gap size and its momentum position are strongly dependent on the given potential modulation. Specifically, its size is twice the second Fourier component of the periodic potential ($2V_G$).

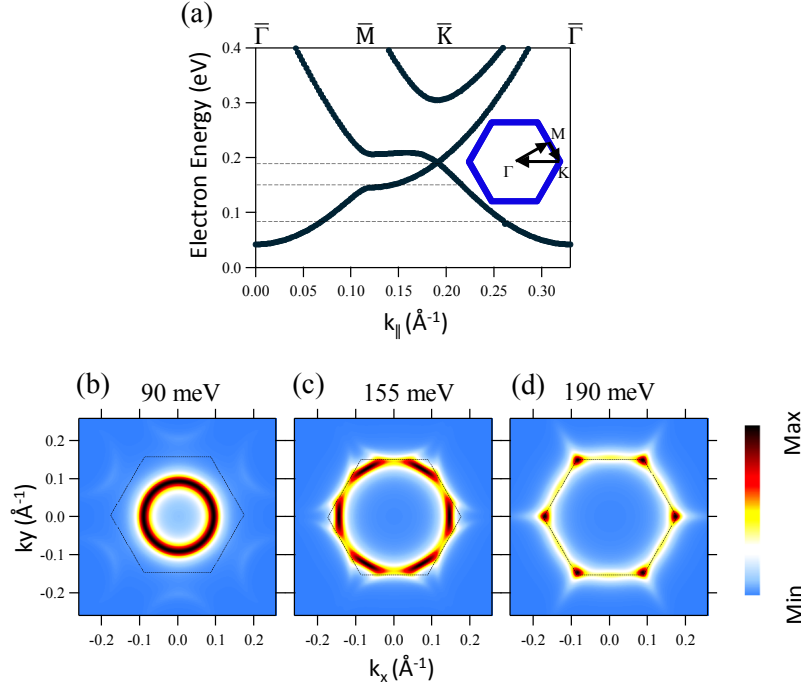


Figure 3.6: Band structure and CES of hexagonal metallic lattice - (a) Calculated band structure using EPWE along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions of first Brillouin zone arrows in (blue hexagon). (b,c,d) Simulated Constant energy surfaces taken at the black dashed lines in (a), i.e. at 90, 155, and 190 meV respectively.

The dispersion of the surface state at $\bar{K}$ -point, on the other hand, does not exhibit any energy gap between the first and second bands, leading to a clear degeneracy at the $\bar{K}$ -point. Using perturbation and group theories one can show that for a positive Fourier component V_G of the potential, the $\bar{K}$ -point (E_k energy) features a doubly

degenerate state associated with E ($E = E_k - V_G$). Therefore, for a repulsive (i.e., positive) superlattice potential with hexagonal symmetry, gap opening at $\bar{K}$ -points is not allowed^[55].

As a matter of fact, the band structure presented in Fig. 3.6(a) gives the electronic characteristics only at the BZ boundary. The full electronic behavior at other k -points within the BZ is contained in the constant energy surfaces (CES) taken at fixed energies. These EPWE-simulated CES taken at the black dashed lines in (a), i.e. at 90 meV, 155 meV, and 190 meV, respectively, are presented in Fig. 3.6(b-d). The black dashed hexagons in each CES define the borders of the first BZ. The CES taken at electron energy $E = 90$ meV, Fig. 3.6(b), exhibits a highly intense circular contour centered at the $\bar{\Gamma}$ -point of the BZ resulting from the main (first) band. A closer inspection revealed the presence of weaker circular contours centered at the $\bar{\Gamma}$ -points of the second BZ, namely unklapp bands. The appearance of six such bands directly mimics the six-fold symmetry of the hexagonal lattice. All CES taken at energies less than the lower edge of the $\bar{M}$ -point gap are characterized by such circular contours, the radius of which increase by going from the $\bar{\Gamma}$ -point energy towards the onset of the $\bar{M}$ -point gap. The second CES in (c) taken at electron energy of $E = 155$ meV, i.e. right at the lower edge of the $\bar{M}$ -gap, shows intensity reduction only at the six equivalent $\bar{M}$ -points, indicating the beginning of energy gaps at these points. Approximately at the center of $\bar{M}$ -gap, i.e., at electron energy of $E = 190$ meV, the intensity was strongly suppressed for all k -points on the surface BZ, except for the six corners defining the $\bar{K}$ -points, Fig. 3.6(d). This observation confirms that the six-fold symmetry prevent the opening of energy gaps at surface BZ corners^[139]. The analysis presented in Fig. 3.6 could be directly compared to existing data taken by ARPES on hexagonal superlattices^[75].

ARPES complementary information are commonly obtained from scanning tunneling spectroscopy (STS), where the site specific, i.e., local density of states (LDOS) could be mapped out in real space. In Fig. 3.7(a), the calculated LDOS obtained from the EPWE at three different cites, i.e. specific points in the real space geometry as marked in the unit cell in (b), are presented. The blue, black and red curves are the LDOS obtained at the corresponding points within the unit cell in (b). It is clear that LDOS calculated at the center of the circular potential barriers (blue) is very low at all energies compared to the other two points (blue and red). This indicates that electrons are likely excluded from regions with relatively large potential values. In contrast, the LDOS taken at the center of the upper (red) and lower (black) triangle of the unit cell, where the potential is zero, are much higher and exhibit strong variation with the

3. RESULTS AND DISCUSSION I

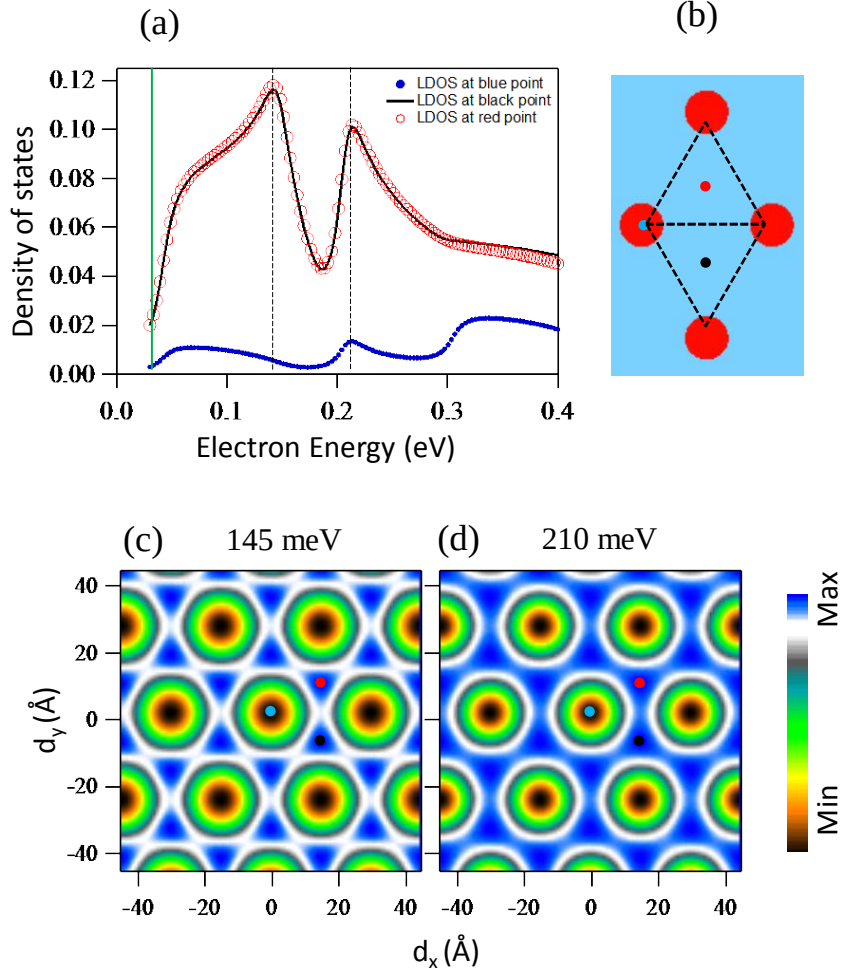


Figure 3.7: LDOS of hexagonal lattice with circular scatterers - The local density of states (LDOS) calculated by EPWE for three points in the unit cell, as shown in (b). (c,d) The 2D-LDOS taken at (b) 145 meV and (c) 210 meV.

electron energy. Furthermore, they are practically identical, confirming the same symmetry of these two points. The LDOS spectra just presented contain information on the surface state band minimum and the presence of energy gap. The inflection points (vertical green line) at ~ 42 meV for all spectra define the $\bar{\Gamma}$ -point energy, in agreement with band structure calculations. Similarly, the strong depletion of the LDOS (red and black) indicates the presence of an energy gap, where the vertical dashed lines define the upper and lower edges of this gap at which LDOS exhibit clear maxima. The fact that there are only two such maxima and the equivalence between the red and black LDOS curves, confirm that this is the $\bar{M}$ -point gap and, additionally, reassure the absence of $\bar{M}$ -point gap, in decent correspondence with band structure calculations^[55].

In order to follow the LDOS at all real-space lattice points, the calculations of 2D-LDOS are performed at the two energies that define the borders of $\overline{M}$ -point gap indicated by the black dashed lines in (a), i.e. at 145 meV (c) and 210 meV (d). The blue, red, and black points are overlaid for direct comparison with Fig. 3.7(a). The LDOS attains a minimum value at all and nearby the blue points, i.e., at the bases, and identical maxima at the red and black points. Most importantly, the full 2D-LDOS before (c) and after (d) the gap are identical, confirming the absence of $\overline{K}$ -point gap based on symmetry considerations^[139]. We, therefore, conclude that in a patterned metal surface with six-fold rotational symmetry the degeneracy of the $\overline{K}$ -point is protected by this symmetry, and could only be lifted otherwise.

3.2.1 The effect of potential:

In the following we investigate the effect of the scattering potential barrier on the electronic properties here discussed. Figure 3.8(a) shows two electronic band structures calculated along the $\overline{\Gamma MK\Gamma}$ directions, by EPWE, for two hexagonal metallic superlattice with different values of scattering potentials. The periodicity for the two systems was set to 30 Å. In the first case (blue), previously displayed in Fig. 3.6(a), the circular potential barrier (V_1) was 700 meV, while for the second band structure (red) (V_1) is changed to 30 eV. It is clear that the two band structures follow the same trend, i.e., parabolic dispersions with band gaps that open up at the BZ boundaries ($\overline{M}$ -points), with no signature of additional gaps at the $\overline{K}$ -points.

The direct comparison between the two electronic bands revealed that, there are mainly two electronic modifications were introduced by increasing the scattering potential barrier, the band minimum of the surface state and $\overline{M}$ -point gap size. The position of $\overline{\Gamma}$ -point energy increased from 42 meV, for the blue band, to 115 meV for red band. This confirms that, the shift of the surface state towards increasing the electron energy is related to the potential inside the circles, being larger for stronger potential. This behavior is now corroborated by recent numerical simulations on similar superlattices^[148].

The second consequence of increasing the height of potential barrier is the enlargement of the $\overline{M}$ -point gap size, which increased from 65 meV to 95 meV for the blue and red bands, respectively. It is axiomatic that the base potential has no effect on the back-folding momentum-positions of the parabolic dispersions, as these are defined by

3. RESULTS AND DISCUSSION I

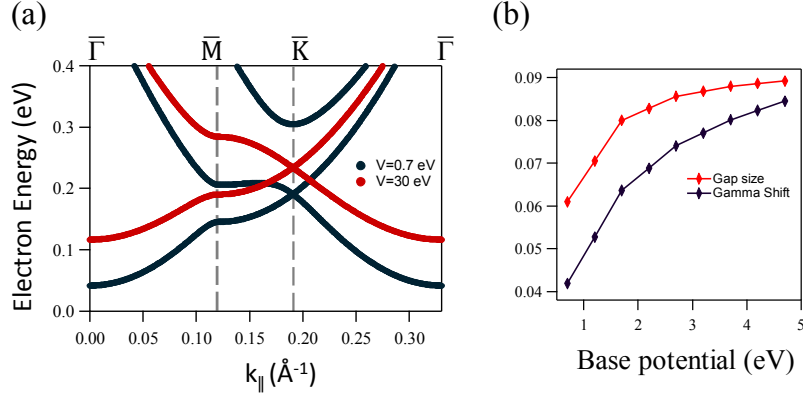


Figure 3.8: Effect of periodic potential on the hexagonal electronic structure

- (a) Calculated band structures using EPWE for two hexagonal metallic superlattices of different potential barrier V_1 , along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions. All calculation parameters were the same in the two cases, except that V_1 is changed from 700 meV (dark blue band) to 30 eV (red band). (b) The $\bar{\Gamma}$ -point energy (blue) and $\bar{M}$ -gap size (red) as a function of the base potential barrier V_1 .

the geometrical construction of the BZ. To discuss both, the surface state shift and the $\bar{M}$ -point gap size as a function of scattering periodic potential, several electronic band structures of patterned metal surface with hexagonal superlattice were calculated by EPWE. All calculation parameters were the same for all cases, except that the periodic potential (V_1) was changed stepwise between 700 and 4700 meV. The resulting $\bar{\Gamma}$ -point energy (blue) and $\bar{M}$ -point gap size (red) as a function of V_1 is presented in Fig. 3.8(b). It is clear that both the $\bar{\Gamma}$ -point energy and $\bar{M}$ -point gap size increase rapidly with the scattering potential, and tend to saturate at higher V_1 values. The highest potential where these saturate sets the infinite potential limit of the given system.

3.2.2 The effect of periodicity:

Based on the inverse relationship between real and reciprocal spaces, the magnitude of folding vectors in the BZ ($|\bar{\Gamma}\bar{M}|$) is determined by $2\pi/\sqrt{3}|d|$, for a hexagonal lattice. Hence, the lattice constant (d) is an effective parameter in any band structure calculations, especially as the Fourier expansion of the potential is performed over this lattice constant. This is the motivation to discuss the effect of superlattice periodicity on surface state of the hexagonal superlattice presented here.

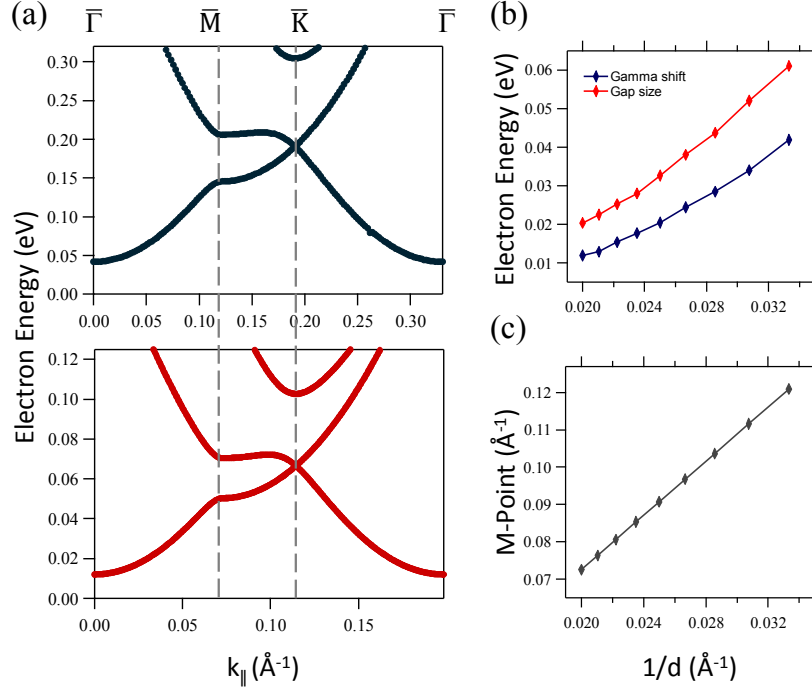


Figure 3.9: Effect of periodicity on the hexagonal electronic structure - (a) Calculated band structures along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions using EPWE for two hexagonal metallic superlattices of scattering potential $V_1=700$ meV. The superlattice constant (d) was set to 3 nm (blue) and 5 nm (red). (b) The $\bar{\Gamma}$ -point energy (blue) and $\bar{M}$ -gap size (red) as a function of the superlattice constant (expressed as $1/d$). (c) The momentum position $\bar{M}$ -point gap as a function of ($1/d$).

The two electronic band structures presented in Fig. 3.9(a) are calculated along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions for two hexagonal metallic superlattices with different superlattice constant d . As the superlattice periodicity increased from 3 nm (blue) to 5 nm (red), three electronic modifications showed up. These are the surface state band minima, $\bar{M}$ -point gap size and its momentum position. The shift of $\bar{\Gamma}$ -point energy decreased from 42 meV (blue) to 12 meV (red), while the $\bar{M}$ -point gap size is reduced from 65 meV (blue) to 20 meV (red). These are obvious consequences from the reduction of both the potential per unit cell and its second Fourier component, respectively, for large superlattice. It is also straightforward to notice that the back-folding position of the parabolic dispersions (i.e., $\bar{M}$ -point momentum position), is also decreased from $0.121 \text{ \AA}^{-1}$ (blue) to $0.072 \text{ \AA}^{-1}$ (red). To discuss both of the surface state shift and the $\bar{M}$ -point gap size as a function of superlattice constant (expressed as $1/d$), several electronic band structures were calculated by EPWE, where the superlattice periodicity (d) is changed gradually from 3 nm to 5 nm. Figure 3.9 3.9(b) showed that, the bigger

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the superlattice periodicity is, the smaller both the surface state shift (blue curve) and the $\overline{M}$ -point gap size (red curve) are. Figure 3.9(c) clearly showed that, the momentum positions of the $\overline{M}$ -point decrease linearly with increasing the superlattice parameter (d), in accordance with the relation $\overline{\Gamma M} = 2\pi/\sqrt{3}|d|$.

3.2.3 The effect of symmetry:

In his marked article “*More is different*”, Anderson, P.W. discussed the broken symmetry and the nature of the hierarchical structure of science^[149]. He indicated that for large systems, the physical states are usually less symmetrical than the laws. A trivial example is crystals, which clearly violate the homogeneity and isotropy of space and physical laws. Such a behavior is an example of the spontaneous symmetry breaking mechanism. More elaborate is the Higgs mechanism and the generation of mass for bosons in particle physics^[150]. Such a mechanism is also useful in condensed matter physics, in particular, for phase transitions^[151].

The electronic properties of solids depend on the chemical nature of the atomic constituents and their bonding. However, periodicity, i.e., the “*translational*” symmetry have shown to play a fundamental role since the Bragg diffraction of the electron states can lead to gaps in the band structure, as shown in the previous section. Additional example for “*atomic*” solids is the semiconducting character of silicon, which is a direct consequence of the symmetry of its Si crystal lattice. Similarly, the singular Fermi surface of graphene resulting from linear dispersions associated with massless Dirac electrons that are intact or degenerate at the $\overline{K}$ -point of the surface Brillouin zone (SBZ)^[152,153] also results from symmetry. Indeed, by changing the symmetry through interaction of a graphene plane with a substrate, the linear dispersion partially disappears and an energy gap opens up at the $\overline{K}$ -point^[154,155].

In the following, we shall discuss the effect of the “*rotational*” symmetry breaking on the electronic structure of 2D metallic superlattices. The existence or absence of gaps at the $\overline{K}$ -points for hexagonal superlattice resulting from symmetry effects shall be examined via changing the shape of the scattering potential, i.e., the base.

3.2.3.1 Three-Fold rotational symmetry

Figure 3.10 presents the simplest case of breaking the “*six-fold*” rotational symmetry in hexagonal superlattice. This system has the same parameters as used in Fig. 3.5(a),

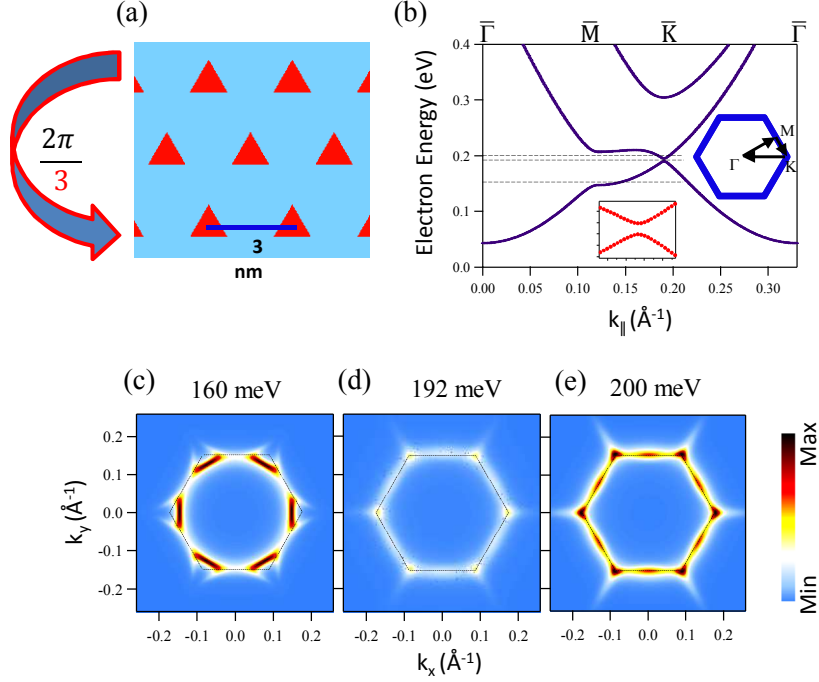


Figure 3.10: Hexagonal metallic superlattice with triangular scatterers - (a) The geometry used in EPWE calculations. The cyan area and red triangles represent a hexagonally patterned metal surface by periodic array of triangular defects. (b) Calculated band structure using EPWE along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions of first brillouin zone arrows in (blue hexagon). (c,d,e) Simulated Constant energy surfaces taken at the black dashed lines in (b), i.e. at 160, 192, and 200 meV respectively.

except the base shape of the scatterers is changed from circular to triangular barrier as shown in Fig. 3.10(a). The cyan area represents the clean region of a metal surface with a potential $V_0=0$, while the red triangular areas define the periodic potential barriers each of height $V_1=700$ meV. The triangular base clearly reduces the symmetry of the hexagonal lattice into “three-fold” rotational symmetry. The characteristic band structure of this system, as obtained from EPWE, along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions (black arrows) of the first BZ (blue hexagon in the inset of this figure) is shown in Fig. 3.10(b). In addition to the band gap opened at the BZ boundaries (i.e., the $\bar{M}$ -points) and the rigid upward shift for the band minimum at $\bar{\Gamma}$ -point, the lowering of symmetry leads to an additional energy gap at the $\bar{K}$ -point in the band structure, no matter what the values of the superlattice, potential barrier V_1 , and constant d are^[55]. The analysis of this band structure yields a $\bar{\Gamma}$ -point energy of ~ 43 meV, $\bar{M}$ -point gap size of ~ 60 meV, and a small gap (~ 10 meV, see zoom-out in (b)) at the $\bar{K}$ -point. Notice

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that both the $\bar{\Gamma}$ -point energy and the size of $\bar{M}$ -point gap are not altered compared to the circular base (Fig. 3.6). In fact, the areas of the triangular and circular bases are, intentionally, set to the same value, in order to decouple Bragg scattering from the symmetry effects. In this context, one clearly assures that $\bar{\Gamma}$ -point energy is only sensitive to the average potential per unit cell (i.e., the area of one base). Additionally, the second Fourier component of the potential V_G is the same for three- and six-fold rotational symmetries. These together strongly indicate that the gap opened at the $\bar{K}$ -point is only induced by symmetry reduction. For such three-fold hexagonal lattice, this gap should exist for any tinny potential and large periodicity, and is only closed for the empty lattice. Interestingly, the presence of this small gap allows a ~ 10 meV energy window where a full gap is defined, similar to those characteristics for semiconductors.

To make sure that the electronic band dispersion has the semiconducting-like full gap, the electronic behavior at all k -points (CESs) within the BZ is calculated at three fixed energies, i.e., at the black dashed lines in (b). The outcome of these calculations is presented in Fig. 3.10(c-e). The first CES taken at electron energy of $E = 160$ meV (c), i.e., barely at the lower edge of the $\bar{M}$ -point gap, exhibits intensity reduction only at the six $\bar{M}$ -points, demonstrating the beginning of energy gaps at these points. At electron energies within this gap, e.g., $E = 192$ meV, the intensity reduction was promoted for all k -points on the surface BZ, including the corners defining the six $\bar{K}$ -points (d). This, when compared to Fig. 3.6(d), clearly indicates the formation of a full energy gap. Since the full gap is only ~ 10 meV wide, the CES (d) taken at slightly higher electron energy ($E = 200$ meV), i.e., right at the upper edge of the $\bar{K}$ -gap, shows intensity at the six $\bar{K}$ -points. Due to the close proximity of the upper edge of the $\bar{M}$ -point gap to 200 meV, this CES exhibits additional intensity at all $\bar{M}$ -points.

A deeper insight into the symmetry of the superlattice and its correlation with the electronic behavior is given through the analysis of the site-dependent LDOS. These can be extremely sensitive to very small gap size, as we show next. The LDOS are calculated by EPWE at three different specific points in the real-space geometry and presented in Fig. 3.11(a). These points in the unit cell, sketched in (b), were chosen as variations in the LDOS are expected to be significant according to symmetry considerations. The blue, black and red curves are the LDOS obtained at the corresponding points within the unit cell. Based on the localization of electrons outside the potential regions, the LDOS at the center of the triangular potential barriers (black) is very low at all energies compared to the other two points (blue and red). On the other hand, the LDOS taken

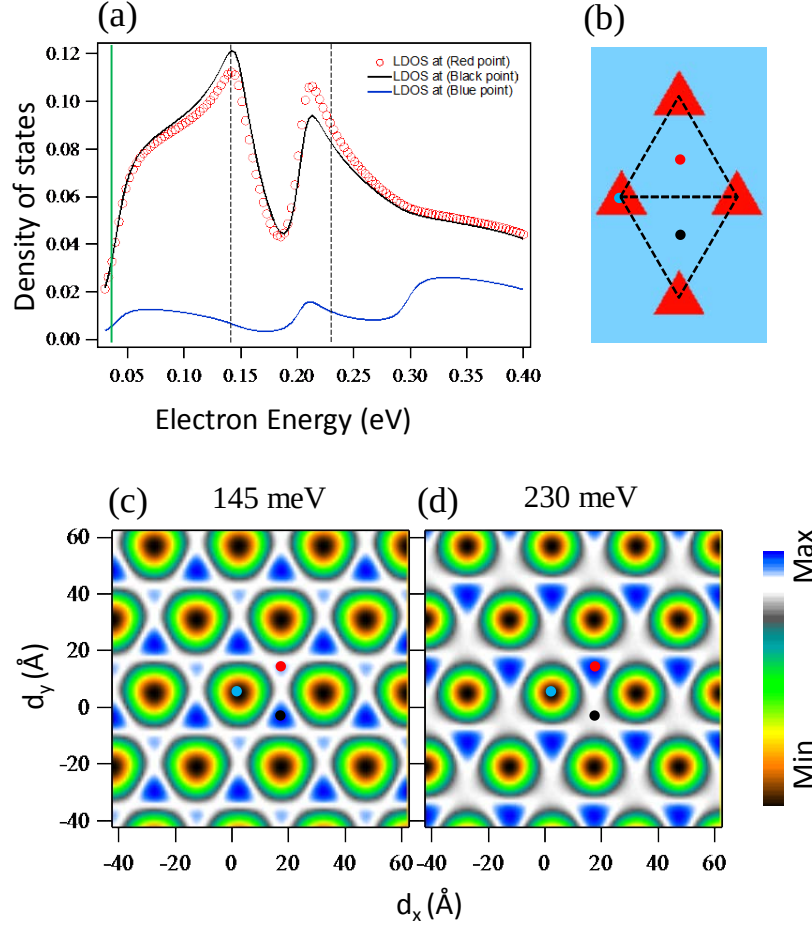


Figure 3.11: LDOS of hexagonal lattice with triangular scatterers - The local density of states (LDOS) calculated by EPWE for three points in the unit cell, as shown in (b). (c,d) The 2D-LDOS taken at (b) 145 meV and (c) 230 meV.

at the center of the upper (red) and lower (black) triangles of the unit cell, where the potential is set to zero, revealed strong variations with the electron energy. In contrary to the case of circular potential basis, these two LDOS are not identical. The depletion in the LDOS for these curves, again, indicated the presence of an energy gap, where the vertical dashed lines define the upper and lower edges of this gap, in agreement with band structure calculations. In fact, they also share the same energetic position of these maxima, but the LDOS at the first maxima is higher for the black curve, and reverses at the second maxima. This is the first consequence of the symmetry reduction introduced by employing triangular scatterers. In fact, one can immediately recognize that symmetry of the potentials surrounding the red and black points in (b) is different. Finally, since the average potential per unit cell is kept the same as for

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the circular base, the inflection points (vertical dotted line) which define the $\bar{\Gamma}$ -point energy were found ~ 43 meV for all spectra.

In order to follow the effect of symmetry breaking through LDOS calculations at all real-space points within the unit cell, the 2D-LDOS are performed at the borders of the full energy gap indicated by the black dashed lines in (a), i.e. at 145 meV for (c) and 230 meV for (d). The blue, red, and black points are overlaid for direct comparison with Fig. 3.11(a). The LDOS exhibits minima at all blue points, i.e., at the bases. These minima are present before (a) and after (b) the energy gap. However, the LDOS at the red and black points change its values considerably below and after crossing the gap. Specifically, the black points have higher LDOS than red points below the gap (a), and this LDOS variation reverses after the gap (b). Since the band structure calculations revealed the presence of full gap, the electron wavefunction must be orthogonal, i.e., $\pi/2$ out-of-phase, at the gap boundaries, according to Bragg reflection at the BZ boundaries. Therefore, the change in the LDOS intensity at every point (red and black) before and after the band gap confirms the existence of a semiconducting-like full gap. Although ARPES measurements give the band structure directly, the signature of energy gaps in scanning tunneling spectroscopy measurements is revealed from such variations in the LDOS.

3.2.3.2 Two-Fold rotational symmetry

Although the hexagonal superlattice with triangular bases, just discussed, owns a three-fold rotational symmetry, the electronic bands, i.e., the CESs present in Fig. 3.10, are still six-fold symmetric. Actually, this additional symmetry is brought by what is called “*time-reversal-symmetry*” as we shall discuss later.

In the following we investigate the electronic behavior when the rotational symmetry of the hexagonal lattice is further reduced into “*two-fold*”. This is made possible by assigning a rectangular shape for the scattering potential as depicted in Fig. 3.12. The rectangular scatterer has the same area and height as the triangular and circular barriers previously studied. For proper comparison among different shape of the basis, the periodicity of the superlattice is kept the same. Figure 3.12(b) depicts the EPWE electronic band structure calculation of the system presented in (a). The calculations are performed along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ directions (black arrows) of the first BZ (blue hexagon in the inset of (b)). The analysis of this band structure allows the extraction of the $\bar{\Gamma}$ -energy (~ 45 meV), the $\bar{M}$ -point gap size (~ 70 meV), and the symmetry-induced $\bar{K}$ -point gap (25 meV). Obviously, the energetic position of the $\bar{\Gamma}$ -point is barely affected

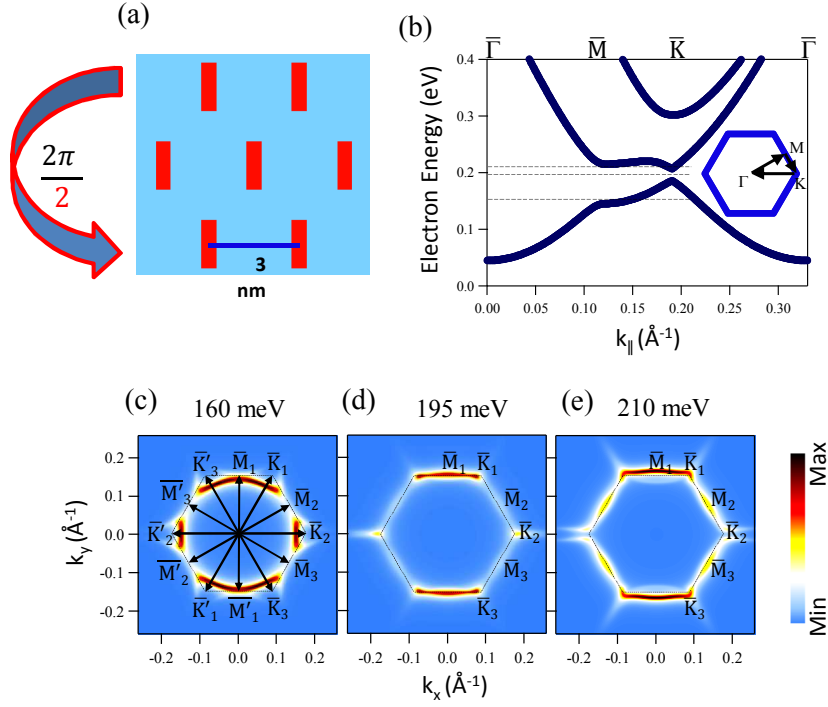


Figure 3.12: Hexagonal metallic superlattice with rectangular scatterers - The geometry used in EPWE calculations. The cyan area and red rectangles represent a hexagonally patterned metal surface by periodic array of rectangular defects. (b) Calculated band structure using EPWE along the $\bar{\Gamma}\bar{M}_2\bar{K}_2\bar{\Gamma}$ directions of first brillouin zone arrows in (blue hexagon). (c,d,e) Simulated Constant energy surfaces taken at the black dashed lines in (b), i.e. at 160, 195, and 210 meV respectively.

given the fixed potential per unit cell in all cases. However, the $\bar{M}$ -point gap is enlarged by ~ 10 meV indicating some variation in V_G . Most importantly, the $\bar{K}$ -point gap also increases (~ 15 meV) compared to the one reported for three-fold symmetric lattice. The side of the full gap reported here is, therefore, ~ 30 meV wide. The overall behavior, apparently, reveals that the transition from three-fold into two-fold superlattice only yields differences in the gap sizes. In the following we show that this is not only the case.

The additional consequences of the two-fold rotational symmetry of this system are highlighted in Fig. 3.12(c-d), which displays the EPWE stimulated CESs are. These CES are taken at the black dashed lines in (b), i.e., at 160 meV, 195 meV, and 210 meV, respectively, while the black dashed hexagons in each CES define the borders of the first BZ. Given the full 30 meV wide gap, one would expect no intensity for the CES taken at 195 meV (d). Obviously, the CES exhibits a strong intensity in a large portion of the

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k -space. The shape of all these CES implies that the set of $\overline{M}$ -points and $\overline{K}$ -points are not all equivalent. Actually, the proper description of this system, and others, requires the incorporation of the intrinsic time reversal symmetry which strictly exist for all electronic bands in the absence of external fields, such as magnetic or electric fields. The time reversal symmetry of electronic bands in solids read; $E(k) = E(-k)$, which is brought by the fact that the parity operator acting on the momentum reverses its direction so that $PpP^{-1} = -p$, where p is the momentum operator. For a hexagonal lattice, there exist two sets of three $\overline{M}$ -points and $\overline{K}$ -points. Time reversal symmetry implies that the other two sets at $-k$, namely $\overline{M}'$ -points and $\overline{K}'$ -points, are equivalent, i.e., $\overline{M}_1 \approx \overline{M}'_1$, $\overline{M}_2 \approx \overline{M}'_2$, $\overline{M}_3 \approx \overline{M}'_3$, $\overline{K}_1 \approx \overline{K}'_1$, $\overline{K}_2 \approx \overline{K}'_2$, and $\overline{K}_3 \approx \overline{K}'_3$.

The three-fold rotational symmetry for the triangular base has kept $\overline{M}_1$, $\overline{M}_2$, $\overline{M}_3$, $\overline{K}_1$, $\overline{K}_2$, and $\overline{K}_3$ equivalent, and the time reversal symmetry enforces their symmetry with the corresponding set of $\overline{M}'$ and $\overline{K}'$ points. The overall symmetry is, thus, 2 (time-reversal-symmetry) $\times$ 3 (three-fold rotational symmetry) = 6, explaining the obvious six-fold symmetry of the electronic bands in Fig. 3.10, although the lattice symmetry is three-fold.

Here the two-fold rotational symmetry keeps $\overline{M}_2$ and $\overline{M}_3$ equivalent, but, $\overline{M}_1$ is clearly different with almost no gap opening. Likewise, $\overline{K}_1$ and $\overline{K}_3$ were found to be equivalent, and differ from $\overline{K}_2$. These differences are easily captured by looking at the CES presented in Fig. 3.12(c-e). An important consequence of such a symmetry breaking is the non-existence of any CES where complete suppressing of photoemission intensity over all k -points takes place. Therefore, the overall system acquires metallic-like bands, and a full semiconducting gap is not defined over the BZ, but rather along specific directions, e.g., $\overline{\Gamma M_2 K_2 \Gamma}$ direction. Finally, the combination of the rotational and time-reversal symmetries renders the CES four-fold symmetric. It is important to note that, the time-reversal symmetry only acts on electronic bands, but the symmetry of the real-space LDOS is only determined by the rotational symmetry. Therefore, although the CES for the triangular lattice exhibit six-fold symmetry, the LDOS presented in Fig. 3.11(c-d) are three-fold.

These EPWE calculated LDOS for the current hexagonal metallic surface with a rectangular scatterer are presented in Fig. 3.13. They were calculated at three specific points chosen at the same real-space positions in the unit cell depicted in (b). The blue, black and red curves corresponding to LDOS spectra at these points are represented in (a). The LDOS at the center of the rectangular potential base (blue) still very

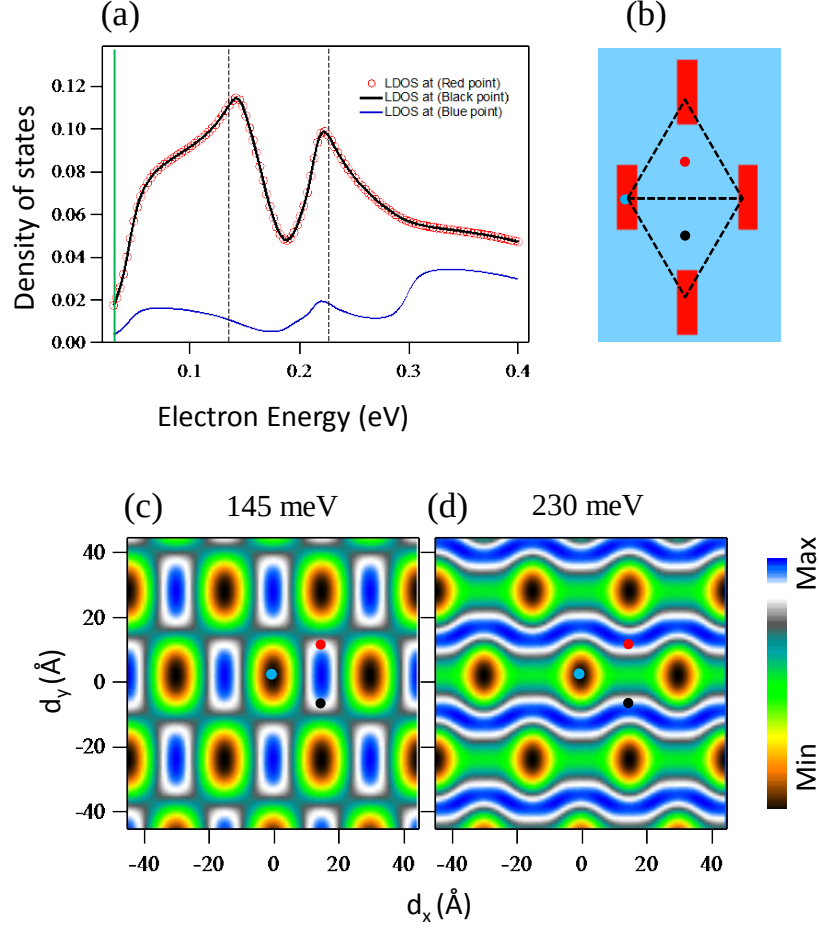


Figure 3.13: LDOS of hexagonal lattice with rectangular scatterers - The local density of states (LDOS) calculated by EPWE for three points in the unit cell, as shown in (b). (c,d) The 2D-LDOS taken at (b) 145 meV and (c) 230 meV.

low at all energies compared to the other two points (black and red). The potentials surrounding the black and red points are symmetric and, therefore, the two curves of the LDOS at these points are identical, masking out the effect of reduced symmetry. They also exhibit intensity depletion indicating the presence of an energy gap. In order to identify possible symmetry effects, the 2D-LDOS are calculated at the borders of the energy gap indicated by the black dashed lines in (a), i.e. at 145 meV for (c) and 230 meV for (d). The LDOS at the two equivalent points (red and black) are the same, but the overall 2D-LDOS clearly changes upon crossing the energy gap. This confirms that the full band gap presented here is direction dependent. Additionally, both LDOS exhibit two-fold symmetry, indicating that real-space maps reflect, only, the “*rotational*” symmetry of the superlattice.

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3.2.3.3 One-Fold rotational symmetry

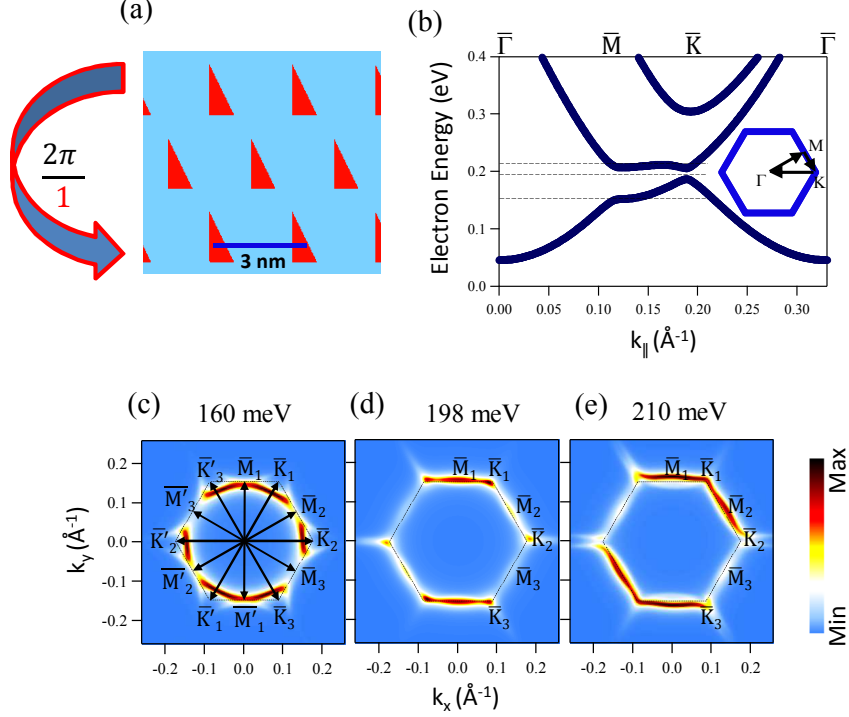


Figure 3.14: Hexagonal metallic superlattice with right-triangular scatterers

- The geometry used in EPWE calculations. The cyan area and red right triangles represent a hexagonally patterned metal surface by periodic array of right triangular defects. (b) Calculated band structure using EPWE along the $\bar{\Gamma}\bar{M}_2\bar{K}_2\bar{\Gamma}$ directions of first brillouin zone arrows in (blue hexagon). (c,d,e) Simulated Constant energy surfaces taken at the black dashed lines in (b), i.e. at 160, 198, and 210 meV respectively.

The above discussion made it easy to anticipate the electronic behavior associated with lattices that have no rotational symmetry, namely “one-fold” symmetry systems. For example, the overall symmetry of the CES is expected to be two-fold, given the time-reversal-symmetry constrain, while the real-space 2D-LDOS should own one-fold symmetry. This is the highest situation of symmetry breaking in a hexagonal superlattice. In order to ensure these anticipations, we changed the shape of the base potential to a right triangle. Figure 3.14(a) displays the corresponding hexagonal metal surface, where the periodicity and potential are kept the same to the previous higher-order symmetry lattices. The corresponding electronic band structure, calculated by EPWE method, along the $\bar{\Gamma}\bar{M}\bar{K}\bar{\Gamma}$ direction (black arrows) of the first BZ (blue hexagon in the inset of (b)) is presented in (b). The information inferred from this band dispersion

are a ~ 46 meV rigid upward shift for the band minimum at $\bar{\Gamma}$ -point, 55 meV gap at the $\bar{M}$ -point, and a smaller gap (20 meV) at the $\bar{K}$ -point, i.e. the metal surface has a semiconductor-like behavior, along this particular direction.

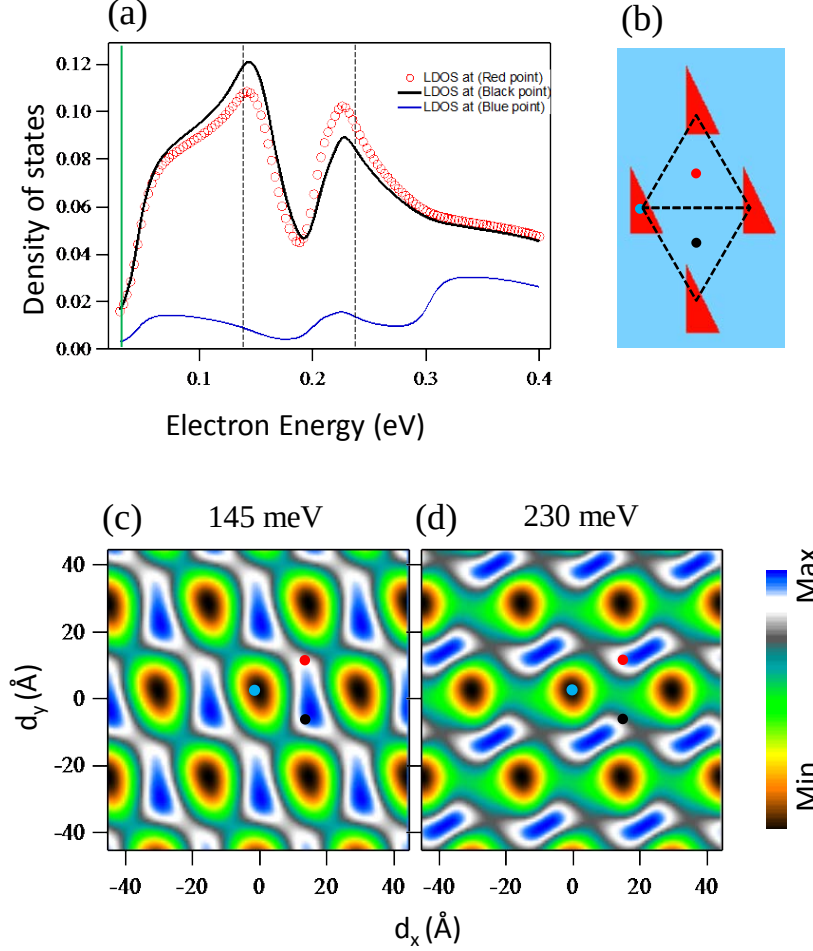


Figure 3.15: LDOS of hexagonal lattice with Right-triangular scatterers - The local density of states (LDOS) calculated by EPWE for three points in the unit cell, as shown in (b). (c,d) The 2D-LDOS taken at (b) 145 meV and (c) 230 meV.

In order to follow this semiconducting gap along all other possible directions and to identify additional symmetry-related effects, three EPWE stimulated CESs were given in Fig. 3.14(c-e). The calculations are performed at the black dashed lines in (b), i.e. at 160 meV, 195 meV, and 210 meV, respectively, while the black dashed hexagons in each CES define the borders of the first BZ. From the CES presented in (c), one immediately recognize the complete symmetry breaking, where the $\bar{M}_1$, $\bar{M}_2$, and $\bar{M}_3$ gaps are different. By following these CES from (c) to (e), it appears that the

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largest gap occurs at $\overline{M}_3$, while the $\overline{M}_1$ -point is almost gapless. Likewise, the largest symmetry related gap is only found at $\overline{K}_2$. In fact, the band structure presented in Fig. 3.14(b) is given along $\overline{\Gamma}\overline{M}_2\overline{K}_2\overline{\Gamma}$ direction. Although the symmetry is completely broken in this system, the fundamental time-reversal symmetry maintains a two-fold symmetry between $\overline{M}_1 \approx \overline{M}'_1$, $\overline{M}_2 \approx \overline{M}'_2$, $\overline{M}_3 \approx \overline{M}'_3$, $\overline{K}_1 \approx \overline{K}'_1$, $\overline{K}_2 \approx \overline{K}'_2$, and $\overline{K}_3 \approx \overline{K}'_3$.

Finally, we present the LDOS calculations for this one-fold symmetric superlattice. These are presented in Fig. 3.15(a) for three different specific points within the unit cell shown in (b). As expected, the LDOS at the center of the right triangular potential base (blue) very low at all energies compared to the other two points (black and red). The LDOS at the black and red points are not identical, but rather reverse their maxima at the borders (vertical dashed lines) of the gap, indicating a full gap along this direction as detailed previously. The inflection points (vertical green line) at ~ 46 meV for all spectra define the $\overline{\Gamma}$ -point energy, in agreement with band structure calculations.

The 2D-LDOS calculated at the borders of the full energy gap, i.e. at 145 meV and 230 meV are presented in Fig. 3.15(c,d), respectively. It is clear that the LDOS changes drastically upon crossing the energy gap. Additionally, the LDOS below and above the gap follows strictly the rotational symmetry of the underlying superlattice, i.e., being one-fold symmetric.

Part II: Square superlattice

3.3 General Overview

We now move to the second Bravais lattice which owns a four-fold rotational symmetry, namely the “*square*” lattice. The electronic behavior of free electrons scattered by a periodic array of defects arranged forming such a square superlattice will be subjected the same analysis presented previously for the hexagonal lattice. Figure 3.16(a) presents the surface potential landscape used as input for EPWE. It simply, describes metal surface (blue) with a periodic array of circular (red) scatterers arranged in a square lattice. Given the known parabolic dispersion for flat metal surfaces, the potential within the blue region is set to constant ($V_0=0$), while the potential barrier inside the circular bases has height of $V_1=700$ meV. This choice of the scattering potential is intended to be within the same order of magnitude for reported studies. Both the circle diameter and superlattice constant were set to 1 nm and 3 nm, respectively, whereas the effective mass used was $0.42 m_e$, i.e., practically the same as reported experimentally for Ag(111) and Cu(111)^[147].

The simulated photoemission intensity for the system presented in (a) is displayed in Fig. 3.16(c,d). The calculations are performed along both $\overline{\Gamma X}$ and $\overline{\Gamma M}$ directions (red arrows) of the first BZ (blue square in Fig. 3.16(b)). One clearly sees in (c) a parabolic dispersion with a band gap opening at the BZ boundary (i.e., the $\overline{X}$ -point) together with a rigid upward shift (~ 35 meV) for the surface state band minimum at $\overline{\Gamma}$ -point. On other hand, the band dispersion in (d) also exhibits a weak band gap opening at the BZ corner (i.e., the $\overline{M}$ -point). The size of each the $\overline{X}$ -point and $\overline{M}$ -point gaps amount to ~ 60 meV and ~ 15 meV, respectively. Additionally, the energetic positions of these gaps are clearly different, indicating an indirect character of the full $\overline{\Gamma X M \Gamma}$ gap.

In order to elucidate more details of the square nanopatterned surface, the electronic band structure along the $\overline{\Gamma X M \Gamma}$ direction and the constant energy surfaces (CES) are

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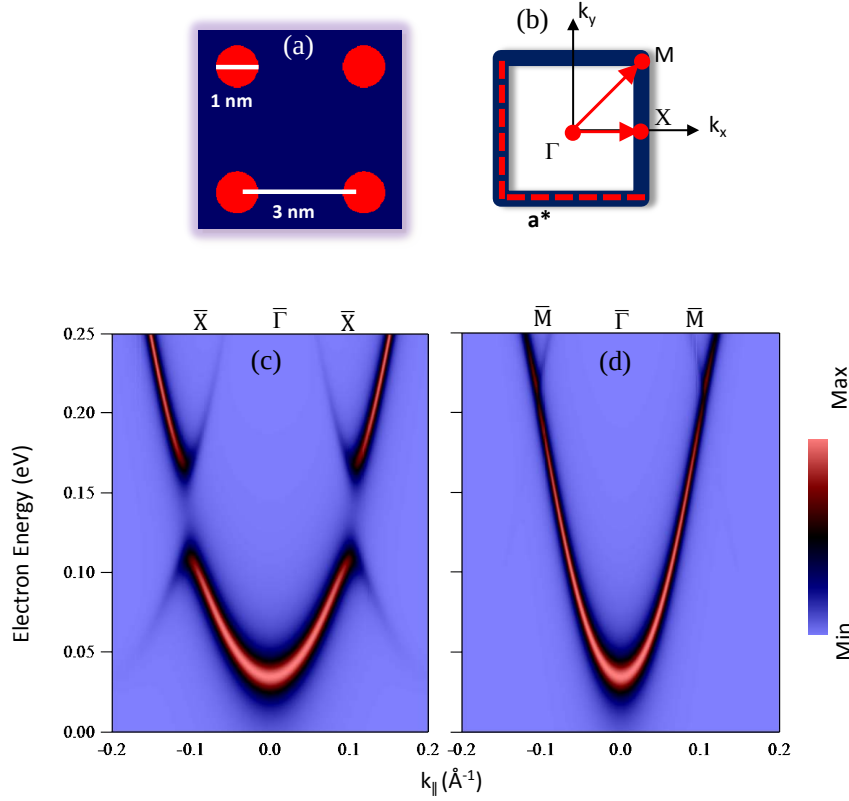


Figure 3.16: Real-Reciprocal geometry of metallic Square lattice - The geometry used in EPWE calculations. The blue area defines a metal surface and the red circles represent square array of defects. (b) The Brillouin zone of the geometry defined in (a). (c,d) The EPWE stimulated photoemission intensity showing the band dispersion for the patterned metal surface in (a) along $\bar{\Gamma}\bar{X}$ and $\bar{\Gamma}\bar{M}$ directions.

calculated and presented in Fig. 3.17. In (a) the band structure calculation confirms the indirect semiconducting band gap of this surface state. The minimum of the conduction band (red line) does not occur at the same k value as the valence band maximum (green line). To trace these electronic details at all k -points inside the BZ the CES were calculated by EPWE at different set of fixed energies.

Figure 3.17(b-e) depicts the simulated CESs taken at the black dashed lines in (a), i.e., at 110 meV, 130 meV, 170 meV, and 210 meV, respectively. The white dashed square in each CES map define the borders of the first BZ. The first CES in (b) taken at electron energy of $E = 110$ meV, i.e., barely at the lower edge of the $\bar{X}$ -point gap, exhibits intensity reduction only at the four equivalent $\bar{X}$ -points, confirming the beginning of energy gaps at these points. At electron energies within this gap, e.g., $E = 130$ meV, the intensity reduction was enhanced for all k -points on the surface BZ,

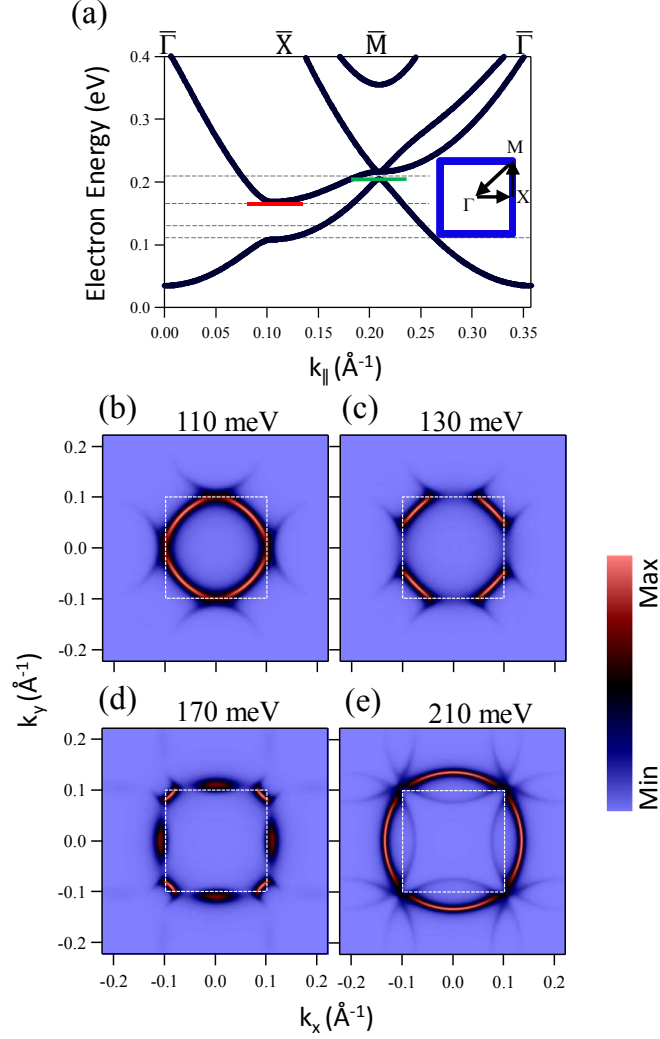


Figure 3.17: Band structure and CES of square metallic lattice - (a) Calculated band structure using EPWE along the $\Gamma\bar{X}\bar{M}\Gamma$ direction, i.e., along the black arrows of the first Brillouin zone (blue square). (b-e) Simulated constant energy surfaces taken at the black dashed lines in (a), i.e., at 110, 130, 170, and 210 meV respectively.

except in the vicinity of the corners defining the $\bar{M}$ -points, Fig. 3.17(c). Furthermore, the CES exhibit a significant deviation from the “circular” contours of free electrons, where the CES clearly contains “flat” segments. The length of these segments is reduced at the expense of the intensity depletion regions for increased electron energies within the gap. Exactly at the conduction band minimum, i.e., $E = 170$ meV, the simulated CES is presented in (d). Four electron pockets showed up at the equivalents $\bar{X}$ -gaps. The last CES in (e) is taken approximately at the center of the $\bar{M}$ -gap i.e., at electron energy of $E = 210$ meV. The relatively low intensity at the BZ corners corresponds to

3. RESULTS AND DISCUSSION I

the four weak gaps at these particular momentum positions.

As defined before, (STS) is an experimental technique that probes the local density of electronic states (LDOS), which in turns contains information on the band gap of nanostructured surfaces and local electronic behavior at the atomic scale^[118]. Using EPWE, the LDOS are calculated at three different cites, i.e., specific points in the real-space geometry in Fig. 3.18(a). These points (in the unit cell, as shown to the right) were chosen as variations in the LDOS are expected to be significant according to symmetry considerations. The LDOS at the center of the circular potential barriers (black) is very low at all energies compared to the other two points (blue and red). This confirms that electrons are possibly excluded from regions with relatively large potential values. Likewise, the LDOS taken at the center of the unit cell (red), and the center of the side length (blue) where the potential is zero, are significantly higher and exhibit strong variations with the electron energy. The inflection points (vertical green line) at ~ 35 meV for all spectra define the $\bar{\Gamma}$ -point energy, in agreement with band structure calculations. The blue and red LDOS, in addition, exhibit maxima and minima which indicate the presence of $\bar{X}$ -point and $\bar{M}$ -point gaps, where the vertical dashed lines define the valence (conduction) band maximum (minimum).

In order to gain information on the LDOS at all real-space lattice points, 2D-LDOS calculations were performed at the three electron energies indicated by the black dashed lines in (a), i.e., at 105 meV (b), 170 meV (c), and 220 meV (d). The blue, red, and black points are overlaid for direct comparison with Fig. 3.18(a). The LDOS has a minimum intensity value at all, and nearby, the black points, i.e., at the bases, at all energies. However, the LDOS values at the red and blue points follow some intensity modulation in accordance with symmetry considerations^[139]. In particular, the reduction of LDOS intensity at the red point directly mimic the influence of the energy gaps, as shown in (b,c,d). The change in the LDOS intensity at every point (red and blue) through the borders of two band gaps confirms the four-fold rotational symmetry of the square metallic supper-lattice, i.e. the existence of an indirect full gap.

3.3.1 The effect of potential:

With the previous analysis, the surface state at metal surface interacting with square array of defects exhibit band gap openings at the symmetry points and LDOS modulations due to these gaps and the symmetry of the structure. In the following, the effect of

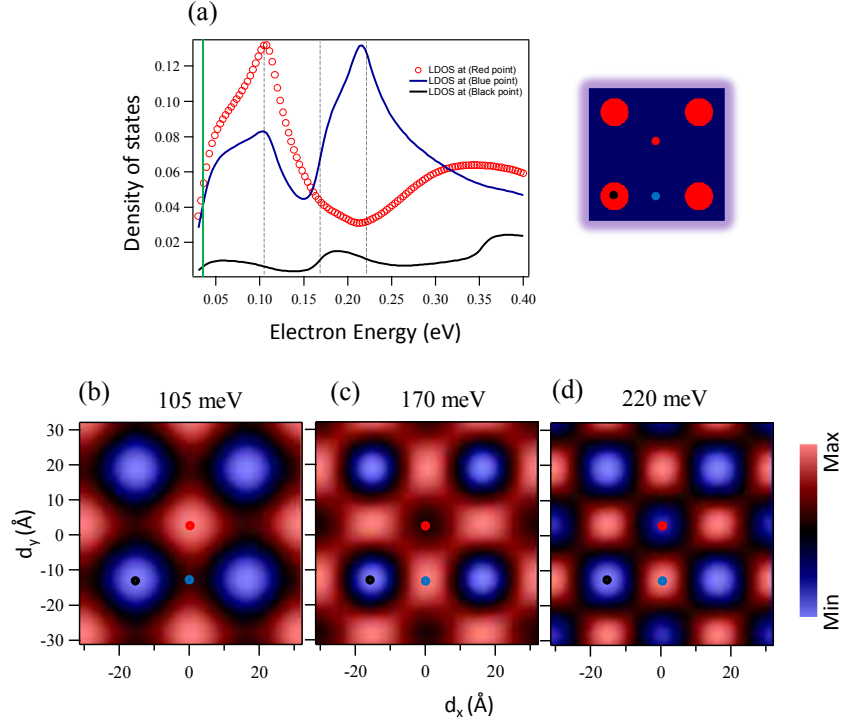


Figure 3.18: LDOS of square lattice with circular scatterers - (a) The energy-dependent local density of states (LDOS) calculated by the EPWE for three real space points in the unit cell, as shown to the right. (b-d) The 2D-LDOS taken at three electron energies.

the scattering potential on the electronic structure of squarely patterned metal surface shall be discussed. Figure 3.19 displays the electronic band structure as calculated by EPWE along the $\bar{\Gamma}X\bar{M}\bar{\Gamma}$ directions (a). All other geometrical and physical parameter in these calculations were identical to those previously discussed except for the height of the potential barrier (V_1), which is changed from 700 meV to 4.7 eV. The surface state exhibits a parabolic dispersion with much wider gaps at the $\bar{X}$ and $\bar{M}$ points, together with a larger rigid upward shift for the band minimum at $\bar{\Gamma}$ -point than the values of band structure depicted in Fig. 3.17(a). The band dispersion has a valence band maximum (green line) at a different k value than its conduction band maximum (red line), again confirming an indirect gap in spite of the order of magnitude potential employed.

Figure 3.19(b-e) present the simulated CESs taken at the black dashed lines in (a), i.e., at 125 meV, 180 meV, 215 meV, and 230 meV, respectively. The CES in (b) is taken at the lower edge of the $\bar{X}$ -gap, and exhibits intensity reduction only at the four

3. RESULTS AND DISCUSSION I

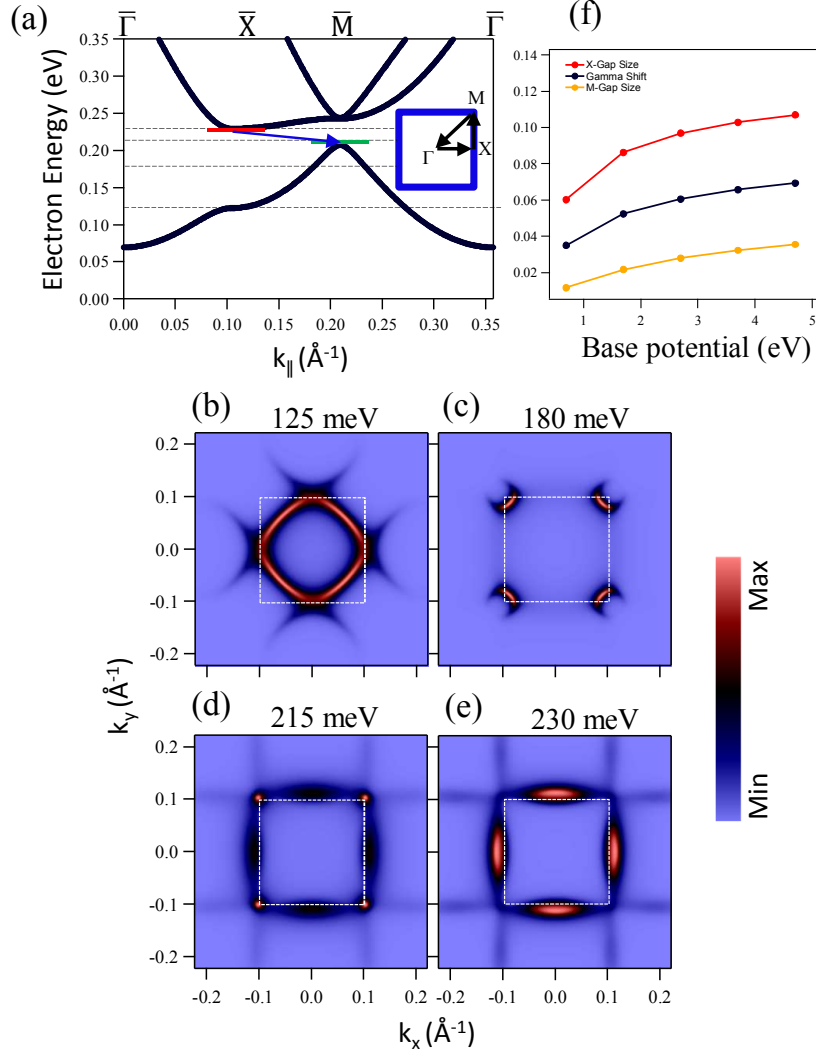


Figure 3.19: Effect of periodic potential on the square electronic structure - (a) Calculated band structure using EPWE along the $\bar{\Gamma}\bar{X}\bar{M}\bar{\Gamma}$ direction for a square patterned metal surface with circular bases potential of 4.7 eV and lattice parameter $d = 3$ nm. (b-e) The simulated constant energy surfaces using EPWE taken at the four electron energies. (f) The variation of $\bar{\Gamma}$ -point shift (blue) and both $\bar{X}$ -point (red) $\bar{M}$ -point (yellow) gaps size (red) as a function of the circular base potential V_1 .

equivalent $\bar{X}$ -points, confirming the beginning of energy gaps at these points. Right at the center of $\bar{X}$ -gap, the intensity reduction was enhanced for all k -points on the surface BZ, except for the corners defining the $\bar{M}$ -points (c). At the electron energy of the conduction band minimum, $E = 215$ meV, the hole pockets close into defined points, with the emergence of weak intensity due to the proximity of the bottom of the conduction (d). The last CES in (e) is taken approximately at the center of the

$\overline{M}$ -gap i.e. the low intensity in the BZ corners is increased. By EPWE, five electronic band dispersions of squarely patterned metal surface were calculated to investigate the effect of periodic potential barrier V_1 on the indirect semiconducting band gap size, and $\overline{\Gamma}$ -point energy. The resulting $\overline{\Gamma}$ -point energy (blue), $\overline{X}$ -point gap size (red) and $\overline{M}$ -point gap size (yellow) as a function of V_1 is presented in Fig. 3.19(f). It is clear that both the $\overline{\Gamma}$ -point energy and $\overline{M}$ -point gap size increase rapidly with the scattering potential, and tend to saturate at higher V_1 values. The highest potential where these saturate sets the infinite potential limit of the given system.

3.3.2 The effect of periodicity:

The Bloch wave function has an exponential proportion with the reciprocal lattice vector (G), i.e. inverse relationship between the effective periodic potential (V) and the superlattice constant (d). Which entails, i.e. the bigger the superlattice periodicity, the smaller the effect of periodic potential. This is the motivation to discuss the effect of superlattice periodicity on surface state of the square superlattice presented here.

The electronic band structure of a squarely patterned metal surface with a potential barrier height $V=700$ meV, and a superlattice constant $d=5$ nm is calculated by EPWE along the $\overline{\Gamma X M \Gamma}$ directions in Fig. 3.20(a). All other geometrical and physical parameter in these calculations were identical to those previously discussed except for the superlattice constant (d), which is changed from 3 nm to 5 nm. The surface state exhibits a parabolic dispersion with the same trend of the earlier.

Figure 3.20(b-e) depict the simulated CESs taken at the black dashed lines in (a). To demonstrate the effect of modified (d), these CESs taken at the edges of the corresponding surface state gaps, i.e. at 40 meV, 50 meV, 65 meV, and 75 meV, respectively. Actually one can determine the $\overline{X}$ -point gap edges and size from these CESs. (a) Confirming the beginning of $\overline{X}$ -point energy gaps, the main and umklapp bands back-folding satisfied at the four equivalents $\overline{X}$ -points. The intensity lowering was developed in (b) for all k -points on the surface BZ, except for the corners defining the $\overline{M}$ -points, i.e. the CES taken at the center of $\overline{X}$ -point energy gap. Clearly seen the high intensity allowed states at the BZ boundaries demonstrated the lower edges of the four equivalents $\overline{X}$ -gaps in (c). Based on this, the $\overline{X}$ -point gap size is reduced from 60 meV in Fig. 3.17, to 25 meV for this surface state. Likewise, the $\overline{M}$ -point gap size is reduced to 2 meV (d), and the $\overline{\Gamma}$ -point energy become 10 meV. To discuss the three preceding dispersion parameters as a function of superlattice constant (expressed as

3. RESULTS AND DISCUSSION I

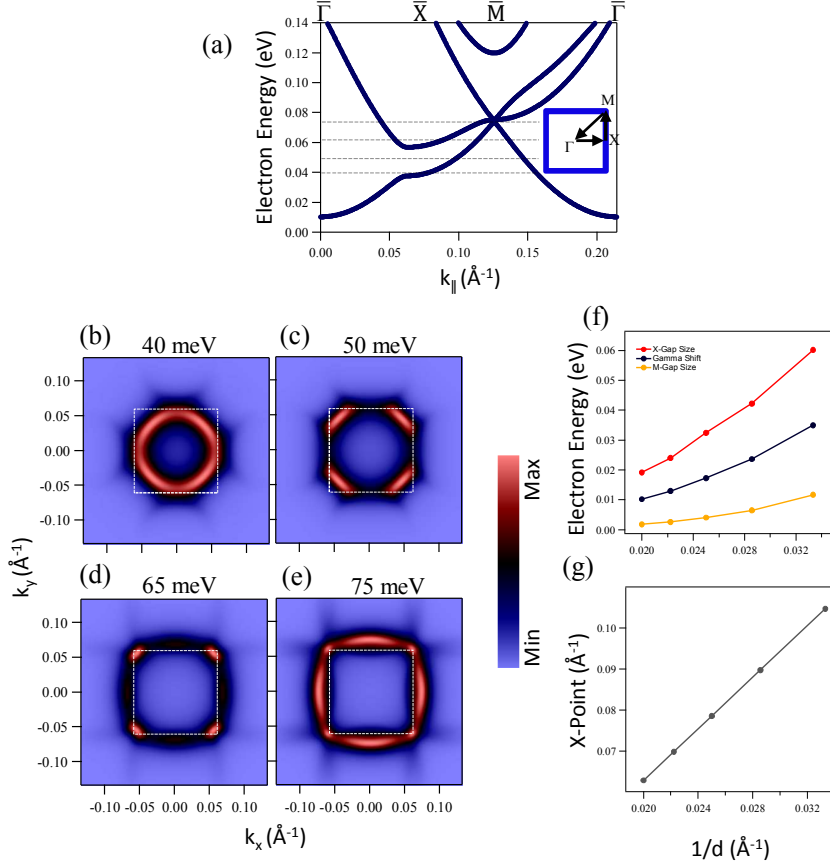


Figure 3.20: Effect of periodicity on the square electronic structure - (a) Calculated band structure using EPWE along the $\bar{\Gamma}\bar{X}\bar{M}\bar{\Gamma}$ direction for a square patterned metal surface with circular bases potential of 700 meV and lattice parameter $d = 5$ nm. (b-e) The simulated constant energy surfaces using EPWE taken at the four electron energies. (f) The variation of $\bar{\Gamma}$ -point shift (blue) and both $\bar{X}$ -point (red) $\bar{M}$ -point (yellow) gaps size (red) as a function of the circular base potential V_1 . (g) The momentum position $\bar{M}$ -point gap as a function of $(1/d)$.

$1/d$), five electronic band structures were calculated by EPWE, where the superlattice periodicity (d) is changed gradually from 3 nm to 5 nm. The three curves in Fig. 3.20(f) confirmed the analogy of the periodicity decreasing and potential barrier increasing. Fig. 3.20(g) clearly showed that, the momentum positions of the $\bar{X}$ -point decrease linearly with increasing the superlattice parameter d .

Part III: Chapter Conclusion

3.4 Conclusion

In conclusion, using Electron Plane Wave Expansion method (EPWE), it has been shown that the band structure of hexagonal metallic superlattice, with circular bases, is characterized by a partial energy gap at the $\bar{M}$ -point and symmetry protected degeneracy at the $\bar{K}$ -point. The lowering of structural symmetry under six-fold rotational symmetry leads to an additional energy gap at the $\bar{K}$ -point in the band structure, no matter what the values of the superlattice, potential barrier V_1 , and constant d is. We further investigated the possible tunability of the semiconducting-like full gap size and the $\bar{\Gamma}$ -point energy through both the barrier height and superlattice periodicity.

The band structure of square metallic superlattices, with circular bases as scatterers, is characterized by an indirect semiconducting energy gap and the surface state exhibit a rigid upward energy shift. Evidence of this semiconducting gap is also seen in the LDOS, where 2D-maps of LDOS taken at energies around this gap were not identical. Furthermore, it has been possible to tune this energy gap size and the $\bar{\Gamma}$ -point energy through both the barrier height and superlattice periodicity.

Using the EPWE (band structure, constant energy surface, and local density of state) calculations we have shown that, the potential barrier manipulation leads to radical change in the electronic structure of the patterned metallic surfaces.

3. RESULTS AND DISCUSSION I

4

Results and Discussion II

4. RESULTS AND DISCUSSION II

Surface State Nanoelectronics Applications

4.1 Introduction and Literature Survey

It seems as though we must use sometimes the one theory and sometimes the other, while at times we may use either. We are faced with a new kind of difficulty. We have two contradictory pictures of reality; separately neither of them fully explains the phenomena of light, but together they do^[156], said Albert Einstein referring to the wave-particle duality. This duality, which was firmly established for light in the beginning of the twentieth century and, later, for electrons, has set great analogue between them and brought together their corresponding physics^[78]. A number of physical properties have, since then, shown to hold for both electrons and electromagnetic waves. The band theory, first, fulfilled for electrons in crystalline solids, has been successfully applied to electromagnetic waves and varieties of artificial periodic 1D, 2D and 3D photonic crystals (PC) have been extensively studied^[79]. Photonic crystals are materials whose electromagnetic (EM) properties vary periodically on a length scale comparable to the wavelength of light, particularly for strong contrast in the dielectric modulations. Within such materials, the propagation of classical EM waves is determined by the detailed structures of the photonic lattice, and is completely governed by Maxwell's equations^[157], i.e., by the photonic band structure^[158]. The propagation of EM waves inside a PC is a consequence of the multiple diffraction through the strongly modulated periodic lattice of dielectric scatterers, which can give rise to ordinary and extraordinary physical phenomenon, such as total internal reflection^[81], positive and/or negative refractions^[82,83], self-collimation^[84], and superprism effect^[85]. Total internal reflection is the phenomenon which occurs when a propagated wave strikes a medium boundary at an angle larger than a particular critical angle with respect to the normal to the surface. If the refractive index is lower on the other side of the boundary and the incident angle is greater than the critical angle, the wave cannot pass through and is entirely reflected. This equally holds for the boundaries between homogeneous media and/or PC interfaces. For incidence smaller than the critical angle, transmitted beam undergoes

4. RESULTS AND DISCUSSION II

refraction governed by the well-known Snell law^[159,160]. In contrast to homogeneous and isotropic materials, the so-called negative refraction can take place in PC^[161] or metamaterials^[162,163]. The similarities between homogeneous media and patterned PC is pointed out by M. Notomi, et al.^[158] who showed that PCs can mimic the behavior of dielectric materials with an effective index of refraction in certain spectral regions for which the so-called equi-frequency contours (EFCs) become rounded, despite the fact that the underlying physical phenomenon is not actually refraction. When an EM wave propagating through air (or another dielectric medium) impinges on a PC interface, the direction of propagation of the EM radiation inside the PC can be obtained by analyzing the EFCs of both media, taking into account the conservation of the wave vector component parallel to the interface^[85]. Within this approach, anomalous optical phenomenon could be well understood. For example, the self-collimation is a consequence of flat portions of the EFCs, which enforce a fixed direction for the group velocity irrespective of the incidence angle^[164,165]. Furthermore, the superprism effects in photonic crystals, in which an entering beam of light follows extremely large angular dispersion are results of EFCs with highly anisotropic shapes^[85,166]. The superprism also modifies the group velocity rather than phase velocity in order to achieve the “*superprism phenomena*”. The ability of the photonic crystal to send optical beams with different wavelengths to considerably different angles in space using superprisms has been used to demonstrate wavelength demultiplexing in these structures^[167]. In fact most of the exotic optical behaviors in PC, or metamaterials, stem from the differences between the group and phase velocities, which in isotropic materials are identical. For example, if certain rounded EFC of PC shrinks with increasing frequency, then the group velocity vector points inwards (i.e., negative group velocity) and a negative refraction is expected at the interface between this PC and effective dielectric medium of positive group velocity^[158]. The previously mentioned phenomena were all observed experimentally at optical frequencies in many literatures. For example, negative refraction in PCs was observed experimentally at optical frequencies by Kosaka and co-workers^[85] and, afterwards, it has been the subject of intense theoretical studies, numerical simulations, and experimental measurements^[158,168,169]. It should be pointed out that all these works have been focused on two-dimensional (2D) PCs as they can be analyzed in a more flexible and straightforward way than their three-dimensional counterparts.

Given the wave-particle duality, the above concepts and discussions could be equally applied to electrons. Indeed, the EFSs in photonic crystals are the analogous to the so

called, constant energy surface (CES) for electron systems, such as the Fermi surface, i.e. the CES at the Fermi energy. In this respect, the physics of photonic crystals are often discussed in conjunction with crystalline solids, in which the periodic scattering potentials provide a similar modification to the quantum-mechanical motion of electrons^[170,171]. For photonics-based applications, however, structures with periodicity of the order of the light wavelength (300-1000 nm) are required and the lithographic techniques were commonly used for this purpose. Noble metal surfaces, for example, host Shockley type surface states characterized by a Fermi wavelength of the order of 1-3 nm. In particular, the energetic position of the surface states band minima is essentially related to the lower edge of such projected band gap of the supporting bulk^[19]. The free-electron parabolic dispersion with different dispersion parameters for the three noble metal surfaces is presented in several surveys^[20,21,132]. Each of them has particular energetic position of the surface states band minima (E_0), effective mass (m_e), and the Fermi wavevector (k_F). Given their ultra small wavelength, much smaller periodic structures are required to fabricate surface state based devices. Techniques, such as self-assembly, made the fabrication of these nanostructures (electronic crystals) possible. Based on this and on the wave-particle duality of electron, all of these previous refractive-reflective phenomena can be satisfied for electrons at the interface between two homogeneous or homogeneous-nanostructured metal surfaces. An important difference between the electronic and the photonic systems is that electrons are present in the crystal, whereas the EM waves couple into the crystal from outside. Consequently, in the photonic crystal the wave vector cannot take any value and is subject to boundary conditions, in contrast to the case of electronic crystals. The observed refractive-reflective effects can be quite complicated, and in many cases, the direction of the propagating beam cannot be interpreted with the use of a simple Snell-like formula. Hence, the formalism of wave vector diagram (or wavevector-space analysis) was proven to be an excellent tool in explaining the unusual refractive properties for, e.g., the diffraction grating system^[172]. In view of the wavevector-space analysis, the reflection-refraction takes place because the two media have Constant Energy Surfaces (CESs) of different k -radius, and the refracted beam follow the direction that conserves the parallel component of the wavevector. Based on wavevector-space analysis, the propagation of electron beam at the interface between two surfaces depends on their band structures and the size/shape of their CESs.

This chapter presents a theoretically study of collections of refractive-reflective phenomena that can possibly occur at the interface separating two-dimensional (2D) elec-

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tronic crystals using a combination of wavevector-space analysis and real-space representations.

The first case study in this chapter is the propagation of electron beam at the lateral interface between two “*different homogeneous*” metal surfaces. At $T = 30$ K, the Au(111), Cu(111), and Ag(111) surface states band minima are 484 meV, 435 meV, and 63 meV^[147], respectively. Therefore, the experimental applicability motivates the investigation of reflection-refraction at the interface between two homogeneous metal surfaces with different surfaces state band minima. The CESs of homogeneous mediums exhibit circular shapes, at all energies, with radii that increase towards higher energy. Based on the difference of the particular energetic position of the surface states band minima for each metal surface, the radii of fixed energy contours will be quite different at the same energy, assuming the same effective mass. This variance motivates, by wave vector analysis, positive refraction or internal reflection according to the beam incidence angle at the interface.

The second case study in this chapter is the scattering of electron beam at the “*homogeneous-nanostructured*” interfaces. A nanostructured metal surface in this case is a 2D lateral superlattice made by combination of two noble metal surfaces with large lattice mismatch. Such combinations commonly self-assemble into moiré superstructures. The 1 ML Ag/Cu(111), in particular, exhibits an irreversible transformation from such moiré pattern into hexagonal lattice of dislocation network (periodicity ~ 2.4 nm)^[53,173]. A recent publication has demonstrated electron guiding and collimation due to the peculiar band structure of this system, however, experiments in this regards are not yet performed^[146]. Here we demonstrate that electron reflection-refraction could be seen at the lateral interface between the dislocation-1 ML Ag/Cu(111) system and the homogeneous Cu(111) surface. Since the scattering potential of Ag steps is very small and the system assembles into one single domain of dislocation network, the experimental realization of electron refraction is far easier. The refraction phenomena in such an interface were found to be strongly energy dependent. At low electron energies, where the CESs are circular, the electrons behave in the same way as in homogeneous-homogeneous interfaces, where positive refraction and internal reflection take place. Additionally, the presence of the dislocation lattice umklapp bands, results in a negative refraction at slightly higher energies, where the CESs are still circular, and different incidence angles. At sufficiently high energy, the CESs are strongly shaped, where contours with large flat segments could be realized. These were used to simulate the possibility for electron self-collimation. Finally, a negative refraction induced by

anisotropic umklapp bands, characterized by negative group velocity, is also simulated at high electron energy. The negative refraction in this case is the analogous to left-handed metamaterials, given the negative group velocity, and superprism effects were simulated.

Throughout this chapter and for didactic purposes, the wavevector space representations of the interface are presented for all cases, i.e., the CESs of each interface at the particular energy. The incident beam energy is fixed to the energy of the concerned CES of the medium of incidence. The construction lines are drawn to define the expected directions of the refracted beams. The direction of the group velocity is obtained from the slope of the dispersion at each medium. Such complete information was all obtained from EPWE calculations. The corresponding real-space representations of refraction/reflection are obtained from EBEM.

4.2 Numerical Considerations

The EBEM real-space simulations in this chapter describe the propagation of a Gaussian electron beam at the interface between two metal surfaces with specific incidence angles. Whilst one usually thinks of a Gaussian beam as a perfectly collimated beam of electron rays with the beam energy uniformly spread across the cross section of the beam. This is not an adequate picture for discussing the propagation of a Gaussian electron beam over any appreciable distance because diffraction causes the electron waves to spread transversely as they propagate^[174]. Therefore, the important issue in the study of the Gaussian beam is to know its width evolution along the direction of propagation z , i.e., $\omega(z)$. This dependence is extracted from the evolution of the amplitude distribution, which provides the following formula:

$$\omega(z) = \omega_0 \sqrt{1 + \left(\frac{z\lambda}{\pi\omega_0^2}\right)^2} \quad (4.1)$$

where ω_0 being the beam width at $z = 0$ (i.e., the beam focal point) and λ is the wavelength in the material where the beam is propagating. A graphical representation of this equation is overlaid on the EBEM simulations in Fig. 4.1(c). The parameter ω_0 , which governs the rest of the beam evolution, is usually named as the beam waist. It should be noted that $\omega(z)$ depends on λ , where large wavelengths lead to wider beams. At each perpendicular plane to z , the beam shows a Gaussian profile. The width reaches the minimum at the waist and then the beam expands. The same

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amount of energy located at the beam waist plane needs to be distributed in each plane. As a consequence, the maximum of irradiance at each z plane drops from the beam waist very quickly, as expressed in the colored-scale intensity distribution in Fig. 4.1. Equation 4.1 has a very interesting behavior when z tends to ∞ . This width dependence shows an oblique asymptote having a slope of:

$$\tan \theta = \frac{\lambda}{\pi \omega_0} \quad (4.2)$$

This parameter (θ) is named the divergence of a Gaussian beam. It describes the spreading of the beam when propagating towards infinity. From the previous equation, we see that the divergence and the waist width are reciprocal parameters. This means that larger values of the width lead to lower values of the divergence, and vice versa. This relation has even deeper foundations, showed in Fig. 4.1, on characterizing generalized beams that made in terms of the parameters already defined for the Gaussian beam cases.

Figure 4.1 represent real-space simulations of four Gaussian electron beam (left panels) of different waists (ω_0) and their corresponding phase propagation (right panels) calculated by EBEM. Gaussian beams in these cases are propagating through a clean metal surface, i.e., a uniform isotropic material of circular CES taken at the same electron energy as the beam (350 meV, to compare with results section). The radius of this CES, i.e., the wavevector $k = 0.19 \text{ \AA}^{-1}$, and consequently the wavelength of a propagated beam in this homogeneous medium is 3.2 nm.

Using Eq. 4.2, the divergence of the electron Gaussian beam with $\omega_0=1.3$ nm was almost 40° as in Fig. 4.1(a). The same behavior is obtained by the circular phase propagation of this beam in Fig. 4.1(b), when the electron beam wavefront spreading in all direction resembling the case of a point source. In Fig. 4.1(c) the divergence decreased to 20° by increasing the waist to $\omega_0=2.65$ nm, which also exhibits quasi-circular phase propagation of this beam as shown in Fig. 4.1(d). Likewise, the simulation of a Gaussian electron beam of $\omega_0=5.25$ nm, displayed in Fig. 4.1(e), revealing a small divergence angle 10° . A beam of this waist is more collimated than the two previous cases, which is further confirmed by the wavefront propagation graph in Fig. 4.1(f), nearly taking the plane wave form. By increasing the waist further ($\omega_0=13.25$ nm), an ideally collimated beam is obtained in the real space simulation and its corresponding phase propagation is exactly a plane wave, as shown in Fig. 4.1(g-h). Based on these discussions we conclude that a high collimation (i.e., low divergence) will be obtained when the beam is wide. On the contrary, a highly focused beam will be

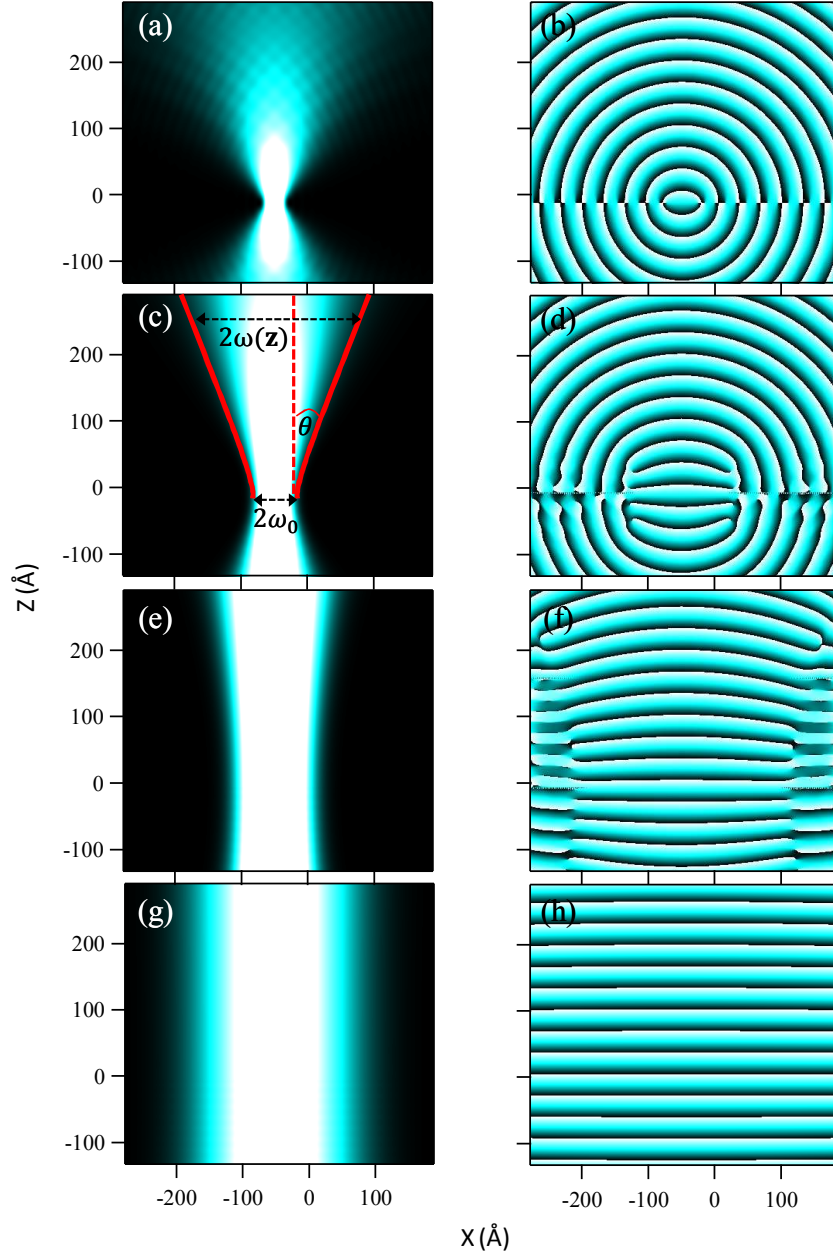


Figure 4.1: Divergence of a Gaussian electron beam - Real-space simulations of Gaussian electron beam (left panels) and its corresponding phase propagation (right panels), using EBEM with four different beam “waist width” ω_0 . ω_0 was set to 1.32 nm in (a,b), 2.65 nm in (c,d), 5.25 nm in (e,f) and 13.25 nm in (g,h).

obtained by allowing a large divergence angle. But in fact, the increasing of the waist width of a Gaussian electron beam in EBEM calculations implies working out with very large real space geometry and, more worse huge number of discretization points,

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leading to exceedingly long simulations. This numerical limitation requires a right balance, where working out with moderate beam waists, to combine both well-defined Gaussian beam and relatively short simulation time, is favorable. The implication of this limitation on the physical outcomes will be shown in the following results sections.

Part I: SSNE via Isotropic Surfaces

4.3 Homogeneous-Homogeneous Interfaces

In this section, the refraction-reflection behavior will be defined and simulated for an electron beam propagating through lateral interface between homogeneous and isotropic clean metal surfaces^[157] each with different potential barrier. The system to be discussed is sketched in Fig. 4.2. The blue and red areas represent two homogeneous metal surfaces, with different potential barriers, $V_1=0$ and $V_2=200$ meV for the blue and red regions, respectively. The problem under investigation is the refraction (dotted black arrow) and reflection (white arrow) at the interface between these two systems for an incident electron beam (black arrow) with fixed energy and different incidence angles. For the band structure, CESs and the real-space simulations, the geometry depicted in Fig. 4.2 is created.

Using EPWE, the calculated band structures for these homogeneous metal surfaces are presented in Fig. 4.3(a-b). Calculations are performed for each system separately with an effective mass of $0.41 m_e$. Figure 4.3(b) depicts the dispersion relation of a clean metal surface (blue area in Fig. 4.2). As expected for the case of free electron gas model, a parabolic dispersion, without any shift of the surface state band minima neither band gap opening, is obtained. However, the dispersion relation of the other isotropic surface with homogeneous potential of 200 meV (red area in Fig. 4.2) is rigidly shifted upward by 200 meV, as shown in Fig. 4.3(a). This shift of the surface state towards increasing the electron energy is correlated with the average potential inside the system, being larger for stronger potential^[175]. It should be noted that, the present of potentials do not induce any gap opening in the band structure as long as the homogeneity condition is satisfied.

In Fig. 4.3(c-d) EPWE-calculated CESs taken at the black dashed lines in (a-b), i.e., at 350 meV, for each system are displayed. The calculations additionally present the spectral weight, i.e., the photoemission intensity, commonly obtained from ARPES

4. RESULTS AND DISCUSSION II

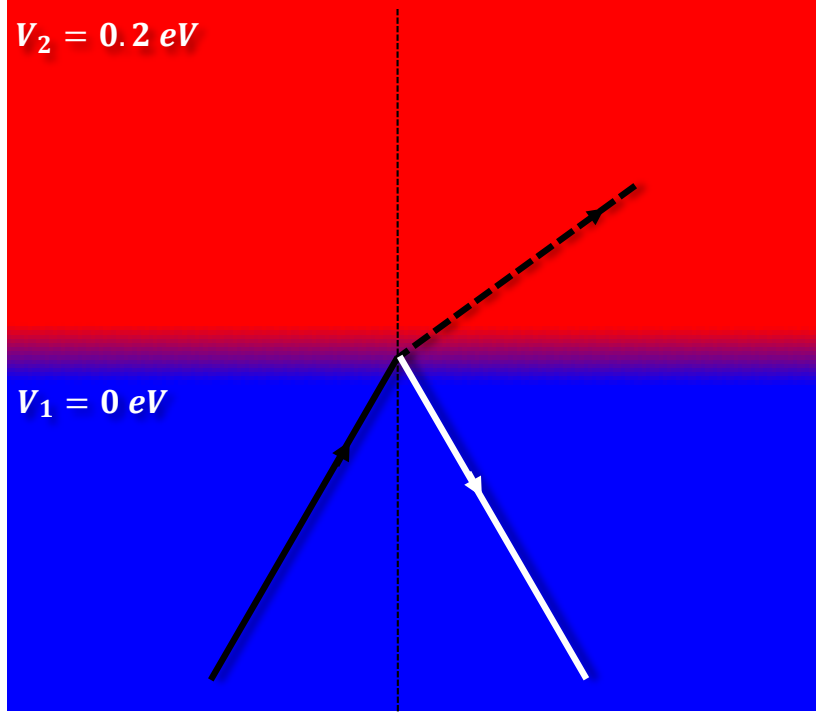


Figure 4.2: Homogeneous-Homogeneous real-space geometry - The geometry used in EBEM and EPWE calculations. The blue and red areas represent two homogeneous metal surfaces with different potential barriers, $V_1=0$ eV for blue and $V_2=0.2$ eV for red. The arrows represent the incidence (solid black), the refracted (dotted black), and the reflected (solid white) beams.

measurements. Both CESs consist of circular contours characteristic for isotropic systems. Generally the CES is a surface (contour in the two-dimensional space) that consists of all allowed electron wavevectors (k) within the first and higher order Brillouin zones at a certain energy. The intensity of two CESs is constant along the circular contours (i.e., no band gap opening), but the wavevectors, i.e., the k -radii of the energy contours are varied from $0.194 \text{ \AA}^{-1}$ for (d) to $0.127 \text{ \AA}^{-1}$ for (c) by the potential-induced shift of Shockley surface in (a).

In order to highlight the electronic differences between the two media, we re-plot in Fig. 4.4(a) their dispersions in a single graph. Both the wavevector difference (Δk) and the large energy offset (ΔE) between the two Shockley surface states, at fixed energy of 350 meV, are obvious.

Using the same representation, Fig. 4.4(b-c) represents the geometry defined in Fig. 4.2 in the wavevector space, for the two sets of CESs shown in Fig. 4.3(c-d).

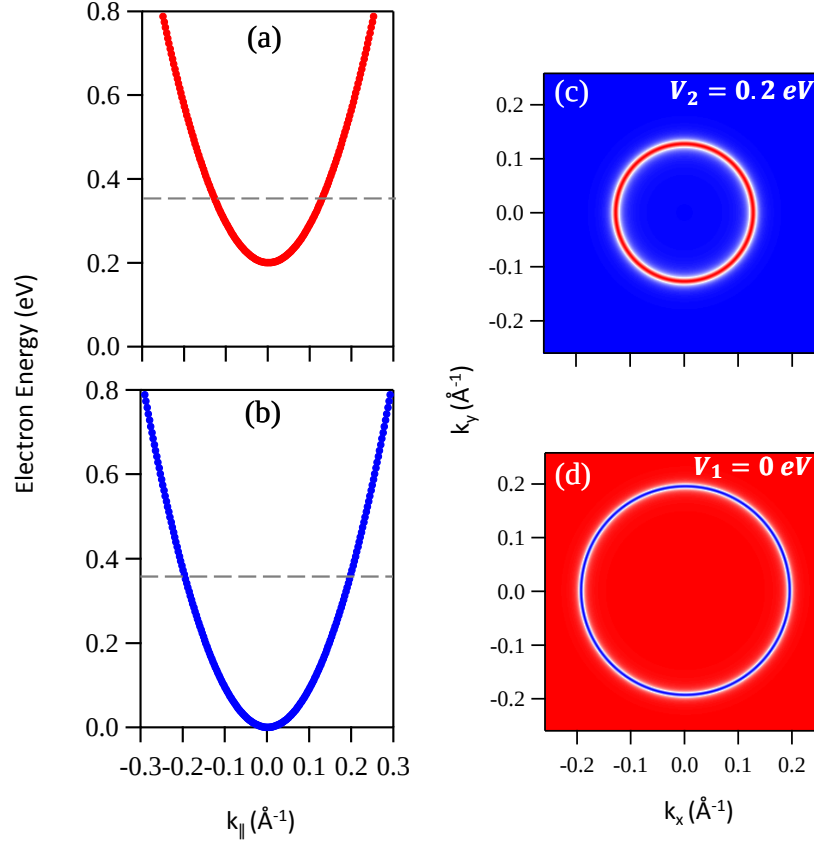


Figure 4.3: The band dispersion of two homogeneous metal surfaces - Calculated band structures using EPWE for two previous homogeneous metal surfaces, (a) for a surface with potential barrier $V_2=0.2 \text{ eV}$ and (b) for the other one. (c,d) Simulated Constant energy surfaces taken at the black dashed lines in (a,b), i.e. at 350 meV, for each system.

Two equivalent formalisms of the wavevector space analysis shall be used through this chapter, where the choice between them depends on the case under investigation. Often we will use the formalism presented in Fig. 4.4(b), because it allows a one-to-one comparison with the real space simulations. The top and bottom halves in (b) are the CESs for the red and blue surfaces, respectively. An incoming electron beam (black solid arrow) from the blue surface shall undergo refraction (white solid arrow) and reflection (white dashed arrow) at the interface with directions and intensities defined by the Snell's law and Fresnel coefficients^[176], respectively. In wavevector space, the direction of the refracted beam is such that the wavevector components parallel to the interface are the same i.e., the wavevector distance from the center of energy contour to the construction lines (dotted vertical lines). A partial reflection also takes place,

4. RESULTS AND DISCUSSION II

where the reflected beam intensity increases for larger incidence angles. The phases are matched at the interface and are π -out-of-phase for the refracted and reflected beams, respectively. At the critical angle (yellow arrow) the refracted beam is now parallel to the interface (yellow horizontal arrow) forming an evanescent wave and, a total internal reflection (green dotted arrow) takes place at higher incidence angle (green arrows), which also follows the parallel-momentum conservation.

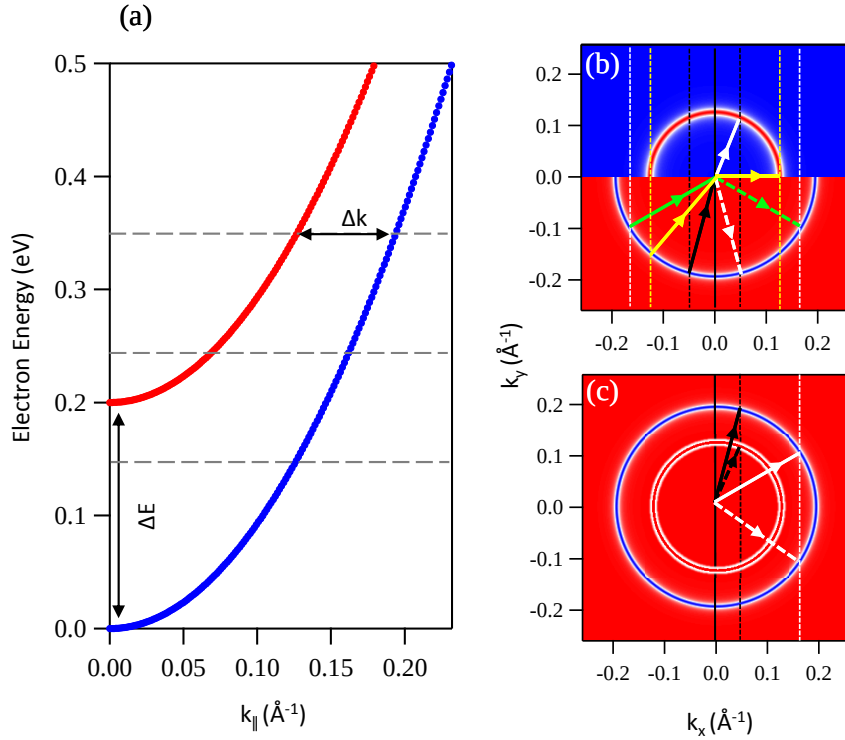


Figure 4.4: The combination of two band dispersions and the corresponding Wavevector-space formalism - (a) The combination of calculated band structures in Fig. 4.3(a,b), Δk represent the difference between two CESs wavevectors. (b) Wavevector-space analysis of the refraction/reflection at the lateral interface of two previous homogeneous metal surfaces performed for electron energy $E = 350$ meV. Upper and lower halves of (b), red and blue circles of (c) are CESs of two metal surfaces, respectively. The arrows represent electron beams with incidence angle (solid black and green), refracted angle (dotted black), and reflected angle (solid green). The dotted lines defines the direction of the interface confined evanescent wave.

Likewise, the second formalism of wavevector space diagram is presented in (c)^[83,177]. The wave vector diagrams in these cases (homogeneous interfaces) are nothing but the graphical representation of Snell's law. The intersection of the CES with the plane

4.3 Homogeneous-Homogeneous Interfaces

of incidence is always circles with different k -radii energy contours as discussed earlier. In Fig. 4.4(c), the blue circle represent the CES of blue surface presented in Fig. 4.2, while the red circle represents the CES of the red surface, and the dotted (white and black) vertical lines represent the conservation of the parallel component of the incident wave vector (i.e., the construction lines)^[82]. Since the present discussion concerns surfaces with positive group velocities, i.e., right handed materials, then the choices for the wave vector inside the red medium of 200 meV potential, easily follow the directions obtained from the previous analysis in (b). However, in the case of left handed-like media, the formalism of (c) is more convenient to determine the direction of negative refracted beams, as we show in more detail in the section describing homogeneous-nanostructured interfaces.

In 2D momentum space, Snell's law reads^[82];

$$|k_{inc}| \sin(\alpha) = |k_{ref}| \sin(\beta) : |\lambda_{ref}| \sin(\alpha) = |\lambda_{inc}| \sin(\beta) \quad (4.3)$$

where α and β are the angle of incidence and refraction, respectively, and k_{inc} and k_{ref} are the respective wavevectors.

We obtained previously the k -radii of the energy contours for electron energy of 350 meV. These were $k_{inc} = 0.194 \text{ \AA}^{-1}$ and $k_{ref} = 0.127 \text{ \AA}^{-1}$ for the blue and red surfaces presented in Fig. 4.2, respectively. Hence the corresponding wavelengths are $\lambda_{inc}=3.2 \text{ nm}$ and $\lambda_{ref}=4.9 \text{ nm}$, while the critical angle ($\alpha(\beta = 90^\circ)$) according to Eq. 4.3 is 41° . Let us proceed now with the determination of the refracted beam (dotted arrow) using the wave vector space analysis of Fig. 4.5(a). In this example, an electron beam of incident angle $\alpha_{inc} = 15^\circ$ (solid arrow) is transmitted through the interface between two homogeneous metal surfaces. Basically, the wave vector diagram method consists of the following steps. First, we draw the upper and lower halves of CESs, taken at the energy of electron beam, in the repeated zone scheme. Secondly, we draw a line perpendicular to the interface from the intersection between lower CES and the incident beam, which is called incident construction line (left dotted white line). Third, we draw the equivalent construction line (right dotted white line), the same distance to the right from the center of the CES, i.e., representing the conservation of $k_{\parallel}$. Then, we take down all intersections between this construction line and the upper CES of the refractive medium. The directions taken from the center of the CES to these interceptions (black dashed arrow) define the direction of the phase propagation of the refracted beams. In the case of isotropic energy contours as the present case, the phase and group velocities have the same direction^[47]. The refracted angle in this particular

4. RESULTS AND DISCUSSION II

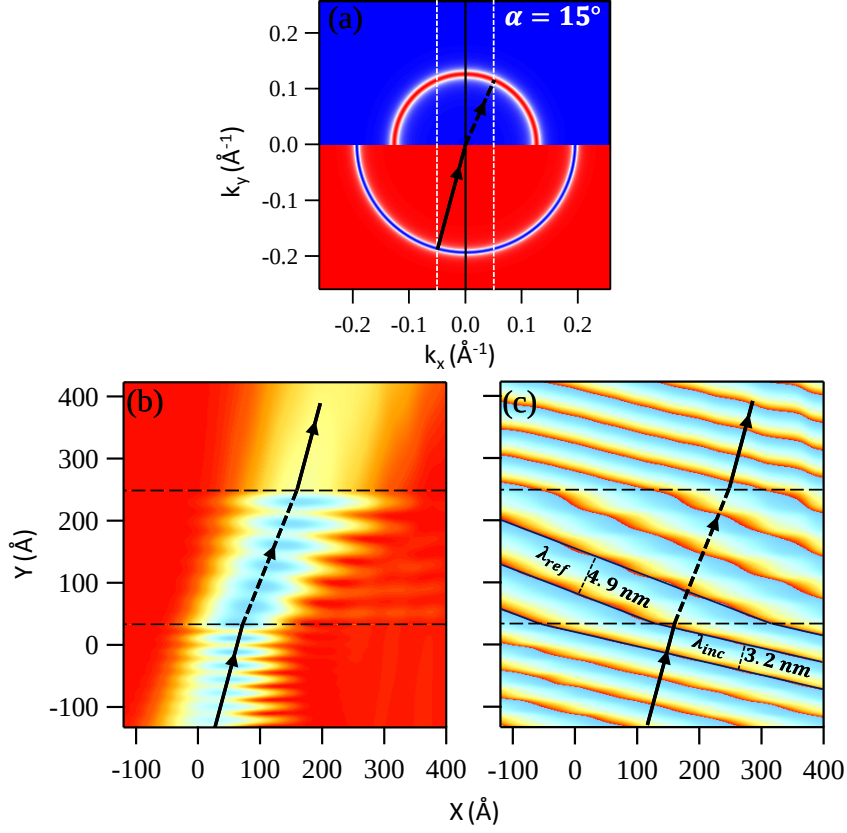


Figure 4.5: Electron refraction at lateral interface of two homogeneous metal surfaces, $\alpha_{inc} = 15^\circ$ - (a) Wavevector-space analysis of the refraction at the lateral interface of two previous homogeneous metal surfaces. (b) Real-space simulations of electron refraction at this lateral interface and (c) its corresponding phase propagation. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 15^\circ$, waist width $\omega_0 = 5.25 \text{ nm}$ and $E_B = 350 \text{ meV}$.

example was found to be $\beta_{ref} = 22^\circ$ according to wavevector space analysis, in nice agreements with Eq. 4.3. Figure 4.5(b) shows the corresponding real-space simulation result, obtained by EBEM, for the problem analyzed in (a). An electron beam with a waist width $\omega_0 = 5.25 \text{ nm}$, and made up of 101 plane waves, hits the interface (lower dashed line) at incidence angles $\alpha_{inc} = 15^\circ$ (solid arrow). According to the Fresnel transmissivity of the incident power, the electron refraction takes place through the refractive medium with $\beta_{ref} = 22^\circ$ (dotted arrow). The upper dashed line defines the end of the refractive slab (the second interface), where the electrons beam is refracted back into the initial incoming direction. The direction of the incident and refracted beams are best estimated from the corresponding phase analysis presented in Fig. 4.5(c). The black lines defined the wavefronts propagation, where the beam directions

4.3 Homogeneous-Homogeneous Interfaces

are perpendicular to these lines. From this phase analysis the refractive angle was found to be 22° . Additional information could be obtained from the presented phase analysis, such as the phase matching condition at the interfaces, where the wavefronts are clearly continuous at the interface. The wavelengths, defined by the separation between two successive wavefronts maxima or minima, i.e., the distance between black dotted lines, changes its value after crossing the interface from 3.2 nm to 4.9 nm. The overall agreement between the wavevector space and real space analysis is perfect in terms of defining the direction of the refracted beams and the corresponding wavelength.

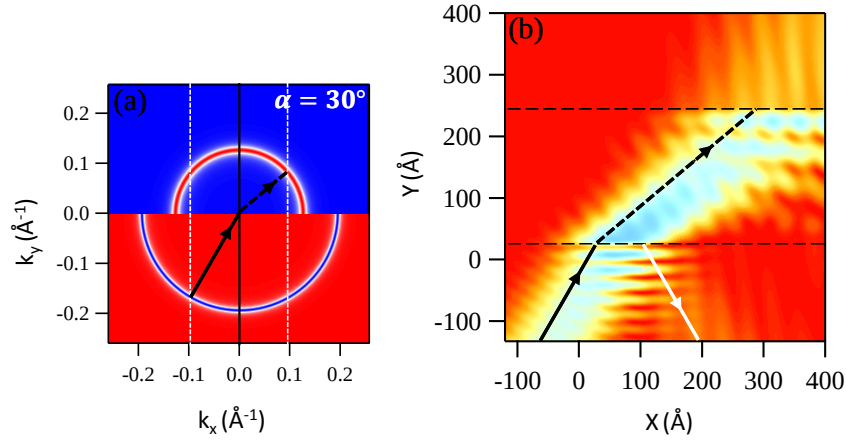


Figure 4.6: Electron refraction at lateral interface of two homogeneous metal surfaces, $\alpha_{inc} = 30^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the lateral interface of two previous homogeneous metal surfaces. (b) Real-space simulations of electron refraction at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 30^\circ$, waist width $\omega_0 = 5.25$ nm and $E_B = 350$ meV.

Likewise, the wavevector space analysis in Fig. 4.6(a) shows that an electron Gaussian beam (solid black arrow) of incidence angle $\alpha_{inc} = 30^\circ$ is refracted (dotted black arrow) by $\beta_{ref} = 50^\circ$ at the momentum intersection between the construction line and the upper CES of refractive media. The corresponding real space simulation obtained by EBEM is presented in (b) for an electron beam (solid black arrow) of waist width $\omega_0 = 5.25$ nm. The angle of refraction was found to be $\beta_{ref} = 50^\circ$ in a perfect agreement with wavevector space representation. Notice that most of the intensity is transmitted to the blue medium, i.e., refracted following the momentum conservation (dotted black arrow). According to Fresnel, the reflectivity of the beam should increase

4. RESULTS AND DISCUSSION II

for large incidence angles. This reflected part of the beam intensity is clearly visible in the incoming medium (white arrow direction), Fig. 4.6(b).

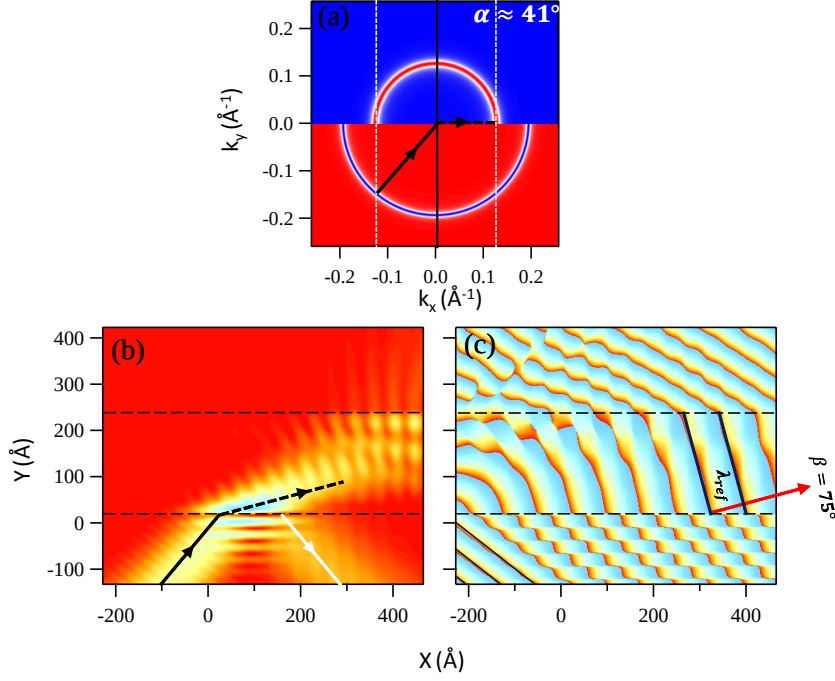


Figure 4.7: Electron refraction at lateral interface of two homogeneous metal surfaces, $\alpha_{critical} = 41^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the lateral interface of two previous homogeneous metal surfaces. (b) Real-space simulations of electron refraction/reflection at this interface and (c) its corresponding phase propagation. Calculation are performed with a Gaussian electron beam of critical angle $\alpha_{inc} = 41^\circ$, waist width $\omega_0 = 5.25$ nm and $E_B = 350$ meV.

According to Eq. 4.3, the critical angle for the present interface system is $\sim 41^\circ$. Figure 4.7(a) presents the corresponding wavevector diagram analysis, where the solid arrow is incident beam with the critical angle, and the refracted beam is now parallel to the interface (dotted arrow). Nevertheless, from the real space simulation results, shown in (b), for an electron Gaussian beam of $\omega_0 = 5.25$ nm (solid arrow) it is clear that the transmitted intensity of this beam is refracted by an angle less than 90° . To precisely define the direction of the refracted beam, the corresponding phase analysis presented in (c) by employing a plane wave source in EBEM simulation. The black lines define the wavefronts, and the refraction angle was found, by constructing a line perpendicular to the wavefronts (red arrow), to be 75° . Here comes the important of the beam divergence discussed in the numerical consideration section. With the beam waist used throughout these calculations, the divergence angle is $\sim 10^\circ$. Later, we

will show that, by the increasing in the beam waist width, the agreement between the wavevector and real space simulations is significantly improved.

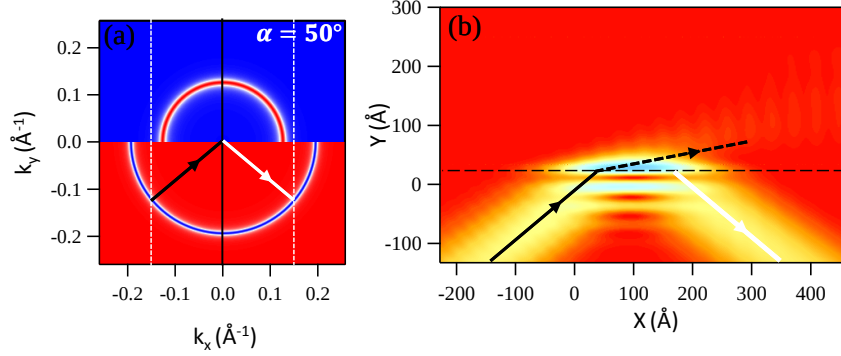


Figure 4.8: Electron refraction at lateral interface of two homogeneous metal surfaces, $\alpha_{inc} = 50^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the lateral interface of two previous homogeneous metal surfaces. (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 50^\circ$, waist width $\omega_0 = 5.25$ nm and $E_B = 350$ meV.

The critical angle is the angle of incidence above which the total internal reflection occurs. Figure 4.8 investigates the case of total internal reflection for an incident angle of 50° . The wavevector space diagram in (a), clearly shows that the construction line (left dotted white) does not intersect the CES of refractive material. However, the construction line in the right side intersects the CES of incidence medium, and therefore, all beam intensity must be reflected back (white arrow). According to this wavevector space analysis, transmitted beam is not allowed. EBEM real space simulation, presented in (b), confirmed that the incident electron beam (solid black arrow) is actually reflected back by the same angle (white arrow), i.e., 50° . Nevertheless, a very low intensity of the beam is transmitted (dotted black arrow), which is due to the beam spreading that allows for a relatively wide range of incidence angles. This transmitted intensity is completely vanished at beam incidence of 60° as depicted in Fig. 4.9. The construction line of the parallel-momentum conservation (vertical dotted white) is far away from the refractive CES as in wavevector analysis of (a). At such high incidence angle, the EBEM real space simulation, Fig. 4.9(b), also showed that the incoming electron beam (solid black arrow) is totally reflected by the same angles (white arrow) without any transmitted intensity.

4. RESULTS AND DISCUSSION II

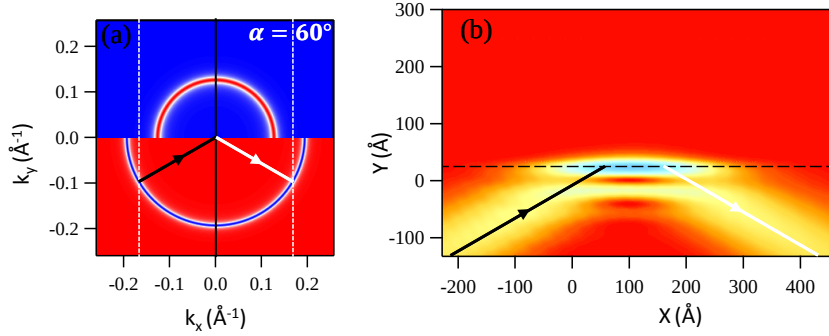


Figure 4.9: Electron refraction at lateral interface of two homogeneous metal surfaces, $\alpha_{inc} = 60^\circ$ - (a) Wavevector-space analysis of the reflection at the lateral interface of two previous homogeneous metal surfaces. (b) Real-space simulations of total internal reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 60^\circ$, waist width $\omega_0 = 5.25$ nm and $E_B = 350$ meV.

The influence of the wavefront distortion and beam divergence on the propagation of electron Gaussian beam is best seen by a direct comparison between the previous simulations and the one presented in Fig. 4.10. The results displayed in Fig. 4.10 are acquired with the exact same parameters of the previous results, except for the beam waist width, which is increased from 5.25 nm to 21 nm (i.e., $\sim$ four order of magnitude wider beam is used). With such a simple optimization, the agreement with the wavevector space analysis and Snell law (Eq. 4.3) is significantly improved. The detailed interpretation of the effect of Gaussian beam waist on the quality of the simulation is discussed in section 4.2. For higher waist width, the electron beam is more collimated and its wavefront has a plane wave form. This translates in the wavevector space into a ray with well-defined incidence angle (i.e., narrow angular distribution). In Fig. 4.10(a), the real space simulation of electron refraction obtained by EBEM for Gaussian electron beam (solid black arrow) of incidence angle $\alpha_{inc} = 15^\circ$, waist width $\omega_0 = 21$ nm and $E_B = 350$ meV is refracted (dotted black arrow) according to the parallel-momentum conservation by $\beta_{ref} = 22^\circ$. Similarly for $\alpha_{inc} = 30^\circ$ the beam is partially transmitted by $\beta_{ref} = 50^\circ$ (dotted black arrow), and the remaining intensity is reflected back (white arrow). For the critical angle incidence (Fig. 4.10(c)), i.e., 41° , the refracted beam is almost parallel to the interface ($\beta_{ref} = 82^\circ$), which is improved compared to the same case in Fig. 4.7. Furthermore, the total internal reflection of the electron beam actually is achieved slightly (4°) above this angle, when the beam hitting the interface by $\alpha_{inc} = 45^\circ$, Fig. 4.10(d). Notice for smaller beam waist, Fig.

4.8, transmitted intensity was present up to 50° incidence, confirming the importance of using wider Gaussian beam for accurate simulations.

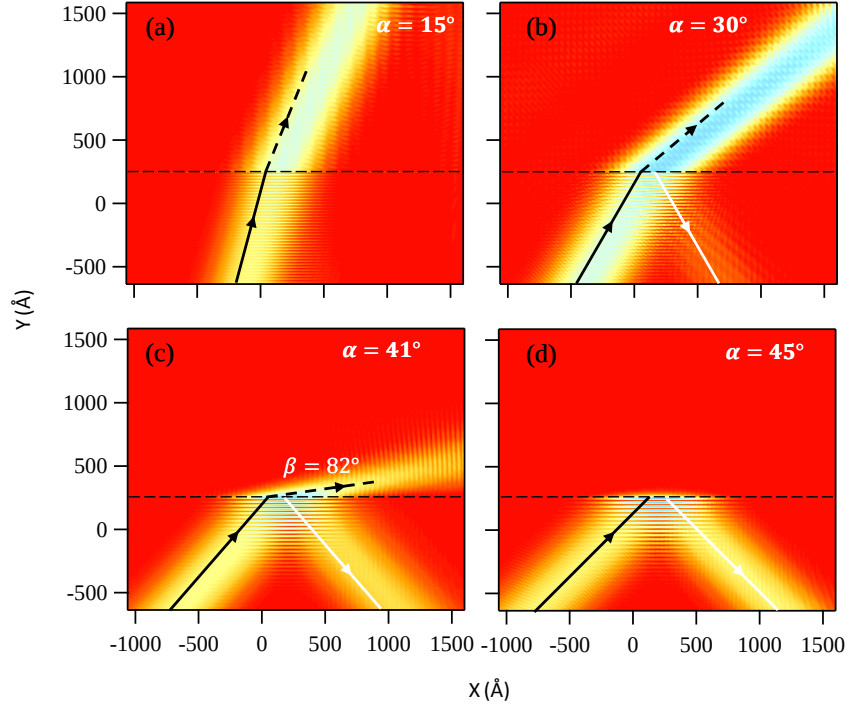


Figure 4.10: Electron refraction at lateral interface of two homogeneous metal surfaces, $\omega_0 = 21$ nm - Real-space simulations of electron refraction/reflection at the lateral interface of two previous homogeneous metal surfaces. Calculation are performed with a Gaussian electron beam of waist width $\omega_0 = 21$ nm and $E_B = 350$ meV. There are four different values of the incidence angle α , in (a) $\alpha_{inc} = 15^\circ$, (b) $\alpha_{inc} = 30^\circ$, (c) $\alpha_{inc} = 41^\circ$ and in (d) $\alpha_{inc} = 45^\circ$.

The above simulations were restricted to single electron energy, being 350 meV. At lower energies, the values of Δk , which are responsible for the refraction, get larger. For example, at electron energy of 250 meV, $k_{inc} = 0.164 \text{ \AA}^{-1}$ and $k_{ref} = 0.073 \text{ \AA}^{-1}$, and the corresponding wavelength are $38.4 \text{ \AA}$ and $85.4 \text{ \AA}$. The larger difference between k_{inc} and k_{ref} at 250 meV indicates a stronger refraction at this energy, but also implies a smaller critical angle. The critical angle in this case, as obtained from Eq. 4.3, is $\sim 27^\circ$, and total internal reflection is expected beyond these angles.

Therefore, electrons with energy of 250 meV also undergo refraction/reflection at the interface, as shown from the wavevector space analysis presented in Fig. 4.11(a-b). However, at 15° , the refraction angle is clearly larger as a result of larger ($\sim 0.09 \text{ \AA}^{-1}$)

4. RESULTS AND DISCUSSION II

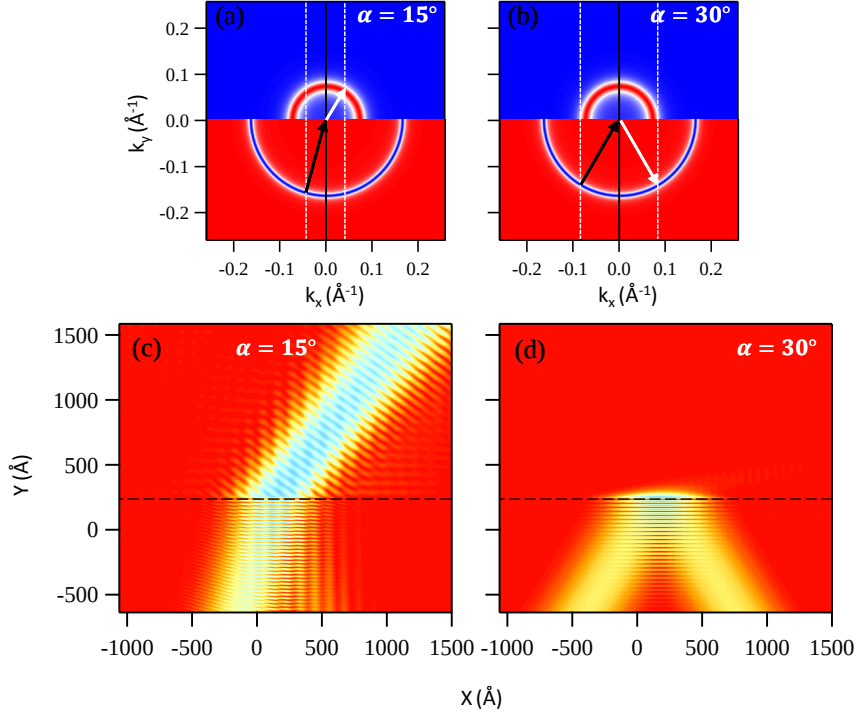


Figure 4.11: Electron refraction at lateral interface of two homogeneous metal surfaces, $E_B = 250$ meV - (a,b) Wavevector-space analysis of the reflection-refraction at the lateral interface of two previous homogeneous metal surfaces performed for electron energy $E_B = 250$ meV. There are two different values of the incidence angle α_{inc} , in (a) $\alpha_{inc} = 15^\circ$, and in (b) $\alpha_{inc} = 30^\circ$. (c,d) Real-space simulations of electron reflection at the lateral interface of two previous homogeneous metal surfaces. Calculation are performed with a Gaussian electron beam of waist width $\omega_0 = 21$ nm and $E_B = 250$ meV. There are two different values of the incidence angle α_{inc} , in (c) $\alpha_{inc} = 15^\circ$, and in (d) $\alpha_{inc} = 30^\circ$.

compared to the 350 meV electron energy ($\sim 0.07 \text{ \AA}^{-1}$) presented in Fig. 4.10(a). The corresponding real space simulation, additionally, showed that the intensity of the reflected beam, which was negligible in Fig. 4.10(a), is observed in Fig. 4.11(c). At 30° , a total internal reflection occurs, Fig. 4.11(d), in contrast to the obvious refraction for the 350 meV electrons presented in Fig. 4.10(b). Therefore, for an interface with the dispersions presented in Fig. 4.4(a), the strength of refractions decreases for higher energy electrons, whereas the critical angle increases.

The exceptions are those electrons with energies between the $\bar{\Gamma}$ -point energies of the two media, which undergo a naturally expected total internal reflections at all incidence angles, because there is an energy offset (ΔE) as shown in Fig. 4.4(a). This case is

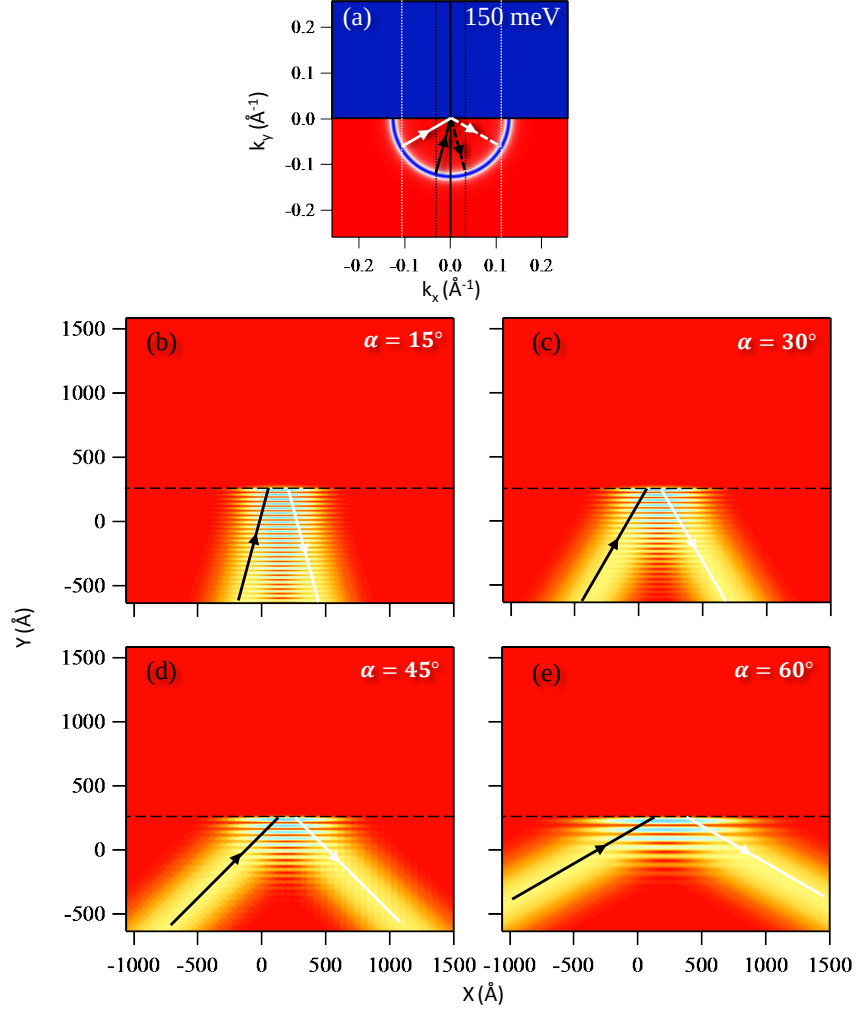


Figure 4.12: Electron internal reflection at lateral interface of two homogeneous metal surfaces, $E_B = 150$ meV - (a) Wavevector-space analysis of the total reflection at the lateral interface of two previous homogeneous metal surfaces performed for electron energy $E_B = 150$ meV. (b-e) Real-space simulations of electron reflection at the lateral interface of two previous homogeneous metal surfaces. Calculation are performed with a Gaussian electron beam of waist width $\omega_0 = 21$ nm and $E_B = 150$ meV. There are four different values of the incidence angle α_{inc} , in (b) $\alpha_{inc} = 15^\circ$, (c) $\alpha_{inc} = 30^\circ$, (d) $\alpha_{inc} = 45^\circ$ and in (e) $\alpha_{inc} = 60^\circ$.

presented in Fig. 4.12, where electron beam of energy 150 meV (i.e., lower dashed line in Fig. 4.4(a)) hits the interface with different incidence angles. At this energy the metal surface (blue), with $V_1 = 0$ meV, exhibits a circular energy contour, whereas the $V_2 = 200$ meV (red) metal surface contains no electronic states at this energy, i.e., no energy contour. The wavevector-space analysis, shown in Fig. 4.12(a), schematically

4. RESULTS AND DISCUSSION II

depicts the electron total reflection for electron beams of $E=150$ meV, and different incidence angles. All electron beams propagating from the incoming media must be reflected back at the lateral interface of these homogeneous metal surfaces irrespective of the incidence angle. In (b-e) the corresponding real space simulations calculated by EBEM performed at four different incidence angles are presented. It is clear from these simulations that electrons with this energy (or, generally, energies between the $\bar{\Gamma}$ -point energies of the two media) must undergo total internal reflections at all incidence angles.

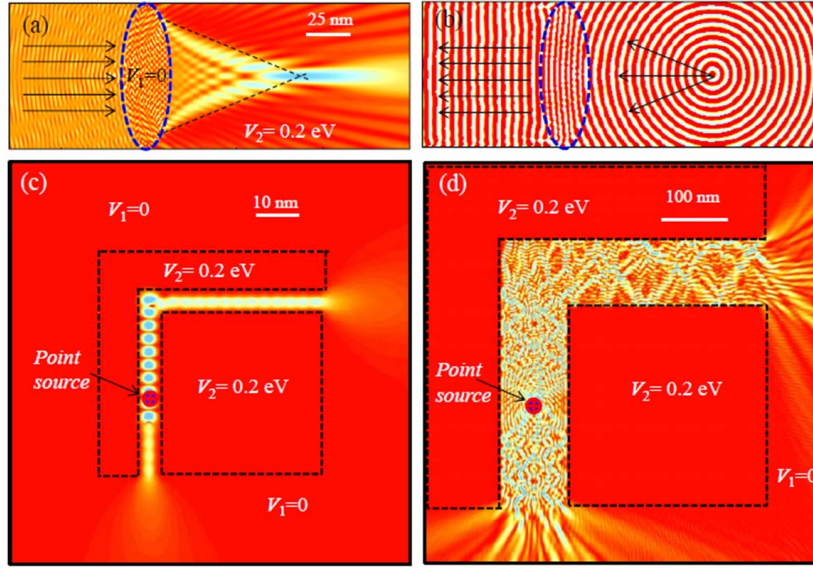


Figure 4.13: Electronic convex lens and waveguides - Real-space simulations of (a-b) an electronic convex lens and (c-d) waveguides. In (a-b) plane wave and Gaussian beam sources, respectively, are used with 350 meV. In (c-d) point sources of energy 150 meV are employed.

With these obvious similarities between electrons and electromagnetic waves, the simulation of analogues optical components and devices is possible. For example, an electronic convex lens could be made by patterning a region inside a homogeneous metal (blue dashed region), where the potentials are different inside and outside this region, Fig. 4.13(a-b). A plane wave source (black arrows) entering such a lens is collimated at the focal point, being $\sim 800\text{\AA}$ for electron energy of $E_B = 350$ meV, Fig. 4.13(a). Likewise, a point source positioned at the focal point is converted into a plane wave behind the lens, as shown from the wavefront plot presented in Fig. 4.13(b). Additionally, the all-angles total internal reflection presented in Fig. 4.12 provides an electron analogy to

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optical mirrors. Figure 4.13(c-d) demonstrates the possibility of using such electronic mirrors for electron guiding, in contrast to Bragg-based waveguides. The channels are defined by setting the potential, inside areas defined by the black dashed lines, to $V_2 = 200$ meV and to zero elsewhere. A point source, with energy 150 meV, positioned inside channels of width (c) 2λ and (d) 10λ is guided through the entire channel, where multiple reflections are easily recognized for the channel of higher width. We note that the fabrication of such optical-like components/ devices, in case of electronic systems, implies atomic layer deposition and subsequent lithographic treatments^[33,34].

4. RESULTS AND DISCUSSION II

Part II: SSNE via Anisotropic Surfaces

4.4 Homogeneous-Nanostructured Interfaces

Materials with nanoscale periodic structures often have unique optical, electronic, or mechanical properties^[175]. Nanostructured materials are a class of artificial composite materials which, in the presence of periodic structure, exhibit electronic band gaps (i.e., energy regimes where the electron waves cannot propagate). This band gap property makes the nanostructured materials very important for the manipulation of electron waves; they can act in the same way as the semiconductor gaps for the propagation of electrons. The realization that gaps can exist in the propagation of waves through a periodic structure actually dates back to 1913^[178,179]. In contrast to the circular energy contours of homogeneous metal surfaces, the nanostructured materials exhibit highly featured isotropic/anisotropic and gapped constant energy surfaces (CESs), i.e., might lead to anomalous behaviors of electron propagation. The electronic band structure of these materials reveals that they can act, in some energy regime, as right handed-like materials, and at other energies exhibit left handed-like behaviors. In optically anisotropic materials, the eigenvalue equation has two roots, i.e., two solutions for the dispersion relation and correspondingly two propagating waves in the anisotropic medium^[180]. The first solution corresponds to a spherical dispersion relation, like the case of homogeneous medium, and is called ordinary. The second solution yields usually an ellipsoid-like dispersion relation and is called extraordinary^[180]. The phenomenon that describes the coexistence of two different contours, obeying two different dispersion relations, is widely known as “*birefringence*”^[180]. In this section, we demonstrate that electron reflection and, both, positive and negative refraction could be seen at the lateral interface between homogeneous and nanostructured metal surfaces.

Owing to the numerical limitation of the electron boundary element method (EBEM) for large geometries and their associated huge discretization points, all real space simulations in this section will be calculated by Gaussian electron beam of $\omega_0 = 5.25$ nm (i.e., $\sim 2\lambda_{Cu}$, where λ_{Cu} is the Cu(111) wavelength^[181] at the concerned energy

4. RESULTS AND DISCUSSION II

contours). With this value of the beam waist the balance between simulation accuracy and speed is achieved.

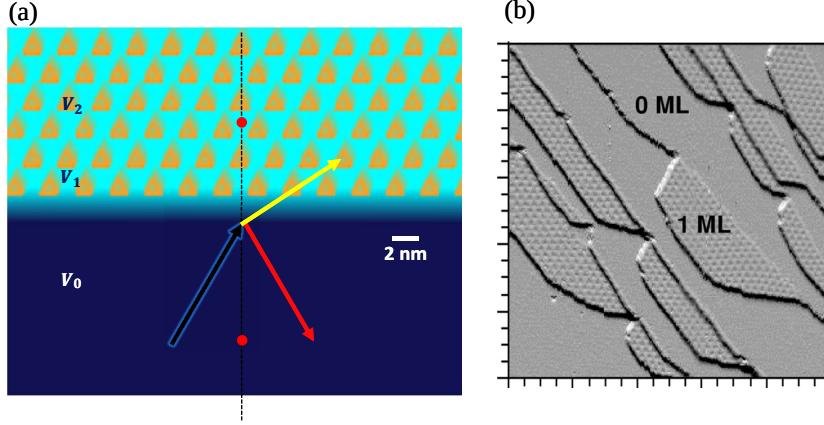


Figure 4.14: Homogeneous-Nanostructured real-space geometry - (a) The geometry used in EBEM and EPWE calculations. The light blue and dark blue areas represent the moiré-1 ML Ag/ Cu(111) with potential barrier V_1 and the clean Cu(111) surface with potential barrier V_0 , while the yellow triangle represents the dislocation lattice characteristic for the annealed moiré-1ML Ag/Cu(111) with potential barrier V_2 . The arrows represent the incidence (solid black), the refracted (solid yellow), and the reflected (solid red) beams. (b) STM image corresponding to the 0.4 ML Ag films on Cu(111), i.e. the Cu-Ag/Cu real interface . Adapted from Ref. ^[141].

The nanostructured metal surface in the present case is a 2D lateral superlattice made by combination of two noble metal surfaces with large lattice mismatch^[19]. Such combinations commonly self-assemble into moiré superstructures. The 1 monolayer (ML) Ag/Cu(111), in particular, exhibits an irreversible transformation from such moiré pattern into hexagonal lattice of dislocation network (periodicity ~ 2.4 nm)^[53,173]. The (8×8) moiré pattern is obtained by low temperature deposition, while the (9.5×9.5) network of triangular dislocation loops within the Cu(111) substrate occurs after annealing to 300 K to allow for relaxation^[56,140].

The present investigation deals with the interface between this “*nanostructured*” 1 ML Ag/Cu(111) dislocation lattice and the “*homogeneous*” Cu(111) surface. Figure 4.14(a) depicts the real space geometry of such a homogeneous-nanostructured interface, which we used as an input in both EPWE and EBEM in order to obtain the band structure, the CESs, and real space simulations, respectively. The dark blue region defines the structureless Cu(111) surface, where the potential V_0 inside this region is

set to zero, and the lactic-yellow region is the 1 ML Ag/Cu(111) dislocation lattice with an effective potential $V = V_2 - V_1 = 650$ meV, where V_1 and V_2 are the potential barrier outside and inside the dislocation triangles. The problem under investigation is the refraction (yellow arrow) and reflection (red arrow) at the interface between the two systems for an incident electron beam (black arrow) with four selected electron energies and different incidence angles.

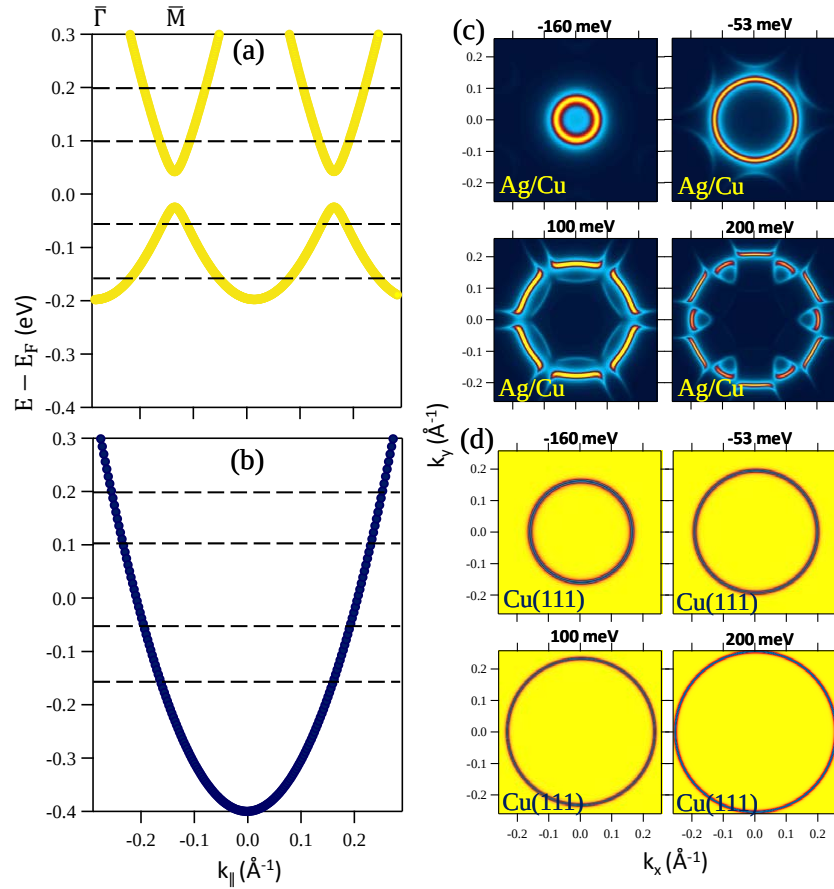


Figure 4.15: The dislocation-1 ML Ag/Cu(111) and the clean Cu(111) surface states - (a,b) Calculated band structures using EPWE for the dislocation-1 ML Ag/Cu(111) (yellow) and the clean Cu(111) (blue) along the $\bar{\Gamma}\bar{M}$ direction. (c) Simulated Constant energy surfaces taken at the black dashed lines in (a,b), i.e. at -160 meV, -53 meV, 100 meV and 200 meV, for both dislocation-1 ML Ag/Cu(111) in (c) and Cu(111) in (d).

In Fig. 4.15(a-b), the calculated band structure along the $\bar{\Gamma}\bar{M}$ directions for 1 ML Ag/Cu(111) and Cu(111) systems are presented. Calculations are performed by EPWE for each system separately. The effective mass of both media was set to (0.41

4. RESULTS AND DISCUSSION II

m_e), which is been reported experimentally for both systems^[20]. Likewise the values of the various potentials being used are ($V_0=0$, $V_1=150$ meV, and $V_2=800$ meV). In Fig. 4.15(b) the free-electron parabolic band structure, without any gaps, is seen and corresponds to the Cu(111) surface state. The energy axis is shifted by the reference energy 400 meV, i.e., the Fermi energy of Cu(111) surface state. The $\bar{\Gamma}$ -point energy of the 1 ML Ag/Cu(111) band structure presented in Fig. 4.15(a) is found at 200 meV, and the $\bar{M}$ -point gap size amount to ~ 60 meV, in agreement with previous experiments and calculations^[53,55,75,146,173]. After the upper edge of $\bar{M}$ -gap the first umklapp band is folded-back into the first Brillouin zone with negative dE/dk slope (i.e., negative group velocity). This means that the radius of the corresponding CES shrinks for increased electron energy, and the group velocity is pointing inwards. The coexistence between the two surface states is experimentally possible at sub-monolayer Ag coverage. In figure 3 of Ref.^[141], the experimental photoemission intensity along the $\bar{\Gamma}MK\bar{\Gamma}$ direction for ~ 0.6 ML Ag/Cu(111) is presented. Both the Cu(111) and the 1 ML Ag/Cu(111) surface states coexist with their binding energy at 400 meV and 200 meV, respectively. At such a reduced coverage, the geometry defined in Fig. 4.14(b) is seen in STM measurements^[141].

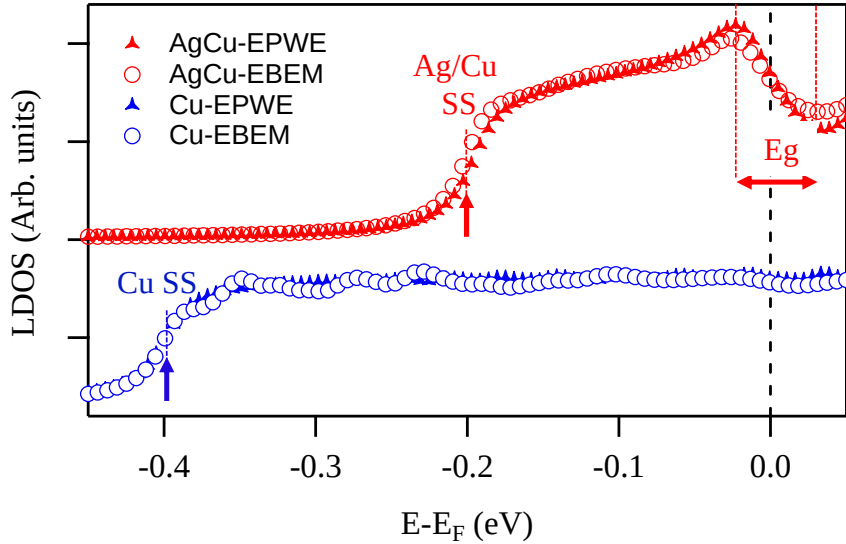


Figure 4.16: The dislocation-1 ML Ag/Cu(111) and the clean Cu(111) LDOSs - Calculated LDOS at Ag/Cu (red) and Cu(111) (blue) using EPWE (triangle symbols) and EBEM (circle symbols). In EBEM, the LDOS is calculated at the red dots in Fig. 4.14.

Electron reflection, positive or negative refraction, and collimation shall be investi-

gated at four different energies (dashed lines in (a-b)), i.e. at -160 meV, -53 meV, +100 meV, and +200 meV. The corresponding constant energy surfaces (CESs) are calculated by EPWE and depicted in Fig. 4.15(c-d) for the two separate systems. At the four fixed energies, all CESs of a clean Cu(111) exhibit circular shapes with different radii shown in (d), in correspondence with the parabolic dispersion relation of Cu(111) presented in (b). In contrast, the shape of the CESs of Ag/Cu(111) at the same fixed energies are highly energy dependent. The first CES taken at -160 meV presents a circular energy contour when the replicas of neighboring zone are quite faraway, Fig. 4.15(c). At slightly lower binding energy, e.g., -53 meV, the main CES band is still circular, and the repeated umklapp bands of the circular CES show up outside, but nearby, the first Brillouin zone at this energy contour. At lower binding energies, the umklapp bands enter the first BZ, leading to highly featured CESs, with inner umklapp contours with negative group velocities ($\frac{dE}{dk}$) inside first BZ boundaries as shown in (a) for the CES which taken at +100 meV. This inner umklapp contour exhibit a hexagonal shape with low intensity compared to the main outer band. Notice, the CES associated with the main band shows 6-fold nearly flat segments. Similarly, the CES taken at +200 meV consists of 6-flat allowed states of the main surface-state band of Ag/Cu(111) superlattice, which could have implication for electron collimation as we show by the end of this chapter.

Before we present a real space representation of reflection-refraction for the finite geometry presented in Fig. 4.14 using EBEM, we ensure that the EPWE calculated band structures of each region are unaltered in the finite geometry. The Local Density of States (LDOS) obtained from EPWE are presented in Fig. 4.16 for Cu(111) (blue) and 1 ML Ag/Cu(111) (red) systems. The corresponding LDOS for the finite geometry are calculated at the two black dots in Fig. 4.14. The agreement between LDOS for the extended and finite geometries is sufficient, i.e. no shift of the surface state energy neither variation in the $\overline{M}$ -point gap size or energetic position.

4.4.1 The 1 ML Ag/Cu(111) Main Circular Contours: Positive Refraction

The first case to be investigated in this section is the reflection-refraction of electrons with binding energy of -160 meV, at the interface between Cu(111) and 1 ML Ag/Cu(111). The calculated band structures along the $\overline{\Gamma M}$ directions for 1 ML

4. RESULTS AND DISCUSSION II

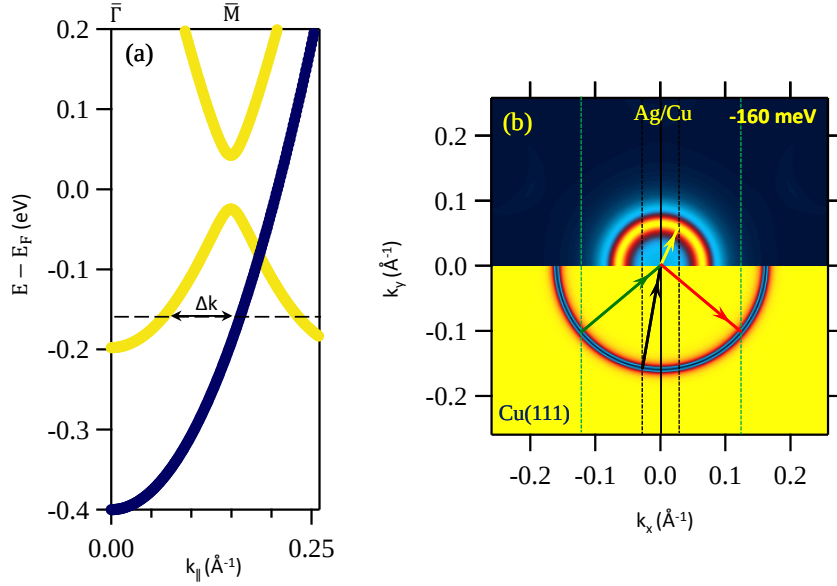


Figure 4.17: The combination of dislocation-1 ML Ag/Cu(111) and the clean Cu(111) surface states and the corresponding wavevector space formalism, $E_B=160$ meV - (a) The combination of calculated band structures in Fig. 4.15(a,b), Δk represent the difference between two CESs radii, i.e. between two wavevectors. (b) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B=160$ meV. Upper and lower halves of (b) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid black and green), refracted angle (solid yellow), and reflected angle (solid red) beams. The dotted lines defines the direction of the interface confined evanescent wave (construction line).

Ag/Cu(111) and Cu(111) systems are combined in Fig. 4.17(a). The required momentum difference of the two surface states (Δk) at the concerned energy, i.e., -160 meV, is clearly satisfied due to the energy shift between the surface states band minima. The top and bottom halves in (b) are the corresponding CESs for 1 ML Ag/Cu(111) and Cu(111) systems, respectively. The circular CESs of the two systems indicates that they are proceeding in this energy regime as uniform isotropic materials^[157,172]. These circular contours with have two different k -radii, k_{inc} or $k_{Cu}=0.158 \text{ \AA}^{-1}$ and k_{ref} or $k_{Ag/Cu}=0.063 \text{ \AA}^{-1}$ for Cu(111) and 1 ML Ag/Cu(111) systems, respectively. Therefore, the corresponding wavelengths are $\lambda_{inc}=3.9 \text{ nm}$, $\lambda_{ref}=9.8 \text{ nm}$, and the critical angle is $\sim 24^\circ$, as calculated by Eq. 4.3. The wavevector space formalism in Fig. 4.17(b) elucidates that an incoming electron beam (black or green arrow) originating from the Cu(111) surface shall undergo refraction (yellow arrow) and reflection (red

arrow) at the interface with directions and intensities defined by the Snell's law and Fresnel coefficients^[176], respectively.

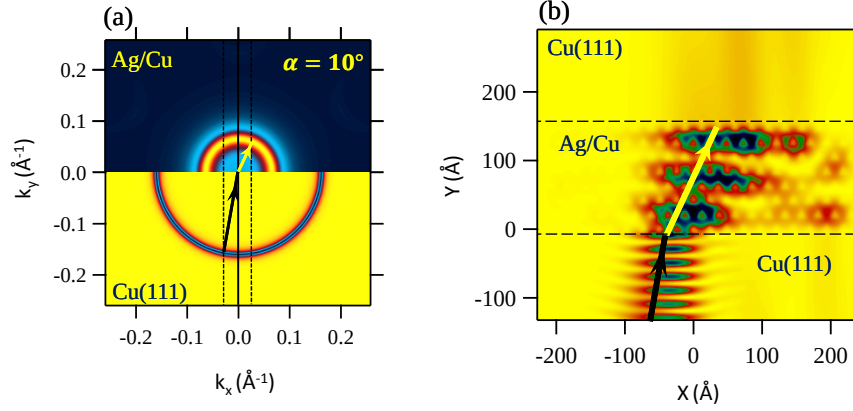


Figure 4.18: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -160$ meV, $\alpha_{inc} = 10^\circ$ - (a) Wavevector-space analysis of the refraction at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -160$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid black), refracted angle (solid yellow). (b) Real-space simulations of electron refraction at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 10^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -160$ meV.

At small incidence angles, e.g., $\alpha = 10^\circ$, and away from the critical one ($\sim 24^\circ$ for this case), the refraction of an electron beam with a binding energy of -160 meV and a waist width of 5.25 nm is detailed in Fig. 4.18. The wavevector space analysis depicted in (a), shows that the incidence beam (black arrow) starting from the left construction line intersects the CES of the incoming medium Cu(111) and is refracted after hitting the interface, due to the smaller isotropic contour of refractive medium (Ag/Cu), by $\sim 26^\circ$ (yellow arrow), according to the momentum conservation and Snell's law. Here the vertical left and right construction lines are at the same distance from the $\bar{\Gamma}$ -point, i.e., the parallel components of the black and yellow arrows are clearly conserved. The corresponding real space simulation as obtained by EBEM is presented in (b). It is clear that most of the intensity of the incidence beam (underneath black arrow) is transmitted i.e., refracted by the momentum conservation angle (underneath yellow arrow). Notice that the perfect match between the positions of the black and yellow arrows and the incidence and refracted beam intensities, which indicates the good correspondence between reciprocal and real space analysis. Additional intensity is seen

4. RESULTS AND DISCUSSION II

at the right side of Ag/Cu slab, which is due to the beam reflection at the second interface. In contrast to homogeneous interfaces, an enhanced intensity modulation is seen, where some of the intensity is missing at the triangular position due to the local scattering potential.

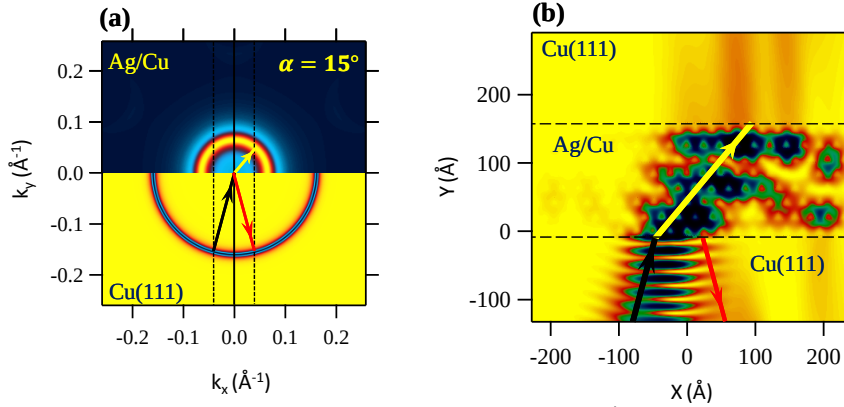


Figure 4.19: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -160$ meV, $\alpha_{inc} = 15^\circ$ - (a) Wavevector-space analysis of the refraction at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -160$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid black), refracted angle (solid yellow). (b) Real-space simulations of electron refraction at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 15^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -160$ meV.

According to Fresnel, the reflection coefficient is expected to increase with the incidence angle of the electron beam. Figure 4.19 presents the reflection-refraction of the same electron beam at the same interface conditions, except that the incidence angle is changed to 15° (black arrow in (a)). The corresponding angle of refraction is changed to $\sim 40^\circ$ (yellow arrow) according to the construction line defining parallel-momentum conservation in the wavevector space analysis depicted in (a). Likewise, beam reflection also takes place (red arrow) with the direction given from the momentum intersection between the CES of Cu(111) and the right construction line (black dotted line). The real space simulation was obtained by EBEM for a Gaussian electron beam with incidence angle of 15° , and is shown in Fig. 4.19(b). Most of the intensity of the electron beam hitting the interface by 15° is transmitted/refracted (yellow arrow) through the 1 ML Ag/Cu slab with a refraction angle of 40° , whereas the remaining intensity is reflected back (red arrow) to the incoming medium Cu(111). Notice that

the first interface reflection was not visible for the smaller incidence angle, i.e., 10° , in accordance with Fresnel formulation, whereas second interface reflections were evident for both incidence angles.

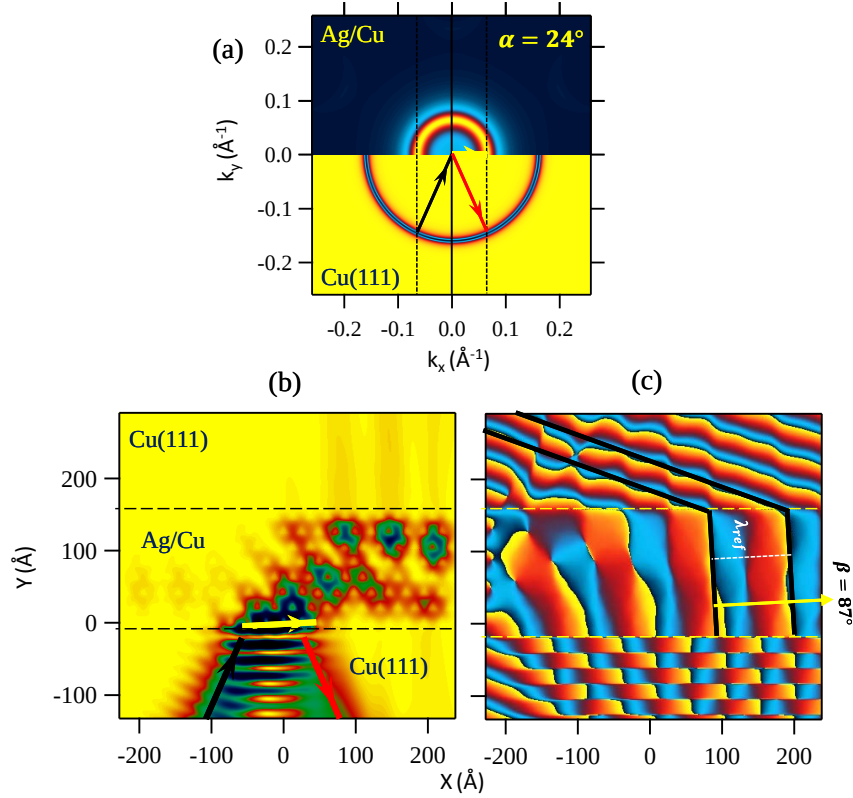


Figure 4.20: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -160$ meV, $\alpha_{inc} = 24^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -160$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with critical angle (black), refracted angle (yellow), and reflected angle (red) beams. (b) Real-space simulations of electron refraction/reflection at this lateral interface, and (c) its corresponding phase propagation. Calculation are performed with a Gaussian electron beam of critical angle $\alpha_{inc} = 24^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -160$ meV.

Figure 4.20 depicts the case of critical angle incidence which for the concerned binding energy and according to Eq. 4.3, is $\sim 24^\circ$. The wavevector diagram analysis in (a) reveals that the electron beam of this incidence angle (black arrow) will be refracted parallel to the interface (yellow arrow) according to the parallel-momentum conservation. In (b), the corresponding real space simulation is presented for an elec-

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tron Gaussian beam of $\omega_0 = 5.25$ nm (black arrow). The transmitted intensity of this beam undergoes a small refractive angle (yellow arrow), and the remaining intensity is reflected back (red arrow) to Cu(111). In fact, the transmitted intensity is broadened through the Ag/Cu slab, rendering the determination of the refraction angle rather difficult.

To precisely determine the direction of the refracted beam, the corresponding phase analysis presented in (c) and obtained by EBEM for a plane wave source, serves as a better alternative. The black lines defined the wavefronts, and the refraction angle as defined the perpendicular to the phase (yellow arrow) was found to be 87° (i.e., almost parallel to the interface). Actually, the apparent transmitted intensity in Fig. 4.20(b) originates from the beam divergence effect discussed earlier, where the Gaussian beam waist width in this case is $\omega_0 = 1.5\lambda_{Cu} = 0.6\lambda_{Ag/Cu}$.

In Fig. 4.21 the case of beam incidence beyond the critical angle is discussed. A total internal reflection^[182] takes place for an incidence angle of 40° . The wavevector space diagram in (a) confirms that the construction lines (dotted black) for the incidence beam (black arrow) do not intersect the CES of the refractive medium (Ag/Cu). This is because the parallel component of the incidence wavevector is greater than the radius of the refractive medium energy contour, i.e., the CES of Ag/Cu slab. Wherefore, the conservation construction line makes, instead, an intersection in the right side with allowed band of the incidence medium, and therefore all the beam intensity must be reflected back to the Cu(111) medium (red arrow). The EBEM real space simulation in (b) clearly showed that, the incoming electron beam with 40° incidence (black arrow) is totally reflected by the same angle (red arrow) into the Cu(111) surface.

From previous results and discussions, the reflection-refraction behavior at the lateral interface between 1ML of Ag/Cu and Cu(111) for electron beam of a binding energy of -160 meV, is practically the same as the homogeneous-homogeneous interfaces. This is because the CES in both cases are circular at the concerned energy, in addition to the absence of the Ag/Cu umklapp bands in the momentum area spanned by the Cu(111) contour at the respective energy. If these two conditions are fulfilled at other binding energies, the behavior of electrons at such homogeneous- nanostructured interfaces do not differ from the corresponding homogeneous-homogeneous interfaces.

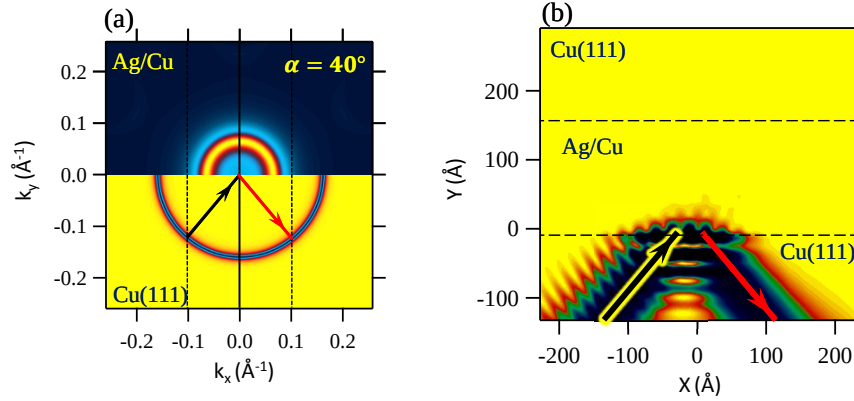


Figure 4.21: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -160$ meV, $\alpha_{inc} = 40^\circ$ - (a) Wavevector-space analysis of the refraction at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -160$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid black), refracted angle (solid yellow). (b) Real-space simulations of electron refraction at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 40^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -160$ meV.

4.4.2 The 1 ML Ag/Cu(111) Main & Outer-Umklapp Circular Contours: Negative Refraction

The intrinsic fingerprint of the 2D “*nanostructured*” media (hexagonal dislocation lattice in our investigation) will be seen in the following three subsections. The electronic crystals (including periodic nanostructure materials) cases are, in general, very complicated because of the repetition of the propagation modes in reciprocal space^[183]. This will become evident when the electron refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface is investigated for electron energies just below the lower edge of the $\overline{M}$ -point gap and at all lower binding energies. The wizard effect of the superlattice replica bands (umklapps) takes place in this energy regime^[82]. This section concerns the reflection-refraction of electron beam with a binding energy of -53 meV, i.e., right below the $\overline{M}$ -point gap of Ag/Cu surface state.

Figure 4.22(a) shows the calculated band structures using EPWE for the dislocation-1 ML Ag/Cu(111) (yellow) and the clean Cu(111) (blue) along the $\overline{\Gamma M}$ directions. The intrinsic momentum difference between the two surface states Δk at the concerned energy (-53 meV) is clearly satisfied, due to the energy offset between two surface states band minima. One can see that, at this energy, the black arrow intersects three bands

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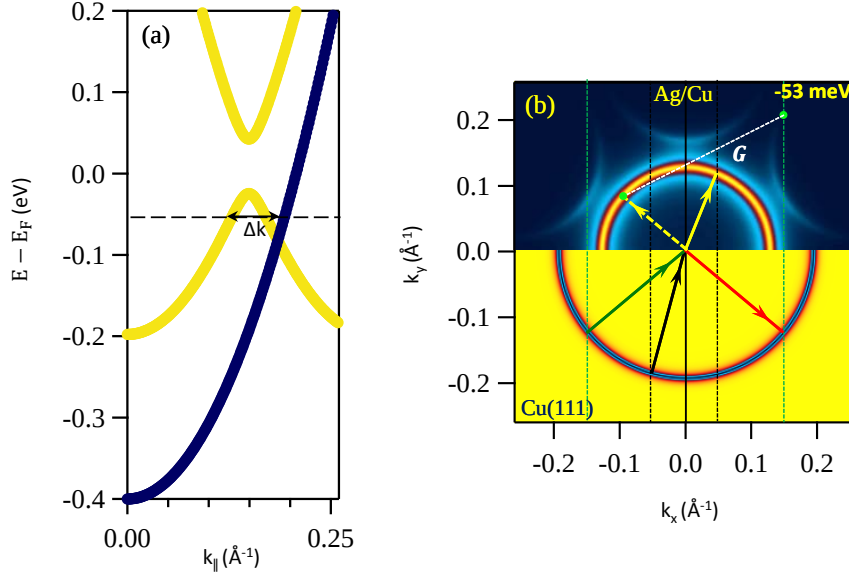


Figure 4.22: The combination of dislocation-1 ML Ag/Cu(111) and the clean Cu(111) surface states and the corresponding wavevector space formalism, $E_B = 53$ meV - (a) The combination of Calculated band structures using EPWE for the dislocation-1 ML Ag/Cu(111) (yellow) and the clean Cu(111) (blue) along the $\overline{\Gamma M}$ direction, Δk represent the difference between two CESs radii, i.e. between two wave vectors. (b) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 53$ meV. Upper and lower halves of (b) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid black and green), refracted angle (solid yellow), negatively refracted angle (dotted yellow), and reflected angle (solid red) beams. The dotted lines defines the direction of the interface confined evanescent wave (construction line).

with different momentum, i.e., different (Δk). Two of these bands correspond to the dislocation Ag/Cu (yellow), while the third band belongs to the Cu(111) medium (blue). It is clear that the k -wavevector of Cu(111) surface state at this energy is greater than the k -wavevector of Ag/Cu umklapp band i.e., the umklapp bands are spanned by the Cu(111) contours and could participate to electrons propagation through the interface^[82]. The possible manifestation of such umklapp band is best seen from Fig. 4.22(b), where the CESs of the 1 ML Ag/Cu(111) and Cu(111) media are presented in the top and bottom halves, respectively. According to wavevector-space analysis, the conservation of the parallel component of the incident wave vector (vertical dotted black line) at any incidence angle smaller than the critical one (solid black arrow) just intersects the main band of first BZ. This shall lead to positive refraction at the

4.4 Homogeneous-Nanostructured Interfaces

interface, as described for the high energy electrons in section 4.4.1. At high incidence angles (green arrow) greater than the critical angle, only umklapp bands at the second BZ satisfy the parallel-momentum conservation (right green point), and the direction of the refracted beam is found by folding it back to the first BZ (left green point) by the appropriate reciprocal lattice vector G (dashed white line). After folding, the direction of the refracted beam follow dashed arrow, i.e., negative refraction takes place^[184]. This finding has no analogues counterpart in homogeneous-homogeneous interfaces, and could not be described by the simple form of the Snell law presented in Eq. 4.3. In fact, Snell's law in the presence of hexagonal periodic structures takes the form^[82];

$$|k_{inc}| \sin(\alpha) + \frac{G}{2} = |k_{ref}| \sin(\beta) \quad (4.4)$$

where G is a reciprocal lattice vector, and is given by; $G = \frac{4\pi n}{a}$ or $G = \frac{4\pi n}{\sqrt{3}a}$ if the interface is oriented along the $\overline{\Gamma K}$ or $\overline{\Gamma M}$, respectively, where $n = 0, \pm 1, \pm 2$, etc. Therefore, umklapps at higher order BZ participate in the refraction process, particularly leading to negative refraction when their CES are circular and approaching the BZ boundaries. We stress that this type of negative refraction, also called anomalous refraction^[82], is fundamentally different from the unusual phenomenon of negative refraction predicted by the Russian physicist V. Veselago in 1967^[185], which requires a medium with a negative refractive index, in optics, or an energy band with negative group velocity for electrons.

In order to investigate the reflection/refraction behavior in the presence of such umklapp bands, as we show in the coming sections, we shall calculated the CESs at different energies, obtain the bands slope (sign of $\frac{dE}{dk}$) from band structure calculations, and subsequently apply the conservation of the parallel component of wavevectors (represented by the construction line). With such a systematic approach, the direction(s) of the refracted beam(s) could be easily determined. We further categorize the refracted beams according to their occurrence within the BZ. For example, a refracted beam corresponding to an intersection lying in the first BZ, i.e., need not to be folded back, is a zeroth-order beam. In the following we shall call this beam “*transmitted*”. On the other hand, beams stemming from intersections at the higher zones, which were subsequently folded back to the first zone, are classified as higher-order beams.

The two circular contours in Fig. 4.22(b) clearly have different k -radii, k_{inc} or $k_{Cu} = 0.19 \text{ \AA}^{-1}$ and k_{ref} or $k_{Ag/Cu} = 0.129 \text{ \AA}^{-1}$ for Cu(111) and 1 ML Ag/Cu(111) systems, respectively. Therefore, the corresponding electron wavelengths are $\lambda_{inc} = 3.2$

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nm, $\lambda_{ref} = 4.8$ nm, and the critical angle (for the main band) is $\sim 42^\circ$ as calculated by Eq. 4.3. The wavevector space formalism in (b) elucidates that an incoming electron beam (black or green arrow) from the Cu(111) surface shall undergo positive or negative refraction (solid or dotted yellow arrow) and reflection (red arrow) at the lateral interface between Cu(111) and Ag/Cu, with directions and intensities defined by the more-general Snell's law (Eq. 4.4.2) and Fresnel coefficients, respectively.

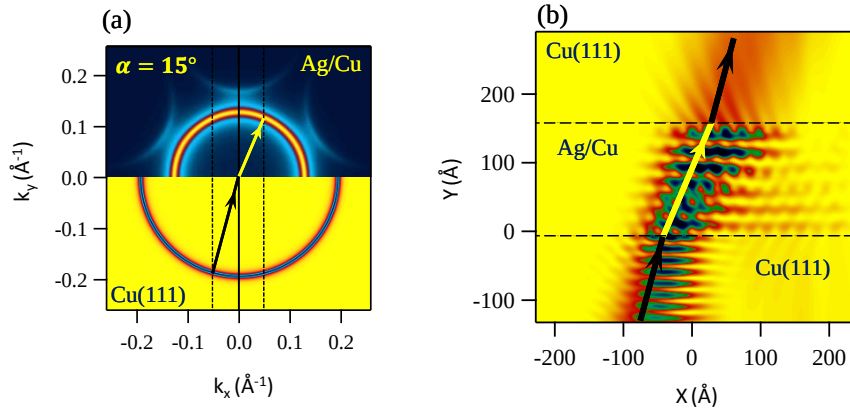


Figure 4.23: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -53$ meV, $\alpha_{inc} = 15^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -53$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (black), and refracted angle (yellow). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 15^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -53$ meV.

At small incidence (e.g., 15°) away from the critical angle ($\sim 42^\circ$ in this case), the effect of higher-order zones vanished as seen in the schematic wavevector-space formalism presented in Fig. 4.23(a). The incoming beam (black arrow) starts from the momentum intersection of the construction line (left dotted line) with CES of incoming medium, i.e., Cu(111). The construction line that sets the parallel-momentum conservation constrains (right dotted line) intersects the main band only of the Ag/Cu medium. Therefore, the zeroth-order beam (yellow arrow) is positively refracted into the refractive medium (Ag/Cu) by 22° according to this parallel-momentum conservation and the simple form of the Snell's law (Eq. 4.3). The same behavior shall be observed for any incidence angle below the critical angle. The difference between

these angles would only show up in the intensity of the reflected beams, which should increase by increasing the incidence angle according to Fresnel, as discussed earlier.

The real space simulation for the 15° incidence beam is displayed in Fig. 4.23(b) and is obtained by electron boundary element method EBEM for an electron beam of binding energy -53 meV and waist width of $\omega_0 = 5.25$ nm. Most of the intensity is transmitted to Ag/Cu medium, i.e., refracted by the angle that satisfy the momentum conservation (yellow arrow). The intensity of the reflected beam is not detectable, at such a small incidence angle.

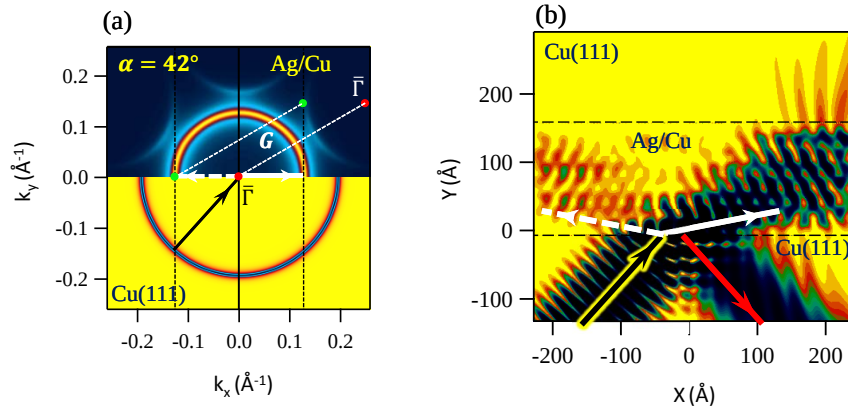


Figure 4.24: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -53$ meV, $\alpha_{inc} = 42^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -53$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. Red points represent the $\bar{\Gamma}$ -points in brillouin zones, while green points represent the equivalent intersection points in replica zones. The arrows represent electron beams with incidence angle (black), and refracted angle (yellow). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 42^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -53$ meV.

The intrinsic effect of outer-umklapp bands will show up in the electron refraction at beam incidence equal to or above the critical angle. In this incidence angles range the construction lines intersect the outer-umklapp contours, and therefore they participate in electron refraction. Figure 4.24 presents the case of the critical angle incidence, i.e., $\sim 42^\circ$, where the construction lines intersect contours within the first BZ. The wavevector-space analysis in (a) schematically shows the incoming electron beam (black arrow) which propagate from the Cu(111) medium with the critical an-

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gle incidence. The parallel-momentum conservation construction line (right dotted) intersects two bands. The first intersection with the main band (first BZ band) leads to the transmitted (zeroth-order) beam to be refracted parallel (solid white arrow) to the lateral homogeneous-nanostructured interface of the Cu(111) and 1ML of Ag/Cu in accordance with Snell law (Eq. 4.3). On other hand, the second intersection with the outer-umklapp band (close to BZ boundary) at the right green point leads to negatively refracted higher-order beam^[186], i.e., it propagates in the opposite direction to the zeroth-order beam (white dotted arrow). The direction of the higher-order beam is determined by folding the intersection point (right green) back to the equivalent momentum position in the first BZ (left green point) by the appropriate reciprocal lattice vector G (white dashed line). According to wavevector-space analysis just discussed, two positive and negatively refracted beams are expected to propagate parallel to the interface.

The corresponding reflection-refraction real space simulation of an electron Gaussian beam hitting the lateral interface of Cu(111) and 1ML of Ag/Cu, at the critical angle, is depicted in Fig. 4.24(b). The binding energy of the electron beam was -53 meV and a waist width of $\omega_0 = 5.25$ nm was used. The real space image contains four regions with detectable electron intensity. These are, the incoming beam (black arrow), the reflected beam (red arrow), the transmitted, i.e., positively refracted, zeroth-order beam (solid white arrow), and the negatively refracted higher-order beam (dotted white arrow). One immediately notice that the refraction angles for the two beams (white arrows) are not precisely 90° , and therefore the beams are not exactly parallel to the interface as predicted from the wave-vector space analysis. This deviation in the refractive angles is of the same value as discussed in Fig. 4.7 for the homogeneous interfaces, and therefore is mainly due to the Gaussian beam waist width, which needs to be larger. From Fig. 4.24(b) it is clear that the intensity of zeroth-order beam is greater than the intensity of higher-order beam. This is found to be in correspondence with the intensity difference between the main and outer-umklapp bands, respectively, shown in the CES presented in (a).

Right above the critical angle, e.g., at incidence angle of 50° , the electron reflection-refraction analyses are given in Fig. 4.25 by EPWE-EBEM numerical techniques. The schematic wavevector-space formalism is presented in (a). The electron beam of 50° incidence (black arrow), start from the momentum intersection of the left construction line (black dotted line) with Cu(111) circular contour. Since, the component of

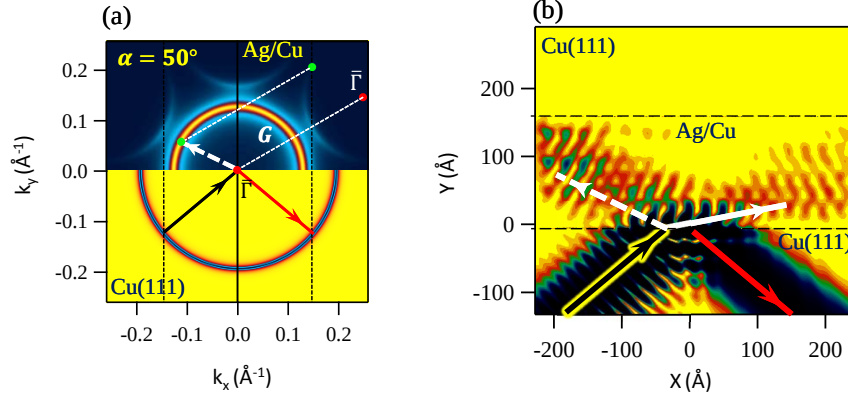


Figure 4.25: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -53$ meV, $\alpha_{inc} = 50^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -53$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. Red points represent the $\bar{\Gamma}$ -points in brillouin zones, while green points represent the equivalent intersection points in replica zones. The arrows represent electron beams with incidence angle (black), and refracted angle (yellow). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 50^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -53$ meV.

parallel-momentum conservation is greater than the k -radius of the Ag/Cu energy contour of, the right construction line only intersects the umklapp bands at the right green point. Hence no zeroth-order transmission due to the main band is expected, and the electron beam must be internally reflected (red arrow) in accordance with Eq. 4.3. In contrast, the outer-umklapp band lead to a higher order transmission exhibiting negative refraction (dotted white) after back-folding the left green point by the appropriate reciprocal lattice vector G (white dashed line). By EBEM, the real space simulation in (b) confirms the previous analysis. The Gaussian electron beam (black arrow) of, an incidence angle of 50° (i.e., close to the critical angle), binding energy of -53 meV, and waist width of $\omega_0 = 5.25$ nm is split into two beams at the interface. One is partially reflected (red arrow) and constitutes most of the beam intensity, and the other is negatively refracted^[187] (dotted white arrow). Traces of positively refracted beam (solid white arrow) which belongs to the zeroth-order transmission by the main band are also observed. It is clearly from simulation that the intensity of zeroth-order is, unusually, lower than the higher-order beam. In fact, the occurrence of this zeroth-order intensity is just related to the numerical limitation imposed by the limited waist width employed

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and the proximity to the critical angle as previously shown Fig. 4.8 for homogeneous interfaces. In other words, with such a relatively small beam waist width, the beam has a finite k -width and, therefore, different incidence angles are simultaneously sampled.

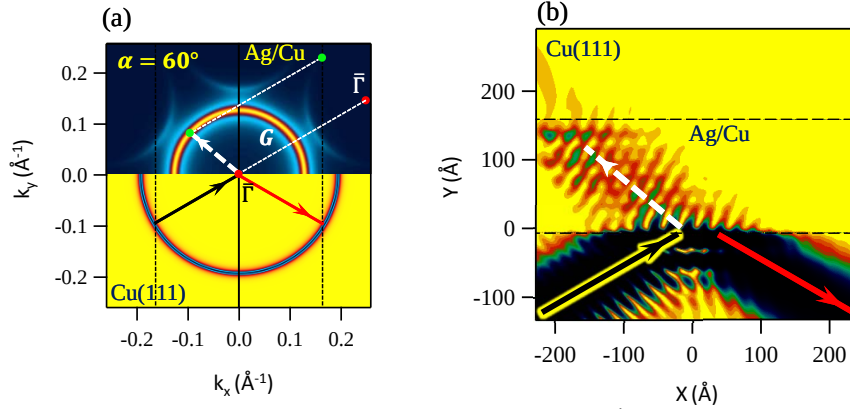


Figure 4.26: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = -53$ meV, $\alpha_{inc} = 60^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = -53$ meV. Upper and lower halves of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. Red points represent the $\bar{\Gamma}$ -points in brillouin zones, while green points represent the equivalent intersection points in replica zones. The arrows represent electron beams with incidence angle (black), and refracted angle (yellow). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 60^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = -53$ meV.

At larger incidence angles, this beam k -width effect is expected to vanish, and the main bands do not contribute to the refraction process. This is clearly shown in Fig. 4.26 which depicts the electron reflection-refraction at the same lateral interface for an electron beam of 60° incidence. At such a high incidence angles (i.e., faraway the numerical limitation), only outer-umkapp band at the second BZ satisfy the parallel-momentum conservation, and the direction of the refracted beam is found by folding it back to the first BZ by the appropriate reciprocal lattice vector (G), according to Eq. 4.4.2. In (a) the wavevector-space formalism demonstrates this arguments, the construction line of the parallel-momentum conservation (right dotted) is extremely away from the first BZ contour of Ag/Cu dislocation lattice. The construction line intersects the outer-umklapp band at the right green point, and should be folded back to the left green one of the first zone to determine the negative refractive higher-order beam direction (dotted white arrow). On the other side, the main band only causes

an internal reflected beam (red arrow). The real space simulation image (b) reveals an excellent agreement with the wavevector space analysis. The incoming Gaussian electron beam (black) of 60° incidence, and binding energy of -53 meV is hitting the lateral interface of Cu(111) and Ag/Cu. In addition to the reflected beam (red arrow), a transmitted “*negatively*” refracted beam (white dashed arrow) is also seen. The intensity of the zeroth-order beam (positive refraction) is totally vanished.

In conclusion, using combined wavevector-space analysis (EPWE) and real-space simulations (EBEM) we showed that outer-umklapp bands associated with higher order zones have great influences on the electron refraction at the lateral interface of Cu(111) and Ag/Cu. More specifically, they open channels for negatively refraction beams, which are otherwise not allowed. The directions of those negatively refracted beam are not given from the simple Snell law, but rather from the more general formulation of Eq. 4.4.2. Generally speaking, when the construction lines defining the parallel-momentum conservation intersect isotropic umklapp contours, with positive group velocity, negative refractions are expected.

4.4.3 The 1 ML Ag/Cu(111) Main & Inner-Umklapp Hexagonal Contours: Negative Refraction and Superprism Effect

As discussed earlier, there are several differences between the homogeneous and periodic nanostructured media. Contrary to the simple isotropic dispersions characteristic for homogeneous media, one of the most intrinsic influences of the periodic nanostructured materials is the diversity of its energy contours. The CESs of these materials are usually not circular and can have quite complicated shapes. Moreover, an important difference is that for the periodic crystal, the CESs repeat themselves periodically in reciprocal space. These repetitions might lead to anisotropic contours within the first BZ, particularly, at sufficiently high electron energies. One such Ag/Cu anisotropic contours will be used in the proceeding section, in order to investigate the electron reflection-refraction at the lateral interface between homogeneous Cu(111) and the periodic nanostructured 1 ML Ag/Cu(111).

Before we present these simulations, we would like to highlight some basics regarding the phase and group velocities, which are identical for homogeneous isotropic media. In nanostructured medium, the spatial average of the time-averaged Poynting vector defines the propagation direction rather than the Poynting vector itself. In fact, the

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latter has the same direction as the energy velocity, i.e., the group velocity. The energy velocity is $V_e = \frac{\langle S \rangle}{\langle U \rangle}$, where S is the poynting vector, U the energy density and $\langle \rangle$ denotes the spatial average within the unit cell of the time averaged quantities^[183]. Moreover, the directional relations between $\langle S \rangle$ and the wavevector (k) are not necessarily parallel or antiparallel. In general $\langle S \rangle$ is at an acute or obtuse angle with k . When the former is the case, $\langle S \rangle \cdot k > 0$, we have forward “*right-handed*” propagation. In the contrary case, $\langle S \rangle \cdot k < 0$, we have backwards “*left-handed*” propagation^[188]. The equality between the group and energy velocity for a general periodic three-dimensional system with non-dispersive components has been shown by Sakoda^[189]. Since the group velocity is given by $\bar{V}_g = \frac{dE}{dk}$, this means it is perpendicular to the slope at a certain k -point of the CESs and points either inwards or outwards depending on the slope. Specifically, the group velocity points outwards for a band with positive curvature and inwards for a band with negative curvature. Since group and energy velocity are equivalent, the group velocity vector (V_g) shows the direction of $\langle S \rangle$, i.e., the direction of the propagating electron.

Figure 4.27 presents the possible reflection-refraction that electrons undergo for the case were the CES features two anisotropic bands. These are the main and its inner-umklapp bands of the Ag/Cu dislocation lattice. The calculated band structures, using EPWE, for the dislocation-1 ML Ag/Cu(111) (yellow) and the clean Cu(111) (blue) along the $(\bar{\Gamma}\bar{M})$ directions, are given in Fig. 4.27(a). The intrinsic momentum difference of between these surface states k at the concerned energy (+100 meV) clearly exhibits two values (k_0, k_1) marked by black arrows for the main and its inner-umklapp bands, respectively. It means that, the k -wavevector of Cu(111) surface state at this energy is greater than the two k_0, k_1 -wavevectors of Ag/Cu (the main and its inner-umklapp bands), i.e., the electron reflection-refraction behavior stems from two different anisotropic bands. This is further clarified by using the wavevector-space analysis, presented in (b), which shows the CESs of (top) 1 ML Ag/Cu(111) and (bottom) Cu(111) systems. This formalism of the wavevector-space analysis is apparently different from the previous formalism where halves of the CESs are presented. Nonetheless, the two formalisms are identical, and this choice is only matter of convenient as we show next. The construction line (vertical dotted red line) for a given beam incidence (solid red arrow, k_{inc}) intersects both the main and inner-umklapp bands of the Ag/Cu dislocation lattice at two points. These intersections occur equally for the upper and lower halves of the CESs (red and white points). There are two choices for the wave

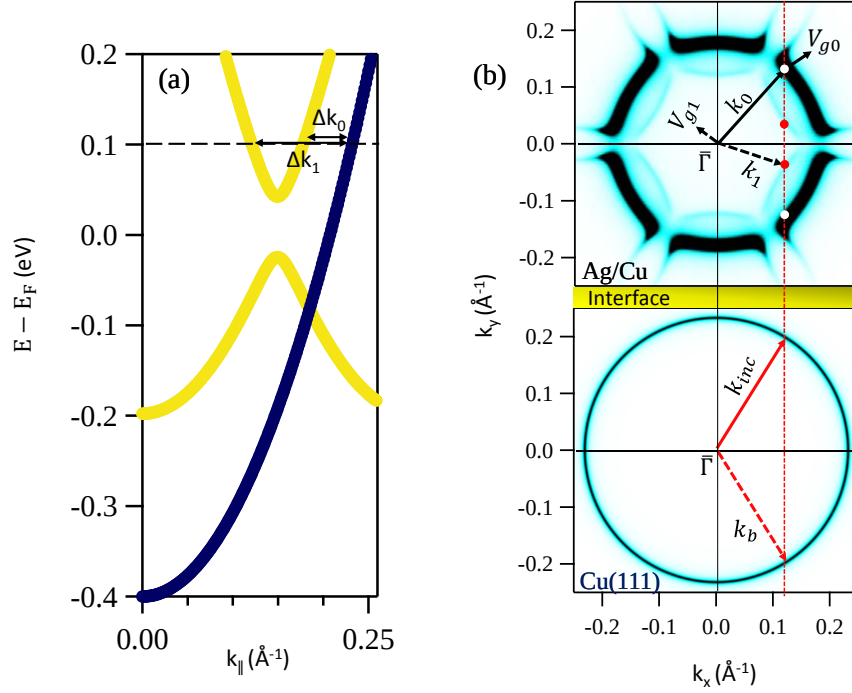


Figure 4.27: The combination of dislocation-1 ML Ag/Cu(111) and the clean Cu(111) surface states and the corresponding wavevector space formalism, $E_B = 100$ meV - (a) The combination of Calculated band structures using EPWE for the dislocation- 1 ML Ag/Cu(111) (yellow) and the clean Cu(111) (blue) along the $\bar{\Gamma}M$ direction, (k_0, k_1) represent the difference between two CESs radii, i.e. between Cu(111) wave vector and the wavevector of Ag/Cu main and umklapp bands, respectively. (b) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 100$ meV. Upper and lower contours of (b) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid red), refracted angle (solid black, V_{g0}), negative refracted angle (dotted black, V_{g1}), and reflected angle (dotted red) arrows. The dotted red lines defines the direction of the interface confined evanescent wave (construction line).

vectors inside the periodic nanostructured medium, either point towards upper intersection or towards lower intersection. The additional constraint comes from requiring the direction of flow inside the periodic medium to be causal, i.e., to propagate away from the source. For the main band case, the medium is right-handed, i.e., $S \cdot k$ is positive. This means that the appropriate choice for the wave vector inside the crystal (solid black arrow, k_0) would be the point at the upper half intersection (upper white point). This black arrow now defines the direction of the phase propagation. The direction of the power or intensity flow for this case with positive curvature should points

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outwards in the perpendicular direction to the tangent line at this point (upper white) i.e. in the direction of group velocity (small solid black, $(\bar{V}_{g0} = \frac{dE}{dk_0})$). The black arrows (long and short) clearly follow different directions, i.e., the phase and group velocities are different in contrast to the isotropic contours.

The opposite scenario occurs for the umklapp band which is characterized by a negative curvature and, therefore, the wavevector (dotted black arrow, k_1) which points towards the lower red point, is the one that gives a causal signal. In this case, and due to the negative curvature, the group velocity points inward and the refracted beam propagates in the perpendicular direction to the tangent line at this point, i.e., in the direction of the group velocity (small dotted black line, $(\bar{V}_{g1} = \frac{dE}{dk_1})$). The momentum conservation wavevectors (k_0 which is antiparallel to k_1) represent the wave front directions i.e., the direction of the phase velocities for the positive and negative refractions by the main and its inner-umklapp bands^[187], respectively. In the following analysis we shall draw k_1 in the antiparallel direction, via the equivalent point in the upper half of the CES. Finally, the direction of the reflected intensity through the wavevector conservation is represented by the dotted red arrow (k_b). These analysis steps shall be repeatedly used throughout this section for different incidence angles.

In Fig. 4.28 we investigate the electron reflection-refraction at the lateral interface between Cu(111) and 1 ML of Ag/Cu for electrons with binding energy of +100 meV and 20° incidence angle. The schematic wavevector-space analysis in (a) confirms that the signature of the inner-umklapp contours will show up at such a small incidence angles, contrary to the case of outer-umklapp bands where incidence beyond the critical angle was required. The incidence electron beam (solid red, $\bar{k}_{inc}$) propagates through the homogeneous metal surface Cu(111) and hits the interface at 20°. The construction line (dotted red line) intersects the circular CES of Cu(111) in the momentum position which define the direction and wavevector of the reflected beam (dotted red, $\bar{k}_b$). Inside the Ag/Cu medium, the construction line intersects both the first and higher order bands in the upper white and lower red points, respectively. The intersection with the main band, with positive group velocity, determines the direction of phase velocity of the zeroth-order refracted beam (solid black arrow, $\bar{k}_0$). The direction of power propagation, however, follows the direction of the group velocity (small solid black, $(\bar{V}_{g0} = \frac{dE}{dk_0})$, i.e., the perpendicular direction to the tangent line at this point (upper white point). Actually at such an incidence angle, the construction line intersects a sharp border of main band, which significantly drive the direction of power propagation, i.e., the group velocity, towards negative refraction. Furthermore, the intersection

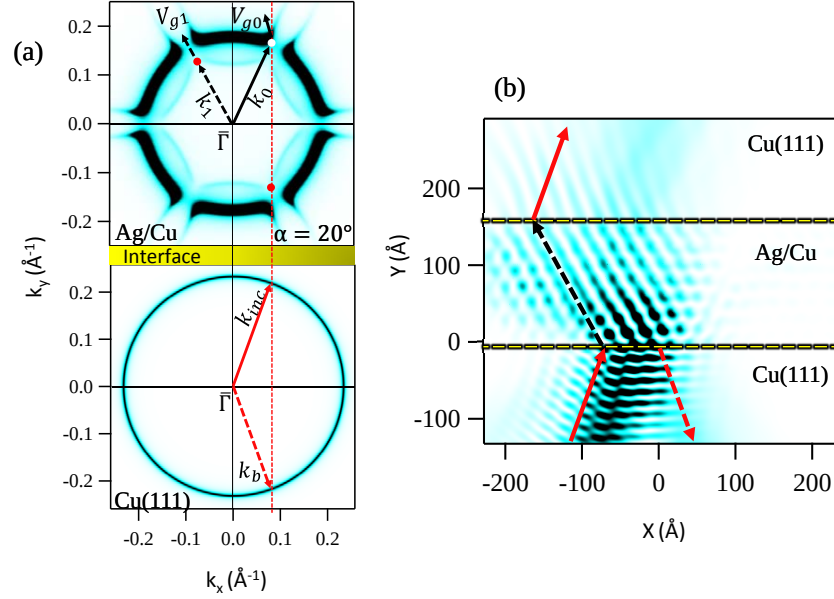


Figure 4.28: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = 100$ meV, $\alpha_{inc} = 20^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 100$ meV. Upper and lower contours of (a) are the 1 ML Ag/Cu and Cu(111) CEs, respectively. The arrows represent electron beams with incidence angle (solid red), refracted angle (solid black, $\bar{V}_{g0}$), negative refracted angle (dotted black, $\bar{V}_{g1}$), and reflected angle (dotted red) arrows. The dotted red lines define the direction of the interface confined evanescent wave (construction line). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 20^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = 100$ meV.

with the umklapp bands define the direction of the phase velocity of the higher-order refracted beam (dotted black, $\bar{k}_1$), whereas direction of the associated group velocity (small dotted black, $(\bar{V}_{g1} = \frac{dE}{dk_1})$ is given by the perpendicular to the tangent line at the upper red point. Accidentally, the combined curvature of the umklapp band and its negative group velocity both lead to a negative refraction^[190], nearly in the same direction as the zero-order refraction. Figure 4.28(b) shows the real-space simulation results of refraction-reflection behavior for an electron beam (solid red), made up of 101 plane waves, coming from the Cu(111), into the periodic nanostructured 1 ML Ag/Cu(111) through the lateral interface, at incidence angles of 20° . By EBEM, the real space simulation in (b) confirms the full agreement with wavevector-space analysis, where most of the transmitted intensity is negatively refracted for both the zeroth-

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order and higher-order beams (dotted black arrows). At the second interface, the electron beam is negatively refracted back to be parallel to the incidence beam direction (upper red arrow). The remaining electron intensity is reflected (dotted red arrow) with low intensity according to Fresnel. We stress that the negative refraction due to the main and umklapp bands are fundamentally different. The former represent a form of anomalous refraction due to the high anisotropy of the main band, and does not stand for metamaterials analogy. The umklapp bands, on the other hand, provide the electron negative refraction analogy to metamaterials, since the negative refraction is mainly derived by the negative group velocity, similar to the case of negative refractive index materials.

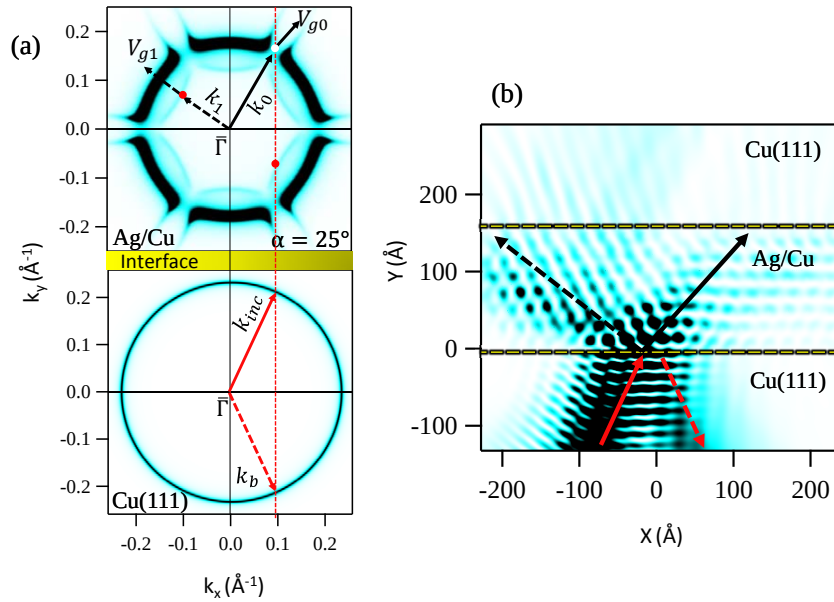


Figure 4.29: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = 100$ meV, $\alpha_{inc} = 25^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 100$ meV. Upper and lower contours of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid red), refracted angle (solid black, $\bar{V}_{g0}$), negative refracted angle (dotted black, $\bar{V}_{g1}$), and reflected angle (dotted red) arrows. The dotted red lines defines the direction of the interface confined evanescent wave (construction line). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 25^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = 100$ meV.

Due to the high anisotropy of both the main and umklapp bands, slight variations in the incidence angle could lead to significant change of the refraction behavior, which might therefore find applications in, e.g., electron superprisms^[191]. For example, by increasing the incidence angle only by 5° , i.e., $\alpha_{inc} = 25^\circ$, the refracted main beam follow different direction as discussed in Fig. 4.29. In this case the intersection of the parallel-momentum conservation line intersects the main band at the right upper gap of the first BZ band. This corner gap has a finite intensity and narrow positive curvature for allowed states. The wavevector-space formalism depicted in (a) shows an incidence electron beam (solid red, $\bar{k}_{inc}$) propagates through the homogeneous metal surface Cu(111) and hitting the interface by 25° . The construction line (dotted red) intersects the CES of Cu(111) in the momentum position which defined the direction of the reflected wavevector (dotted red, $\bar{k}_b$). Within the Ag/Cu slab, the same construction line intersects the first and higher-order zone bands in the white and lower red points, respectively. The intersection with the main band (white point) is actually in the center of its narrow gap where finite allowed states are present. The direction of the phase propagation associated with this zeroth-order beam is determined by this intersection (solid black, $\bar{k}_0$). However, the transmitted power of this band takes the direction of the group velocity at this momentum intersection point (small solid black, $(\bar{V}_{g0} = \frac{dE}{dk_0})$), i.e., the zeroth-order beam undergoes a positive refraction in this direction. Likewise, the construction line intersections with the umklapp band (of left-handed type)^[192] determines the phase direction of the higher-order beam (dotted black, $\bar{k}_1$). The group velocity direction (small dotted black, $(\bar{V}_{g1} = \frac{dE}{dk_1})$) is the perpendicular direction to the tangent line at the equivalent point (upper red point) which, accidentally, takes the same direction as the corresponding phase propagation. This special directionality match between phase and group velocity is brought by the band curvature at this specific incidence angle, and not to be confused with the isotropic materials. An excellent agreement between the real-space analysis presented in Fig. 4.29(b) and the momentum space formulism is achieved. A Gaussian electron beam (solid red arrow) of a binding energy +100 meV and waist width of $\omega_0 = 5.25$ nm, hitting the Cu-Ag/Cu interface at an angle of 25° , is split into three different beam. These are, partially reflected (dotted red arrow), positively refracted (solid black arrow), and negatively refracted (dotted black arrow). A close insight into the intensity profiles of those beams revealed that the intensity of zeroth-order is lower than that of higher-order beam. In fact, the main bands always contribute more to the intensity than the weak umklapp process. The observed contrary behavior is reasoned to the shrinks of the photoemission intensity of main band due to the proximity to the gap borders, while

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the spectral weight of umklapp band at this particular incidence is higher, as shown in Fig. 4.29(a).

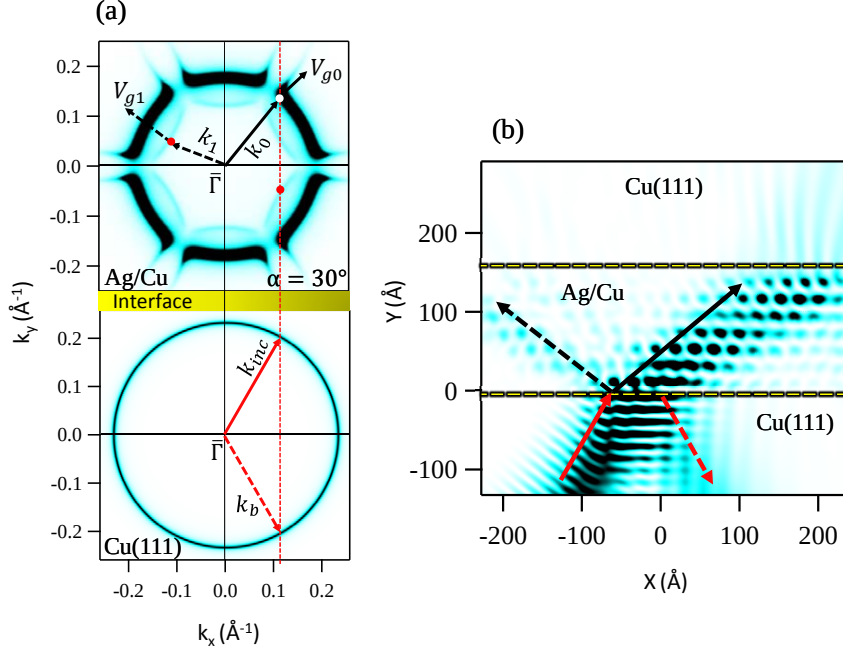


Figure 4.30: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = 100$ meV, $\alpha_{inc} = 30^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 100$ meV. Upper and lower contours of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid red), refracted angle (solid black, $\bar{V}_{g0}$), negative refracted angle (dotted black, $\bar{V}_{g1}$), and reflected angle (dotted red) arrows. The dotted red lines defines the direction of the interface confined evanescent wave (construction line). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 30^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = 100$ meV.

The intensity of the zeroth-order beam (positive refraction) should, in principle, be enhanced by crossing the energy gap associated with main band, i.e., by slightly increasing the incidence angle. Figure 4.30 depicts the electron reflection-refraction at the same lateral interface for an electron beam of 30° incidence. Aside from the usual electron reflection (dotted red arrow, $\bar{k}_b$), the wavevector-space analysis in (a) show that the construction line intersects, both, the first and higher-order bands of the Ag/Cu CES in the white and lower red points, respectively. The intersection with the first

BZ main band (white point) leads to positive refraction with significantly enhanced intensity compared to Fig. 4.29. The phase and group (i.e., power) propagation of this zeroth-order are indicated by the large and small solid black arrows ($\bar{k}_0$, $\bar{V}_{g0} = \frac{dE}{dk_0}$), respectively. Likewise, the intersection with the umklapp band, with negative curvature, determines the higher-order beam phase and power propagations (dotted black, $\bar{k}_1$) and (small dotted black, $\bar{V}_{g1} = \frac{dE}{dk_1}$), respectively.

The expected results according to this wavevector analysis are well captured in the real space simulation in Fig. 4.30(b) as obtained by EBEM. The Gaussian electron beam (solid red arrow) hitting the lateral interface of Cu(111) and Ag/Cu is split into three beams, the reflected beam (dotted red arrow), the zeroth-order positively refracted beam (solid black arrow), and the higher-order negatively refracted beam (dotted black arrow). The beam intensity of the zeroth-order beam is clearly enhanced compared to Fig. 4.29(b), while the umklapp associated beam is relatively suppressed.

In fact a complete suppression of the umklapp contribution to the refraction could be obtained at higher incidence angles. In such a case, the parallel momentum of the incoming electron beam will be greater than the k_1 -radius of the Ag/Cu replica contours (i.e., these left-handed bands do not participate in refraction). Figure 4.31 depicts the electron reflection-refraction at the same lateral interface for an electron beam of 50° incidence. At such a high incidence angles, only first BZ main band satisfy the parallel-momentum conservation. The wavevector-space formalism, shown in (a), demonstrates this argument, where the construction line of the parallel-momentum conservation (red dotted) is clearly far away and not intersecting the replica contour of Ag/Cu dislocation lattice. Only intersection with the main band, of a positive slope, is obtained, which determines the phase velocity direction of the zeroth-order refracted beam (solid black, $\bar{k}_0$). The direction of power or group velocity propagation (small solid black, $\bar{V}_{g0} = \frac{dE}{dk_0}$) is, accidentally, the same as the corresponding phase propagation. Since the CES of Ag/Cu shows nearly flat segments for the main bands, this matching between the phase and group velocity is only possible at this specific angle. Notice, for nearly flat contours, the direction of the group velocity is independent on the incidence angle, as we shall show in the last subsection. The real space simulation image in (b) reveals the presence of only one transmitted positively refracted beam (black solid arrow), in addition to the usual reflection (red dashed arrow), in decent agreements with the wavevector space analysis. The intensity of the higher-order negatively refracted beam, commonly present at lower incident angles, is totally vanished at 50° incidence.

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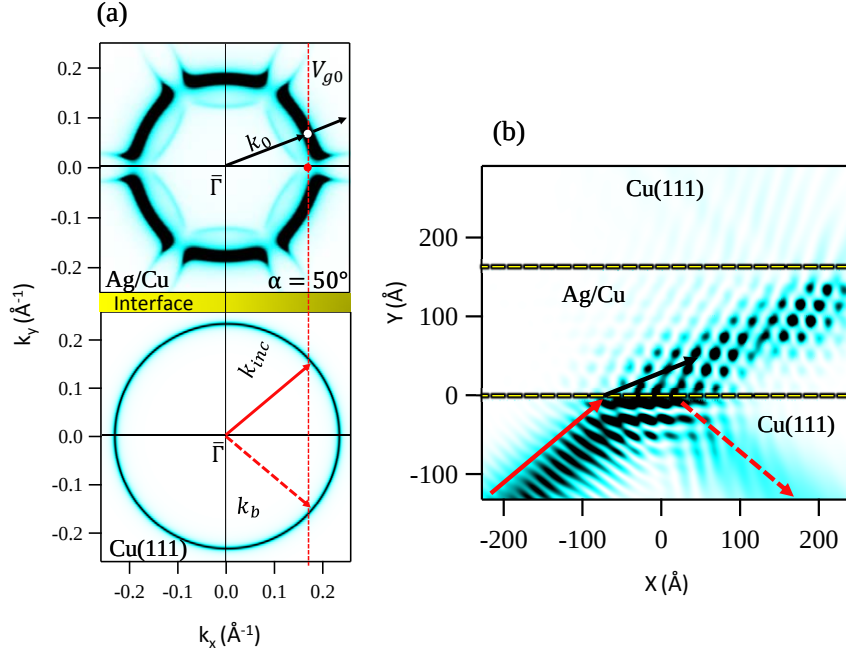


Figure 4.31: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = 100$ meV, $\alpha_{inc} = 50^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 100$ meV. Upper and lower contours of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid red), refracted angle (solid black, $\bar{V}_{g0}$), negative refracted angle (dotted black, $\bar{V}_{g1}$), and reflected angle (dotted red) arrows. The dotted red lines defines the direction of the interface confined evanescent wave (construction line). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 50^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = 100$ meV.

In contrast to homogeneous media where total internal reflection occurs beyond a critical angle, the umklapp bands associated with a nanostructured medium appear throughout the entirely momentum space with small spectral weights at higher order BZ. The achievement of total internal reflection at nanostructured interfaces, thus, requires very high incidence angles, almost grazing incidence. Figure 4.32 displays the possibility of total internal reflection behavior for incidence angle of 75° . The wavevector space diagram, in (a), confirms that the construction line (dotted red) for an incidence beam (red arrow) does not intersect either bands (main or umklapp) of the CES of refractive medium (Ag/Cu). Since the parallel-momentum component of this incidence beam is greater than the two wavevectors $(\bar{k}_0, \bar{k}_1)$ of the refractive medium,

they do not participate in the refraction process. In fact, the construction line intersects bands at the second BZ, but these have almost no spectral weight. In such a situation, the construction line just intersects the CES of Cu(111) in the momentum position which defined the direction of the reflected wavevector (dotted red, $\bar{k}_b$). Therefore, all intensities of this beam must be totally reflected back to the incidence medium. EBEM real space simulation presented in (b) for 75° incident electron beam (solid red arrow), clearly showed perfect agreements with the previous analysis, where the beam is totally reflected back to Cu(111) by the same angle (dotted red arrow).

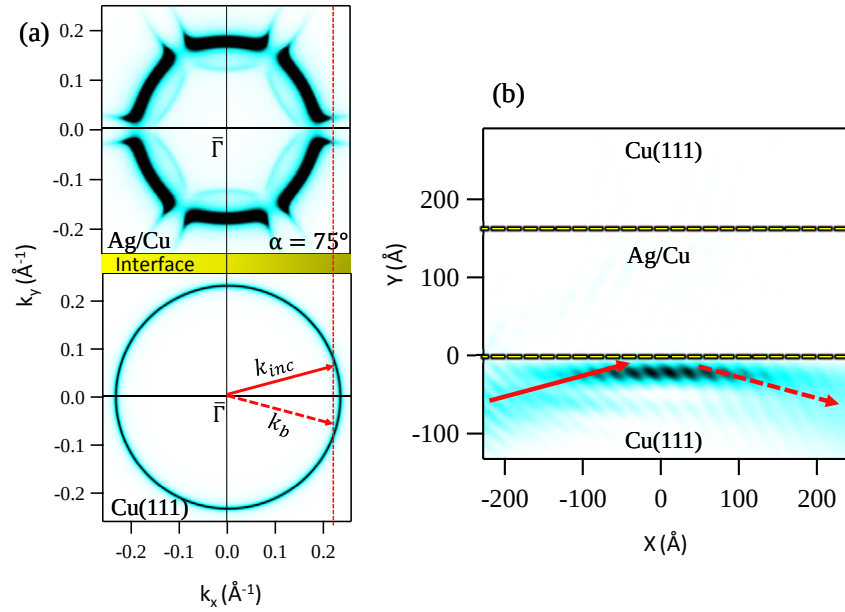


Figure 4.32: Electron refraction at 1 ML Ag/Cu(111)-Cu(111) interface, $E_B = 100$ meV, $\alpha_{inc} = 75^\circ$ - (a) Wavevector-space analysis of the refraction/reflection at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 100$ meV. Upper and lower contours of (a) are the 1 ML Ag/Cu and Cu(111) CESs, respectively. The arrows represent electron beams with incidence angle (solid red), refracted angle (solid black, $\bar{V}_{g0}$), negative refracted angle (dotted black, $\bar{V}_{g1}$), and reflected angle (dotted red) arrows. The dotted red lines defines the direction of the interface confined evanescent wave (construction line). (b) Real-space simulations of electron refraction/reflection at this lateral interface. Calculation are performed with a Gaussian electron beam of incidence angle $\alpha_{inc} = 75^\circ$, waist width $\omega_0 = 5.25$ nm, and $E_B = 100$ meV.

In conclusion, using combined wavevector-space analysis (EPWE) and real-space simulations (EBEM) we showed that the CESs of nanostructured metal surfaces (Ag/Cu)

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are usually not circular and can have quite complicated shapes, particularly at high electron energies. These anisotropic energy contours, frequently exhibit significant influences to the electron reflection-refraction at the homogeneous-nanostructured lateral interface between Cu(111) and 1 ML of Ag/Cu. The right-handed main band of this CES leads to either positive or negative refractions depending on the incidence angle, while their left-handed umklapp bands constantly produce the negative refraction^[188,192].

4.4.4 The 1 ML Ag/Cu(111) Main Flat Contours: Self-Collimation

As mentioned before, electron propagation through a nanostructured metal surface is governed by its dispersion and the full shape of its constant energy surfaces (CES). Generally speaking, incident electron waves propagate in directions normal to the dispersion surface^[84]. That define the direction of the group velocity, i.e., the power or intensity flow. The usual curved dispersion contours can produce beam divergence or convergence, whereas straight CES contour segments leads to the so-called self-collimation phenomenon, as shown in Fig. 4.33. Self-collimation, also known as auto-collimation or self-guiding, allows a narrow beam to propagate in the electronic crystal without any significant broadening or change in the beam profile, and without relying on a band gap or engineered defects, such as the waveguide shown, e.g., in Fig. 4.13. This property can be used for wave-guiding and dense routing of electronical signals. Tailoring the dispersion characteristics of a photonic or electronic crystal structure to create unique devices has been an important area of research^[193,194]. By analogy to photonics and plasmonics, Ref.^[146] anticipated self-collimation, focusing, wave-guiding, and energy-filtering of electronic surface states can provide the basic building blocks for future surface state nanoelectronics (SSNE) devices. In this section, the electron self-collimation through an Ag/Cu nanostripe will be discussed. These nanostripes could be produced experimentally as reported for the Ag/Cu(111) system^[56].

Figure 4.33(a) depicts the wavevector-space analysis for electron collimation at the lateral interface between homogeneous Cu(111) and nanostructured Ag/Cu metal surfaces. An electron point source positioned at the interface (at $\bar{\Gamma}$ -point) injects electrons of energy 600 meV, in all directions. The point source is placed inside the Cu(111) medium, right at the interface. The white arrows ($\bar{k}_{inc}$) stands for two arbitrary incidence directions from the point source. The construction lines enclosing these directions are represented by the vertical dashed red lines. The two construction lines intersect the Ag/Cu contour at the borders of its characteristic flat segment, leading to two

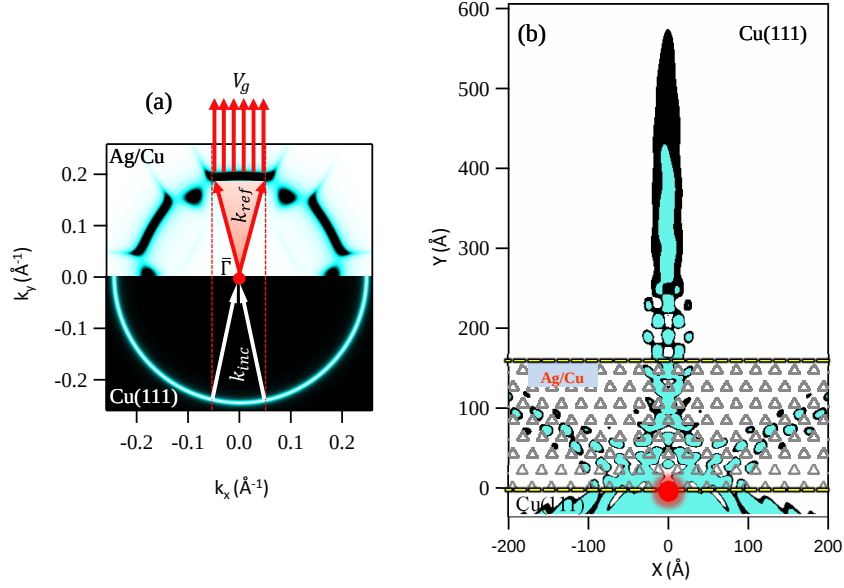


Figure 4.33: Electronic Self-Collimation - (a) Wavevector-space analysis of the electron collimation at the 1 ML Ag/Cu(111)-Cu(111) interface performed for electron energy $E_B = 200$ meV. Upper and lower halves are the 1 ML Ag/Cu and Cu(111) CEs, respectively. (b) Real-space simulations of electron collimation, a point source at the interface (red circle) injects an electron (200 meV) into the Ag/Cu that is collimated after transmission through a lattice stripe

transmitted rays with their phase directions, according to the parallel-momentum conservation, follow the red arrows ($\bar{k}_{ref}$). Since the main bands contours exhibit perfectly flat segments, the group velocity at the two beam incidence, and all incidence angles in between, points to the same direction, that is perpendicular to the flat segment (vertical solid red arrows). Therefore, all incoming wavevectors are collimated in the vertical direction after crossing the interface. Figure 4.33(b) presents the real-space simulation result for an electron point source (red circle) of energy 600 meV injected right at the interface between Cu(111) and 1ML of Ag/Cu. Although the point source emits in all direction inside the Cu(111) medium, a finely collimated beam is clearly seen in the vertical direction. The beam is highly collimated and do not diverge after crossing the second interface towards Cu(111). Actually, due to the six-fold symmetry of the Ag/Cu lattice, the other two flat segments also collimate electrons in directions making an angle of 60° with the vertically collimated beam. The most attractive properties of such nanoscale electron-collimation are that they diffract weakly along the propagation trajectories and have the amazing ability to heal them due to periodic modulation of

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the propagating medium, i.e., the 1ML of Ag/Cu dislocation triangular lattice^[195].

Part III: Chapter Conclusion

4.5 Conclusion:

In conclusion, using combined wavevector-space analysis (EPWE) and real-space simulations (EBEM) we showed that the propagation of material waves (electrons) at the interface between two surfaces is solely determined by their band structures and the size/shape of their CESs.

At the lateral interface between two dissimilar homogeneous metal surfaces of CESs with circular contours, electron refraction takes place following Snell's law, and total internal reflections occur beyond energy-dependent critical angles. Additionally, we established the analogue with geometrical optics by simulating an electronic convex lens and mirror-based electron guiding. Given the parabolic dispersion of the electronic surface state, we speculate the possibility of defining an electronic interface where no refraction is possible at all incidence angles, a finding that has no optical counterpart. With the widely available experimental means of manipulating the electronic dispersions, the electron refraction-reflection at metallic interfaces could be precisely tuned.

On other hand, nanostructured metal surfaces of CESs with quite complicated shapes affected on electron propagation as a photonic crystal-like behavior. The Ag/Cu nanostructured metal surface featured isotropic/anisotropic and gapped constant energy surfaces (CESs), i.e., achieved anomalous behaviors of electron propagation via its both main-umklapp bands. It acted, in some energy regime, as right handed-like materials, and at other energies exhibit left handed-like behaviors. Using EBEM real space simulation many varieties of ordinary and extraordinary optical phenomena were observed for a Gaussian electron beam hitting the Cu-Ag/Cu interface, such as total internal reflection, positive and/or negative refractions, self-collimation, and superprism effect.

4. RESULTS AND DISCUSSION II

5

Arabic Abstract

5.1 Arabic Abstract

الملخص العربي

التركيب الشريطي (التوزيع الإلكتروني) داخل البلورات الأحادية للمعادن كثيراً ما يحتوي على فجوات للطاقة عند نقاط التماثل العالية داخل الخلية الأولية للبلورة (Brillouin zone). ومن ثم، فإن بنية النطاق السطحي تُبرهن على وجود فجوات طاقة داخلية ذات مساقط مُحددة، على غرار أشباه الموصلات، حيث تُوجد حالات إلكترونية جديدة. وتُعرف هذه الحالات الإلكترونية "بالحالات السطحية" والتي لها نطاق تشتت يقع بكامله داخل فجوات الطاقة ذات المساقط المُحددة سالفة الذكر. أسطح المعادن النبيلة، على سبيل المثال تمتلك مثل هذه الحالات الإلكترونية السطحية من نوع شوكللي (Shockley) والتي توصف بتشتت الإلكترون الحر والذي يكون على هيئة القطع المكافئ. مثل هذه الإلكترونات السطحية الحرة ذات طول موجي فيرمي يتراوح بين 1-3 نانومتر، وعليه فإن إنشاء أسطح معدنية دورية ثنائية الأبعاد لها معامل تكرار يساوي القيمة السابقة للطول الموجي تُمكننا من الحصول على المكونات السطحية ذات التركيبات الإلكترونية المطلوبة في الصناعة. تقنيات، مثل التجميع الذاتي، جعلت تصنيع هذه التركيبات الدورية النانومترية ممكناً.

باستخدام الطريقة الحسابية التي تُعرف باسم توسيع الموجة الإلكترونية المُستوية (EPWE) نُقدم هنا تحليلاً نظرياً مفصلاً ومُمنهجاً لتشتت الإلكترون بسبب وجود حواجز جهود فوق أسطح المعادن تأخذ الشكل المُسدس المُنتظم تارةً، والشكل المُربع تارةً أخرى. وقد أوضحت النتائج أنه في كلتا الحالتين تحدث إزاحة لأعلى للتركيب الإلكتروني الشريطي بالكامل تتوافق أو تتناسب مع حجم جهد التشتت للإلكترون. والأهم من ذلك، فتحت فجوات طاقة عند حدود الخلية الأولية للبلورة (Brillouin zone) اعتماداً على كل من جهد التشتت، ومُعامل الدورية، وشكل حاجز الجهد. هذه النتائج قد تمت دراستها ومناقشتها هنا من خلال تحليلات كاملة لحسابات كل من التركيب الشريطي (التوزيع الإلكتروني)، وكذلك حسابات أسطح الطاقات الثابتة (CES)، وأيضاً حسابات الكثافة المحلية لمستويات الطاقة الإلكترونية (LDOS).

انكسار الموجات الكهرومغناطيسية من خلال مرورها عبر حد فاصل بين وسطين كليهما متجانس و لكنهما مختلفين من الناحية البصرية يُحكم ويُحدد من خلال قانون سنيل (Snell's law). هذا القانون الذي يُعتبر حالة خاصة من القاعدة العامة المعروفة ببقاء كمية الحركة (momentum conservation) عند السطح البيني الفاصل والتي تقضي بتساوي المركبات المتوازية لمتجهات الموجة عند السطح الفاصل. واستناداً إلى الطبيعة المزدوجة فإن موجات المواد أيضاً سوف تنكسر عند مرورها عبر السطح الفاصل بين مادتين مختلفتين من الناحية الإلكترونية. ومن ثم فإن إلكترونات الأسطح يمكن أن تنكسر بعبورها الحد الفاصل بين سطحي مادتين مختلفتين إلكترونياً. الكمية الهامة لحدوث مثل هذا الانكسار هي وجود فرق بين متجهي الموجة (Δk) عند طاقة معينة، والتي تُحددها فقط علاقات التشبث والتركيب الإلكتروني.

عن طريق الدراسة المشتركة الناتجة عن كل من التحليل الفراغي لمتجه الموجة بواسطة الطريقة الحسابية (EPWE) ومحاكاة الفراغ الحقيقي بواسطة طريقة الحدود للعناصر الإلكترونية الحسابية (EBEM) ندرس هنا كل من الانعكاس والانكسار للإلكترون عند مروره عبر حد فاصل بين سطحين متجانسين وأيضاً عند مروره بين سطحين أحدهما متجانس والآخر له تركيب دوري نانومتري. في كلتا الحالتين سالفتي الذكر للحدود الفاصلة، برهنت النتائج أن الشعاع الإلكتروني ينكسر وفقاً لقانون سنيل، وأنه ينعكس كلياً عندما تكون زاوية السقوط أكبر من الزاوية الحرجة. ومن المثير للاهتمام، أنه قد تم إثبات حدوث ظواهر بصرية استثنائية إضافية عند مرور الإلكترون عبر الحد الفاصل بين سطحين أحدهما متجانس والآخر له تركيب دوري نانومتري، مثل الانكسار السالب، والتوجيه الذاتي، وتأثير المنشورات الفائقة (superprism effect). الحد الفاصل بين سطحين أحدهما متجانس والآخر له تركيب دوري نانومتري الذي تمت الدراسة من خلاله هنا عبّاره عن الحد الفاصل بين سطح النحاس و سطح آخر مكون من طبقة واحدة (IML) من الفضة (Ag) والتي نمت على بلورة أحادية من النحاس [Cu(111)] وهو نظام حقيقي قد أُجري تحضيره معملياً من خلال عديد من الأبحاث التجريبية.

5. ARABIC ABSTRACT

Appendix 6

Acronyms

BZ Brillouin Zone
SBZ surface Brillouin zone
SAMs Self-Assembled Monolayers
STM Scanning Tunneling Microscopy
ARPES Angle Resolved Photoemission Spectroscopy
DFT Density Functional Theory
EPWE Electron Plane Wave Expansion method
EBEM Electron boundary element method
CES Constant Energy Surface
LDOS Local Density of States
OPW Orthogonalized Plane Wave

LCAO Linear Combination of Atomic Orbitals
STS Scanning Tunneling Spectroscopy
DOS Density of States
PE photoemission
XPS X-ray Photoemission Spectroscopy
VUV vacuum ultraviolet
UPS Ultraviolet Photoemission Spectroscopy
SSNE surface-state nanoelectronics
PC Photonic Crystals
EM electromagnetic
EFC Equifrequency Contours

6. ACRONYMS

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